ESE 3207 Complete Solutions

Solar Photovoltaic Systems · KUET · past papers 2019, 2020, 2021, 2022, 2023 · answers from the teachers' material (Rahaman sir: Parts I-III; Muhit sir: lectures L1-L11) and the books they follow (Smets et al. *Solar Energy*, Messenger *Photovoltaic Systems Engineering*)

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 4 h for Part A and 2 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 ...) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×5, 2022 ×4, 2021 ×3, 2020 ×2, 2019 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Retake tip: read the Concept Primer first (30 minutes). Part B is the easier half: on-grid/off-grid and net metering (B5), temperature and irradiance effects (B1), bypass diodes and shading (B2) and the solar home system sizing numerical (B6) come every year. In Part A, learn the I-V curve, FF and efficiency (A4, with a calculator), losses and the S-Q limit (A5) and quartzite-to-wafer (A6). Junction physics numericals (A3) are the hardest: use the calculator to practise them last.

Section A: Solar cell physics and technology

Block 1: very likely and easy · 1.2 h

Topic	Chance	Level	Covered
A6 Silicon wafer and c-Si cell production: quartzite to wafer, cell fabrication, lithography, TCO, supporting layers	Very likely 100%	Easy	18%
A5 Losses and efficiency limits: optical and electrical losses, design rules, back surface field, thermodynamic and Shockley-Queisser limits	Very likely 100%	Easy	31%
A4 Solar cell parameters and equivalent circuit: I_{sc}, V_{oc}, FF, efficiency, R_s and R_{sh}, quantum efficiency	Very likely 100%	Easy	42%

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
A8 Multi-junction, perovskite and third-generation cells; generations of solar cells	Very likely 100%	Medium	58%
A1 Semiconductor basics: bands, doping, Fermi level, direct/indirect gap, drift and diffusion, Poisson and continuity equations	Very likely 100%	Medium	70%

A7 Thin-film cells: III-V, CdTe, CIGS, thin-film silicon (a-Si:H), materials criteria	Very likely 87%	Medium	82%
✓ Pass line: 82% of Section A marks covered, enough to choose 3 of the 4 questions (about 3.5 h)			
A2 Generation and recombination: radiative, SRH, Auger, surface; diffusion length; photocurrent numerical	Very likely 87%	Medium	90%

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
A3 Junctions: p-n junction solar cell, p-i-n, metal-semiconductor, diode equation; junction numericals	Very likely 100%	Hard	100%

Section B: PV systems

Block 1: very likely and easy · 1.7 h

Topic	Chance	Level	Covered
B5 PV system types: standalone (SHS), grid-connected, with storage, hybrid; net metering; grid interfacing; standalone applications; water pumping; SHS installation and testing	Very likely 100%	Easy	25%
B1 PV cell, module, panel and array; series and parallel; effects of irradiance, temperature, area, orientation and tilt	Very likely 100%	Easy	45%
B7 Economics and ecology: payback time, energy payback time, LCOE, life-cycle cost, types of cost, carbon footprint, recycling	Very likely 100%	Easy	58%
B2 Shading and mismatch; hot spots; bypass and blocking diodes	Very likely 93%	Easy	67%

Block 2: very likely, medium effort · 1.5 h

Topic	Chance	Level	Covered
B6 Designing and sizing a PV system (solar home system numericals)	Very likely 100%	Medium	78%
✓ Pass line: 78% of Section B marks covered, enough to choose 3 of the 4 questions (about 2.4 h)			
B3 Battery, charge controller (set points, PWM, shunt and series regulators), MPPT (P&O, incremental conductance), DC-DC converters	Very likely 87%	Medium	96%

Only if time is left · 0.4 h

Topic	Chance	Level	Covered

B4 Inverters: types by waveform and by mode, THD, converter efficiency and inverter selection numericals	Possible 47%	Easy	100%

CONTENTS

1. One-shot notes
2. **Section A: Solar cell physics and technology**: A6 Silicon wafer and c-Si cell production: quartzite to wafer, cell fabrication, lithography, TCO, supporting layers, A5 Losses and efficiency limits: optical and electrical losses, design rules, back surface field, thermodynamic and Shockley-Queisser limits, A4 Solar cell parameters and equivalent circuit: I_{sc}, V_{oc}, FF, efficiency, R_s and R_{sh}, quantum efficiency, A8 Multi-junction, perovskite and third-generation cells; generations of solar cells, A1 Semiconductor basics: bands, doping, Fermi level, direct/indirect gap, drift and diffusion, Poisson and continuity equations, A7 Thin-film cells: III-V, CdTe, CIGS, thin-film silicon (a-Si:H), materials criteria, A2 Generation and recombination: radiative, SRH, Auger, surface; diffusion length; photocurrent numerical, A3 Junctions: p-n junction solar cell, p-i-n, metal-semiconductor, diode equation; junction numericals
3. **Section B: PV systems**: B5 PV system types: standalone (SHS), grid-connected, with storage, hybrid; net metering; grid interfacing; standalone applications; water pumping; SHS installation and testing, B1 PV cell, module, panel and array; series and parallel; effects of irradiance, temperature, area, orientation and tilt, B7 Economics and ecology: payback time, energy payback time, LCOE, life-cycle cost, types of cost, carbon footprint, recycling, B2 Shading and mismatch; hot spots; bypass and blocking diodes, B6 Designing and sizing a PV system (solar home system numericals), B3 Battery, charge controller (set points, PWM, shunt and series regulators), MPPT (P&O, incremental conductance), DC-DC converters, B4 Inverters: types by waveform and by mode, THD, converter efficiency and inverter selection numericals

★ One-shot notes

One-shot notes: ESE 3207 Solar Photovoltaic Systems

Constants

q 1.602e-19 C; h 6.626e-34; c 3e8; hc = 1240 eV·nm; kT/q 0.0258 V; ε_{Si} = 11.7 × 8.854e-14 F/cm; Si E_g 1.12 eV, n_i 1.5e10, N_C 3.22e19, N_V 1.83e19 cm^{-3}. STC 1000 W/m^2, AM1.5, 25 °C.

Doping and bands (A1)

P (V) donor → n, E_D below E_C, E_F up. B (III) acceptor → p, E_A above E_V, E_F down. Purpose: conductivity, majority carrier, junction field. np = n_i^2.

Fermi = 50% occupancy; DOS g(E) ∝ √(E − E_C); Poisson $d^2\psi/dx^2 = -\rho/\varepsilon$ (field from charge); continuity: change = flow + G − R; L = √(Dτ).
Direct: same k, photon only, α high (GaAs, CdTe). Indirect: needs phonon (Si, Ge). Drift J = qnμE; diffusion J = qD dn/dx; D/μ = kT/q.
Quantum: quantised energy (E = hν), wave-particle duality, uncertainty, Pauli/allowed states.

Recombination (A2)

Radiative (photon, direct gap); SRH (trap in gap, phonons, defects; dominant in Si); Auger (third carrier, heavy doping); surface (dangling bonds; cure SiN_x/Al_2O_3, BSF). G = R in equilibrium.

2021 1c: Φ_0 = Pλ/hc = 2.52e21; ×0.62 = 1.56e21; αd = 333 so all absorbed; **J = 250 A/m^2** (book 248).

Junction (A3)

p-n: diffusion → ions → field → light pairs diffuse to edge → field separates → contacts collect. p-i-n: field across i layer, drift device (a-Si, perovskite). Schottky: $\varphi_m > \varphi_s$ (n) barrier, rectifying; ohmic: no barrier / heavy doping.

$V_{bi} = kT/q \ln(N_A N_D/n_i^2)$; $W = \sqrt{(2\varepsilon(V_{bi} - V)(1/N_A + 1/N_D)/q)}$.

2022 3d: **0.81 V, 0.325 µm, 5e4 V/cm**. 2021 3b: p = 1e16, n = 2.25e4; n = 1e18, p = 225. 2020 3b: L_n 25.7, L_p 22.4, W 0.11 µm, **I_L = qAG(L_n + W + L_p) = 11.6 A**.

Parameters and circuit (A4)

I_{sc}, $V_{oc} = kT/q \ln(I_L/I_0 + 1)$, MPP, FF = $V_m I_m / V_{oc} I_{sc}$, η = P_m/(GA), EQE/IQE.

Real: $I = I_L - I_0(e^{q(V+IR_s)/nkT} - 1) - (V + IR_s)/R_{sh}$; two-diode n = 1 and 2. R_s (bulk, emitter, fingers, contacts) tilts near V_{oc}; R_{sh} (edge leaks, defects) tilts near I_{sc}; both cut FF.

2020 1a: **FF 0.781, η 13.0%**. 2021 5d: **FF 0.724, η 17.8%** (P_{in} 280.6 W).

Losses and limits (A5)

Spectral: non-absorption (~19%), thermalisation (~33%). Optical: shading, reflection, parasitic, transmission. Electrical: recombination, R_s, R_{sh}; voltage factor, FF.
Rules: texture + ARC, fewer/finer fingers, back reflector; passivate, BSF, selective emitter; heavy doping under contacts. BSF: p/p^+ field repels electrons from rear, +V_{oc}, +red response.

Thermodynamic 86.8% (conc.), 68% (1 sun, ∞ junctions); Carnot 95%. SQ: one junction, radiative only → **33.7% at 1.34 eV**; Si practical 29.4%.

Si wafer and cell (A6)

Quartzite: > 98% SiO_2, lumpy, stable, cheap. SiO_2 + 2C → Si + 2CO (arc 1900 °C, MG 98%) → Si + 3HCl → $HSiCl_3$ + H_2 → distil → Siemens 1100 °C → poly 9N → CZ/FZ ingot → square → wire saw → lap, etch, polish.
Cell: damage etch, texture (KOH), $POCl_3$ 850 °C, PSG off (HF), edge isolate, SiN_x PECVD, print Ag/Al, co-fire (Al-BSF), test.
Litho: clean, spin resist, soft bake, align + UV expose, develop, hard bake, etch, strip. TCO: ITO/FTO/AZO, E_g > 3 eV, > 80% T, conductive front electrode for thin films.

Thin film and advanced (A7, A8)

Criteria: E_g 1.1-1.7, high α, long L, dopable, stable, abundant, non-toxic, cheap. III-V: GaAs, InP, GaInP, AlGaAs; thin because direct gap (1-3 µm), epitaxial; 29%/47%.

CdTe: glass / TCO / n-CdS / p-CdTe (1.45 eV, $CdCl_2$) / p^+ back contact. a-Si:H: H passivates dangling bonds; p-i-n drift; Staebler-Wronski.

MJ: GaInP 1.86 / GaAs 1.42 / Ge 0.67, series + tunnel junctions; challenges: bandgap, current, lattice matching, tunnel junction, cost. Lattice 5.65 Å all three. Perovskite ABX_3 ($MAPbI_3$): tunable direct gap, solution processed, > 26%; n-i-p FTO/TiO_2/pvk/spiro/Au; unstable, Pb.

Gen I wafer Si; II thin film; III beyond SQ: MJ, up/down conversion, MEG, hot carrier, intermediate band, perovskite, DSSC, OPV.

Module and array (B1)

Cell 0.5 V; module = 36/60/72 in series (Wp at STC); array = strings in parallel. Series V adds; parallel I adds. $I_{sc} \propto G$ and area; $V_{oc} \propto \ln G$; V_{oc} −2.3 mV/°C; P −0.45%/°C. Face south, tilt ≈ latitude (23°).

2022 5c: 75 Wp → **68.25, 64.9, 61.5 W**. L3: 19 cells (0.765 V at 60 °C); 6 parallel. Cooling: air gap, PV/T, water, low-γ tech.

Shading (B2)

Series current = weakest cell; shaded cell reverse biased → hot spot. Bypass diode: parallel, reverse, per 18-24 cells. Blocking diode: series, stops night discharge and reverse string current. Mismatch: manufacturing/operating; current (series) vs voltage (parallel). Fix: bypass, binning, multi-MPPT, optimisers, micro-inverters, global MPPT.

Controller, MPPT, DC-DC (B3)

Functions: overcharge, deep discharge (LVD), reverse current, load control. Set points (12 V): VR 14.4, ARV 13.2 (VRH), LVD 11.5, LRV 12.6 (LVDH). Shunt: shorts array + blocking diode. Series: opens line.

PWM: bulk → pulses narrow as SOC rises → float. MPPT: DC-DC + algorithm, impedance matching R = V_m/I_m, +10-30%. P&O: step, keep if P up, else reverse (oscillates). IncCond: dI/dV = −I/V at MPP (stops, fast). Buck $V_o = DV_i$; boost $V_i/(1 - D)$.

SG 1.265 full; autonomy days; lm/W; THD. L5: 5 A, 20 A; Ah 83.3%, Wh 69%.

Inverters (B4)

Square (THD ~45%), modified sine (~20%), pure sine (< 5%, SPWM + LC). Standalone, grid-tied, bimodal. Cut THD: SPWM, filters, multilevel, SHE.

2022 6c: **264 W out, 36 W loss**. 2021 6c: 420 W → input 466.7 W, 38.9 A DC → **600 VA 12 V pure sine**. L9: 4000 W, 16.7 A.

Systems and net metering (B5)

Off-grid: PV → CC → battery → inverter → load. On-grid: PV → GTI → DB + net meter → grid. With storage: hybrid inverter, backup, self-consumption. Hybrid: PV + diesel/wind + ECC.

Grid conditions: V, f, phase sync, THD < 5%, DC < 0.5%, pf, anti-islanding 2 s, 5 min reconnect, lockable switch, meter.

Net metering: pay import − export. BD 2018: 3-phase consumer, ≤ 70% sanctioned load, own premises, no dues, approved inverter + bi-directional meter, agreement.

Pump direct: cheap, no battery, day only. With battery: anytime, costly, losses.

SHS sizing (B6)

E → PV = E/(η·PSH) → Ah = E·days/(V·DOD) → CC = 1.25I → inverter = 1.25P → switch 1.25I → A = 2ρLI/ΔV.

2023/2019: E **3280 Wh**, load 1060 W; 4373/5 = 875 → **900 Wp**; **1822 Ah** (10 × 200); CC **100 A**; inverter **1500 VA**; switch **125 A**; cables 25, 50, 16 mm^2 (lengths assumed).

2022 6d: E **3180**; **700 Wp**; **1060 Ah**; CC 80 A; inverter 800 VA-1 kVA.

Economics and ecology (B7)

PBT = money back (5-10 yr); EPBT = energy back (1-3 yr). Energy yield ratio 10-15. LCOE = $\Sigma(I + M + F)/(1 + r)^t \div \Sigma E/(1 + r)^t$; grid parity, socket parity.

Costs: capital, O&M, replacement, fuel, finance, salvage. x = 1.03/1.10; P_a 11.50, P_{a1} 10.76.

2020 6c: PV **$6,726** (3000 + 300 + 900 + 648 + 466 + 336 + 1076); generator **$25,727** (fuel $1.25/gal assumed).

CO_2: PV 20-50 g/kWh vs coal ~900; from Si purification. Recycle frame, glass, Si, Ag, Cu; dumping leaches Pb, Cd.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
a-Si:H	hydrogenated amorphous silicon
Ah	ampere-hour
ALCC	annualised life-cycle cost
AM	air mass
ARC	anti-reflection coating
ARV	array reconnect voltage
AZO	aluminium-doped zinc oxide
BOS	balance of system
BSF	back surface field
CB	conduction band
CIGS	copper indium gallium diselenide
CZ	Czochralski
CZTS	copper zinc tin sulfide
DOD	depth of discharge
EPBT	energy payback time
EQE	external quantum efficiency
ETL	electron transport layer
EVA	ethylene vinyl acetate (encapsulant)
FA	formamidinium
FF	fill factor
FTO	fluorine-doped tin oxide
FZ	float zone
HIT	heterojunction with intrinsic thin layer
HTL	hole transport layer
HVD	high-voltage disconnect
IBC	interdigitated back contact
IEEE	Institute of Electrical and Electronics Engineers
IncCond	incremental conductance
IQE	internal quantum efficiency
ITO	indium tin oxide
LCA	life-cycle assessment
LCC	life-cycle cost
LCOE	levelized cost of electricity
LRV	load reconnect voltage
LVD	low-voltage disconnect
LVDH	low-voltage disconnect hysteresis

MA	methylammonium
MBE	molecular beam epitaxy
MEG	multiple exciton generation
MG-Si	metallurgical-grade silicon
MOCVD	metal-organic chemical vapour deposition
MPP	maximum power point
MPPT	maximum power point tracking (tracker)
nc-Si:H	hydrogenated nanocrystalline silicon
NOCT	nominal operating cell temperature
P&O	perturb and observe
PBT	payback time
PECVD	plasma-enhanced chemical vapour deposition
PERC	passivated emitter and rear cell
PERL	passivated emitter, rear locally diffused
PID	potential-induced degradation
PSG	phosphosilicate glass
PSH	peak sun hours
PV	photovoltaic
PV/T	photovoltaic-thermal (hybrid collector)
PWM	pulse width modulation
SHS	solar home system
SOC	state of charge
SPV	solar photovoltaic
SPWM	sinusoidal pulse width modulation
SQ	Shockley-Queisser
SREDA	Sustainable and Renewable Energy Development Authority (Bangladesh)
SRH	Shockley-Read-Hall (trap-assisted recombination)
STC	standard test conditions (1000 W/m^2, AM1.5, 25 °C)
TCO	transparent conducting oxide
TCS	trichlorosilane ($HSiCl_3$)
THD	total harmonic distortion
TOPCon	tunnel oxide passivated contact
VA	volt-ampere
VB	valence band
VFD	variable frequency drive
VR	voltage regulation (set point)

VRH	voltage regulation hysteresis
Wp	watt-peak (power at STC)

Section A: Solar cell physics and technology

A6 Silicon wafer and c-Si cell production: quartzite to wafer, cell fabrication, lithography, TCO, supporting layers

Pass set · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q3(d)** [12] Discuss the production of silicon wafer from quartzite.
- **2022 Q2(a)** [13] What are the key characteristics of quartzite that make it a suitable source material for silicon wafer production? Discuss the production of silicon wafer from quartzite.
- **2022 Q4(a)** [10] Why is TCO (transparent conducting oxide) used in a solar cell? Also explain why III-V material is considered for thin-film solar technology. (III-V part: A7)
- **2022 Q2(d)** [7] Why are supporting layers used in solar cells and what considerations are taken into account when selecting this crucial component?
- **2021 Q2(c)** [12] Briefly explain the production of c-Si wafer from quartzite.
- **2021 Q2(b)** [11] What is meant by lithography? Write down the steps of lithography with a proper diagram.
- **2020 Q4(b)** [18] Describe in detail how to make a polished wafer of Si from the starting materials.
- **2020 Q4(a)** [12] What is meant by lithography? Write down the steps of lithography with a proper diagram.
- **2020 Q3(c)** [6] Why are supporting layers used in solar cells? Discuss.
- **2019 Q3(a)** [14] Discuss the production of silicon wafer from quartzite.
- **2019 Q3(b)** [14] Discuss the fabrication of a c-Si solar cell from a silicon wafer.
- **2019 Q3(c)** [7] What is TCO? Explain.

★ REMEMBER IT

- Quartzite to wafer: **Q-M-S-C-W**: Quartzite (SiO_2) reduced with carbon in an arc furnace → **M**etallurgical-grade Si (98–99%) → **S**iemens process (HCl → trichlorosilane, distil, deposit on hot rods) → polysilicon 9N → **C**zochralski pulling (single crystal ingot) or casting (multi) → **W**ire-saw into wafers, etch, polish.
- Cell fabrication: saw-damage etch and texturing → phosphorus diffusion (emitter) → edge isolation → SiN_x anti-reflection coating → screen-print Ag front and Al back → firing (Al forms the BSF (back surface field)) → testing and sorting.
- Lithography: clean → coat photoresist → soft bake → expose through a mask (UV) → develop → etch → strip resist. TCO (ITO, FTO (fluorine-doped tin oxide), AZO (aluminium-doped zinc oxide)): transparent and conductive front contact for thin-film cells.

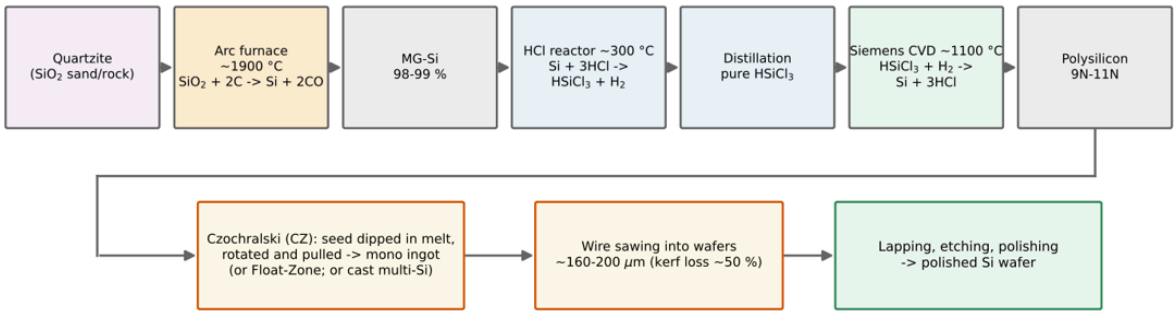
Before you start: from sand to wafer

Silicon is the second most abundant element in the Earth's crust, but only as SiO_2 (quartz) and silicates. A solar cell needs it 99.9999% (6N) pure or better and as a single crystal, so the material goes through a long chain: reduction, chemical purification, crystal growth, sawing. Source: Rahaman sir Part III sec. 12.2-12.4 (Smets ch. 12); details from Smets and standard texts.

Why quartzite is a suitable source (2022 2a, first part)

- Very high SiO_2 content (> 98-99%) with low impurity levels (Fe, Al, Ti, Ca, B, P), so less purification is needed.
- Hard, dense, lumpy rock that stays strong when heated: gives a permeable charge in the arc furnace so CO gas can escape (fine sand would choke the furnace).
- Thermally stable, does not decrepitate (break into powder) at high temperature.
- Abundant and cheap, widely distributed, easy to mine and wash.

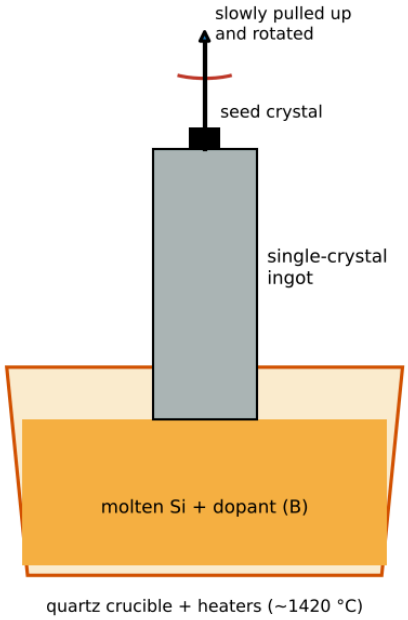
Production of silicon wafers from quartzite (2023 3d, 2022 2a, 2021 2c, 2019 3a, 2020 4b)



From quartzite to polished silicon wafer

1. **Reduction to metallurgical-grade Si (MG-Si):** quartzite + carbon (coal, coke, wood chips) in a submerged-electrode **arc furnace** at about 1900 °C: **$SiO_2 + 2C \rightarrow Si + 2CO$**. Product: MG-Si (metallurgical-grade silicon), 98-99% pure, tapped as liquid and crushed into powder. (Most MG-Si goes to aluminium and silicone industries.)
2. **Conversion to trichlorosilane:** MG-Si powder reacts with HCl in a fluidised-bed reactor at ~300 °C: **$Si + 3HCl \rightarrow HSiCl_3 + H_2$**. $HSiCl_3$ (TCS) is a liquid boiling at 31.8 °C.
3. **Purification by fractional distillation:** impurities form chlorides with different boiling points ($FeCl_3$, BCl_3, PCl_3), so repeated distillation gives ultra-pure TCS (ppb impurity level).
4. **Siemens process (chemical vapour deposition):** TCS (trichlorosilane ($HSiCl_3$)) + H_2 are passed over thin silicon rods heated electrically to ~1100 °C in a bell-jar reactor: **$HSiCl_3 + H_2 \rightarrow Si + 3HCl$**. Pure Si deposits on the rods, which grow to 15-20 cm diameter: **polysilicon**, 9N-11N (electronic grade). (Cheaper alternatives: fluidised-bed reactor with silane, upgraded MG-Si for solar grade 6N.)
5. **Crystal growth:** polysilicon chunks are melted with dopant (boron for p-type, or phosphorus).
   - **Czochralski (CZ):** melt in a quartz crucible at ~1420 °C; a small seed crystal is dipped in, rotated and slowly pulled up; the melt freezes on it in the same crystal orientation, giving a cylindrical **monocrystalline ingot** (up to 2 m long, 200-300 mm diameter). Some oxygen from the crucible enters.
   - **Float zone (FZ):** a polysilicon rod is melted by an RF coil in a narrow zone that moves along the rod; impurities stay in the melt: purest crystals, no crucible.
   - **Casting (directional solidification):** melt solidified in a large crucible → **multicrystalline** block (cheaper, grain boundaries).
6. **Ingot shaping:** the round ingot is cut into (pseudo-)square bricks (156-210 mm side); ends and slabs are recycled.
7. **Wafering:** multi-wire saw (steel wire with SiC slurry or diamond-coated wire) slices bricks into wafers of 150-200 µm; about 40-50% of the silicon is lost as kerf.

8. **Lapping, etching and polishing** (for a polished wafer, 2020 4b): lapping with abrasive removes saw marks and makes faces parallel; chemical etching (NaOH/KOH or HF-HNO_3) removes the damaged layer; chemical-mechanical polishing gives a mirror-flat surface (electronics wafers); cleaning (RCA clean) and inspection. Solar wafers are usually only damage-etched and textured.



Czochralski crystal pulling

> **Memory trick: Q-A-T-D-S-C-S-P**: **Q**uartz, **A**rc furnace, **T**richlorosilane, **D**istil, **S**iemens, **C**zochralski, **S**aw, **P**olish. "Quiet Ants Take Dinner So Cats Sleep Peacefully."

Fabrication of a c-Si solar cell from a wafer (2019 3b)

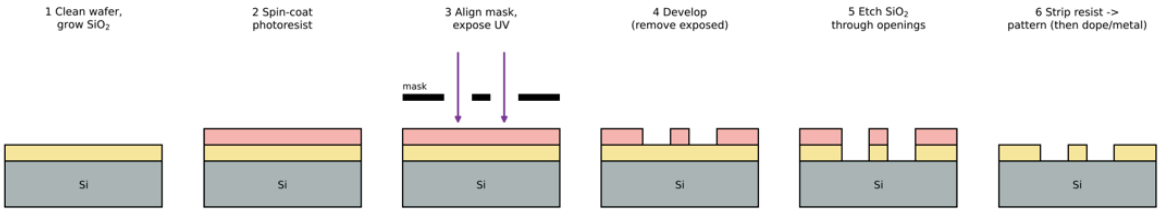
1. **Saw-damage removal and cleaning:** etch ~10 µm from each side in hot NaOH/KOH (or acid) to remove the cracked surface layer.
2. **Texturing:** anisotropic KOH + isopropanol etch forms random pyramids on (100) mono-Si (acid texture for multi-Si): light hits the surface at least twice, reflection falls from ~35% to ~11%.
3. **Emitter diffusion:** $POCl_3$ furnace at ~850 °C diffuses phosphorus ~0.3-0.5 µm into the p-type wafer, forming the n^+ emitter and the p-n junction; a phosphosilicate glass (PSG) forms on top.
4. **PSG (phosphosilicate glass) removal:** dip in dilute HF.
5. **Edge isolation:** the emitter also forms on the edges and back, shorting front and back; removed by plasma etching, laser grooving or single-side wet etch.
6. **Anti-reflection coating:** ~75 nm SiN_x:H by PECVD ($n \approx 2.0$): reduces reflection and the hydrogen passivates the surface and bulk defects.
7. **Metallisation by screen printing:** Ag paste grid (fingers + busbars) on the front, Al paste over the back (with Ag/Al pads for soldering).
8. **Co-firing** in a belt furnace at ~800-850 °C for seconds: Ag fires through the SiN_x to contact the emitter; Al alloys with Si and forms the p^+ **Al-BSF**.
9. **Testing and sorting:** I-V measurement under a flash solar simulator; cells are binned by current/power for module making.

High-efficiency variants (Part III 12.5): **PERC (passivated emitter and rear cell)/PERL** (passivated emitter and rear, local rear contacts, inverted pyramids; 25%), **IBC (interdigitated back contact)** (all contacts at the back, no shading, 26.8%), **HIT (heterojunction with intrinsic thin layer)/heterojunction**

(thin a-Si:H (hydrogenated amorphous silicon) layers on n-type c-Si, 26.3-26.8%), TOPCon (tunnel oxide passivated contact).

Lithography and its steps (2021 2b, 2020 4a)

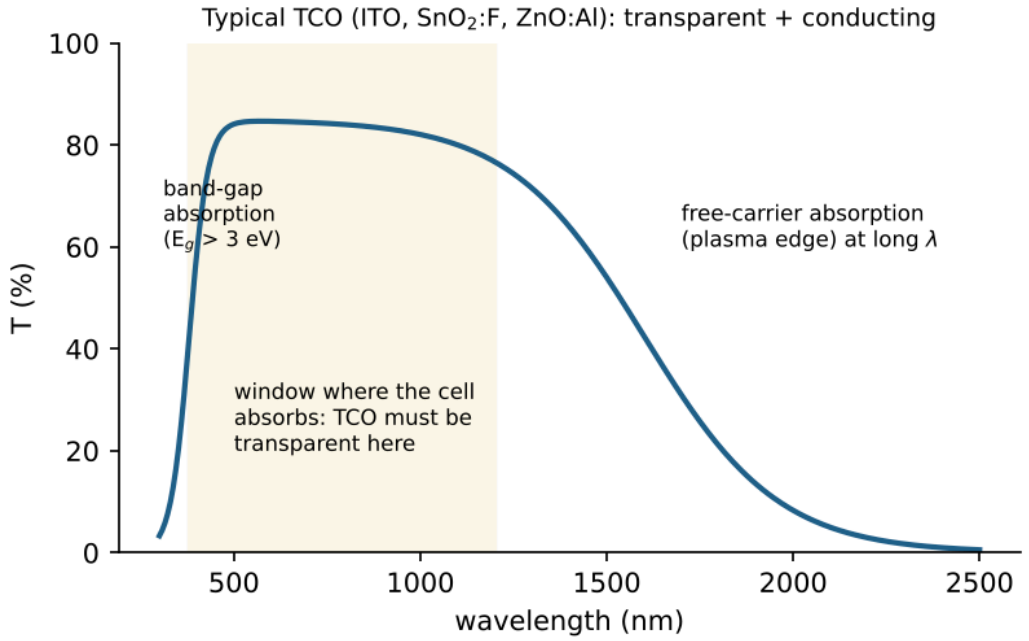
Photolithography is the process of transferring a geometric pattern from a **photomask** to a light-sensitive layer (**photoresist**) on a wafer, and then into the wafer by etching or deposition. In PV it is used for very fine front fingers, local openings in passivation layers (PERL point contacts, inverted pyramids) and selective doping in high-efficiency and laboratory cells.



Steps of photolithography (positive resist)

1. **Wafer cleaning and preparation:** remove contamination (RCA clean), grow or deposit a thin oxide (SiO_2) or other layer to be patterned; dehydration bake and adhesion promoter (HMDS).
2. **Photoresist coating:** spin coating at 1000-5000 rpm gives a uniform 0.5-2 µm layer.
3. **Soft bake (pre-bake):** ~90 °C to drive out solvent.
4. **Mask alignment and exposure:** the mask is aligned to the wafer and UV light (g/i-line, 365-436 nm) shines through its clear areas, changing the solubility of the exposed resist (contact, proximity or projection printing).
5. **Development:** developer dissolves the exposed resist (positive resist) or the unexposed resist (negative resist), leaving the pattern.
6. **Hard bake:** ~120 °C hardens the remaining resist.
7. **Etching (or deposition):** the open areas of the oxide are etched (wet HF or dry plasma); or metal is evaporated over the pattern (lift-off).
8. **Resist stripping:** solvent or oxygen plasma removes the resist, leaving the patterned layer for the next step (doping, metallisation).

Transparent conducting oxide (2019 3c, 2022 4a)



A TCO is transparent in the visible and near-IR and still conducts

A **TCO** is a thin film (~100-1000 nm) of a wide-bandgap oxide (E_g > 3 eV) that is heavily doped so it is both **optically transparent** (> 80% in 400-1100 nm) and **electrically conducting** (sheet resistance 5-20 Ω/sq). Examples: ITO (In_2O_3:Sn), FTO (SnO_2:F), AZO (ZnO:Al), IO:H.

Why used: thin-film absorbers (a-Si, CdTe, CIGS (copper indium gallium diselenide), perovskite) and HIT emitters have low lateral conductivity, so metal fingers alone cannot collect current without large shading; the TCO serves as a transparent front electrode over the whole area; it also acts as anti-reflection coating, diffusion barrier, and its textured surface scatters light into the absorber (light trapping). Trade-off: higher doping increases conductivity but also free-carrier absorption in the IR, so TCOs are optimised between transparency and resistance; indium is scarce.

Supporting layers in solar cells (2022 2d, 2020 3c)

Besides the absorber and junction, a cell needs layers that support, protect and help the carriers: **substrate/superstrate** (glass, metal or polymer foil that carries thin films), **buffer/window layers** (CdS, ZnO: form the junction, protect the absorber, let light through), **electron and hole transport layers** (ETL TiO_2/SnO_2, HTL (hole transport layer) spiro-OMeTAD/PEDOT:PSS: selective contacts), **TCO and back contacts**, **passivation and ARC (anti-reflection coating) layers** (SiN_x, Al_2O_3), **back reflector** (ZnO/Ag), and encapsulation.

Selection considerations: mechanical strength and flatness; thermal stability and matched thermal expansion during processing; chemical compatibility (no diffusion of impurities, e.g. Na from glass for CIGS is useful but harmful elsewhere); optical transparency (if light passes through); electrical conductivity and band alignment (ohmic contact, carrier selectivity, low interface recombination); lattice matching for epitaxial layers; cost, weight, flexibility, abundance and non-toxicity; long-term stability against moisture, UV and heat.

Remember: SiO_2 + 2C → Si + 2CO (arc, 1900 °C); Si + 3HCl → $HSiCl_3$ + H_2; distil; $HSiCl_3$ + H_2 → Si + 3HCl (Siemens, 1100 °C); CZ (Czochralski)/FZ ingot; wire saw; etch, polish. Cell: etch, texture, $POCl_3$, PSG off, edge isolate, SiN_x, print, fire, test. Litho: coat, bake, expose, develop, etch, strip.

A5 Losses and efficiency limits: optical and electrical losses, design rules, back surface field, thermodynamic and Shockley-Queisser limits

Pass set | Very likely · 100% | Easy

PAST-PAPER QUESTIONS

- **2023 Q3(a)** [10] Discuss the different types of losses in a solar cell.
- **2023 Q3(b)** [6] Why is the back surface field important in a high-efficiency solar cell? Give a proper diagram.
- **2023 Q3(c)** [7] Briefly explain the thermodynamic limit and the Shockley-Queisser limit.
- **2022 Q3(c)** [9] Briefly explain the design rules for a solar cell with respect to loss.
- **2022 Q1(c)** [13] Discuss in detail the balance limit of solar cell efficiency according to the Shockley-Queisser limit.
- **2021 Q4(b)** [10] Briefly explain the efficiency losses of a solar cell; define IQE (internal quantum efficiency) and EQE (external quantum efficiency).
- **2021 Q3(a)** [8] "Performance of a solar cell depends on irradiance." Justify the statement. (answered in B1)
- **2020 Q2(b)** [15] How can transparency loss and excess excitation loss be minimized in a solar cell? Explain in detail. What are the main challenges in multi-junction solar cells? (MJ part: A8)
- **2019 Q1(c)** [10] Discuss the different types of losses in a solar cell.
- **2019 Q2(c)** [15] Discuss in detail the balance limit of solar cell efficiency according to the Shockley-Queisser limit.

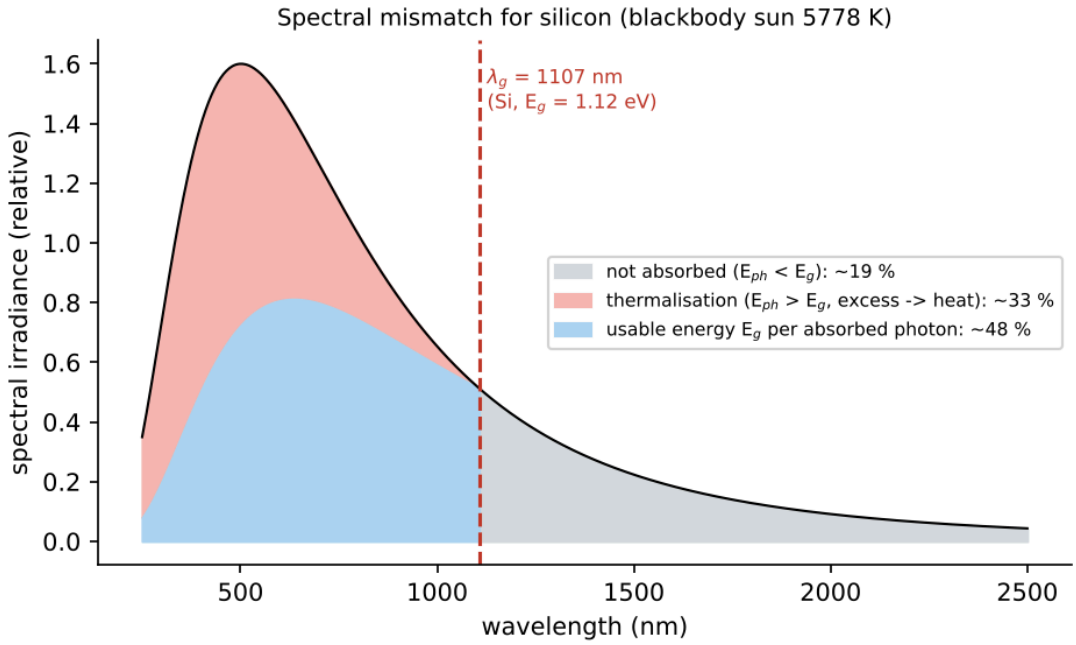
★ REMEMBER IT

- Losses: **optical** (reflection, shading by front fingers, transmission of long-wavelength light) and **electrical** (thermalization of excess photon energy above E_g, non-absorption below E_g, recombination, resistive). Biggest two: thermalization and non-absorption.
- Design rules: anti-reflection coating and texturing (less reflection), thin fingers, back reflector/light trapping (more absorption), surface passivation and **back surface field** (a heavily doped p+ layer at the back repels minority electrons from the back contact, cutting recombination).
- **Shockley-Queisser limit**: about 33% for one junction (best $E_g \approx 1.34$ eV) at 1 sun, from detailed balance (only photons above E_g count, excess energy lost as heat, radiative recombination unavoidable). Thermodynamic (Carnot-like) limit about 86% with infinite junctions and full concentration.

Before you start: where the sunlight energy goes

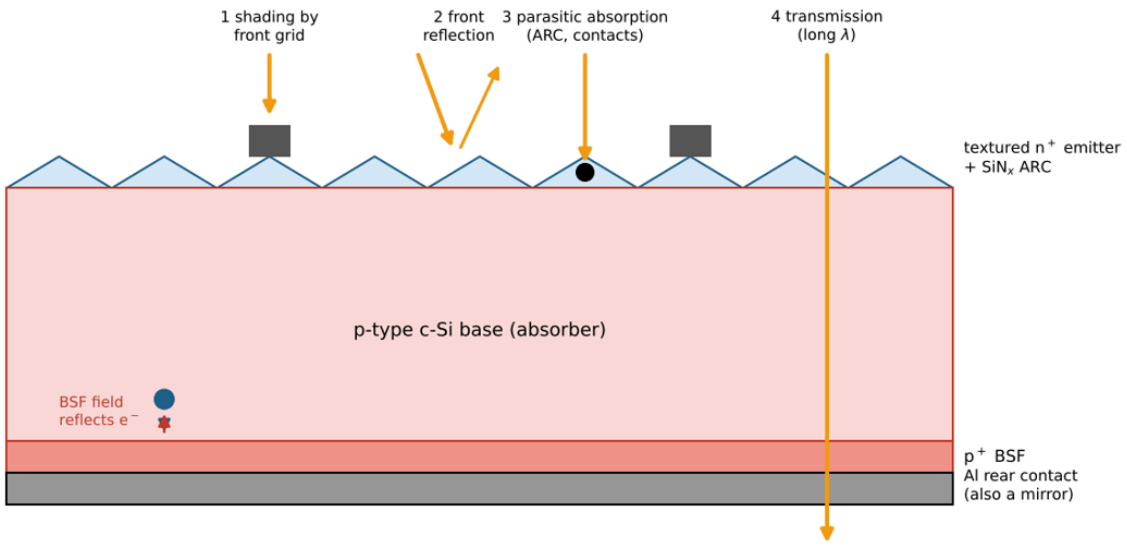
Even a perfect cell cannot turn all sunlight into electricity. Photons below the bandgap pass through; photons above the bandgap waste their extra energy as heat. On top of these fundamental (spectral) losses come practical losses: reflection, shading, recombination and resistance. Source: Rahaman sir Part II ch. 10 and Part III sec. 12.3 (Smets ch. 10, 12).

Types of losses in a solar cell (2023 3a, 2019 1c, 2021 4b)



Spectral mismatch: non-absorption and thermalisation for silicon

Group	Loss	Cause and cure
Spectral (fundamental)	**Non-absorption** (transparency)	$E_{ph} < E_g$ photons ($\lambda > 1107$ nm for Si) are not absorbed: ~19% for Si; cure: lower E_g, tandem/multi-junction, up-conversion
	Thermalisation (excess energy)	$E_{ph} > E_g$: the extra energy $E_{ph} - E_g$ is lost as heat within ~10^{-12} s: ~33% for Si; cure: multi-junction, down-conversion, hot-carrier, MEG (multiple exciton generation)
Optical	Shading	front fingers and busbars cover 3-8% of the area; cure: thin fingers, back-contact (IBC) cells
	Reflection	bare Si reflects 30-38%; cure: texturing (pyramids) + anti-reflection coating (SiN_x, $n \approx 2.0$, $d = \lambda/4n$) → < 3%
	Parasitic absorption	light absorbed in ARC (anti-reflection coating), TCO (transparent conducting oxide), contacts, which does not make carriers
	Transmission / incomplete absorption	weakly absorbed red/IR light leaves the back of a thin wafer; cure: back reflector, light trapping
Electrical (collection)	Recombination	bulk (SRH, Auger), surface and contact recombination: lower J_{sc} and V_{oc}; cure: pure material, passivation, BSF (back surface field)
	Ohmic (series) loss	emitter, finger, busbar, contact resistance: lower FF (fill factor)
	Shunt loss	leakage across the junction (edges, defects): lower FF and V_{oc}
Voltage and FF factors	Voltage factor $qV_{oc}/E_g < 1$ and FF < 1	even ideal cells lose because $V_{oc} < E_g/q$ (radiative recombination, entropy)



Optical losses in a c-Si cell: shading, reflection, parasitic absorption, transmission

The teacher's loss chain: η = (spectral use) × (1 − R) × IQE_{op} × η_G × IQE_{el} × coverage × (qV_{oc}/E_g) × FF, i.e. losses A non-absorption, B thermalisation, C reflection, D incomplete absorption, E recombination, F metal coverage (shading), G voltage factor, H fill factor.

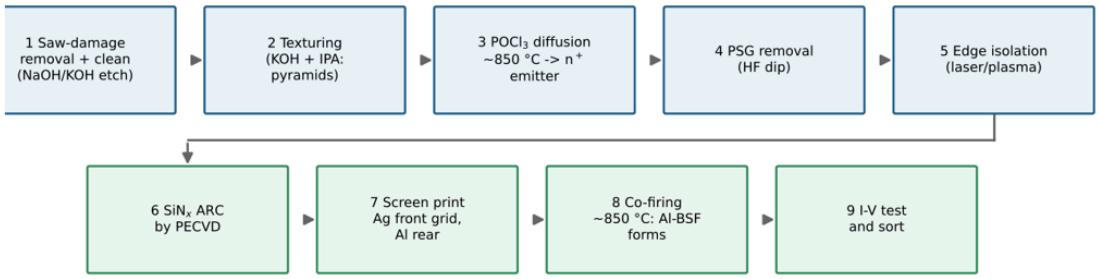
> **Side note: IQE and EQE (2021 4b):** EQE = number of carriers collected / number of incident photons at a given λ; IQE = number collected / number of photons absorbed. EQE = IQE × (1 − R − T); IQE shows the electrical (recombination) quality, EQE includes optical losses too.

How to minimise transparency (non-absorption) and excess-excitation (thermalisation) loss (2020 2b)

- **Transparency loss:** choose an absorber with a suitable (lower) bandgap or use a bottom cell of low E_g (Ge 0.67 eV) in a **multi-junction** stack; **light trapping** (textured front, back reflector, rough TCO) to make weakly absorbed light travel several times through the cell; thicker absorber; **up-conversion** (two sub-gap photons → one usable photon); intermediate-band cells.
- **Excess excitation (thermalisation) loss: multi-junction cells** (each photon absorbed in a junction whose E_g is just below its energy); **down-conversion** or **multiple exciton generation** (one high-energy photon → two pairs); **hot-carrier cells** (extract carriers before they cool); concentration also raises V_{oc}.

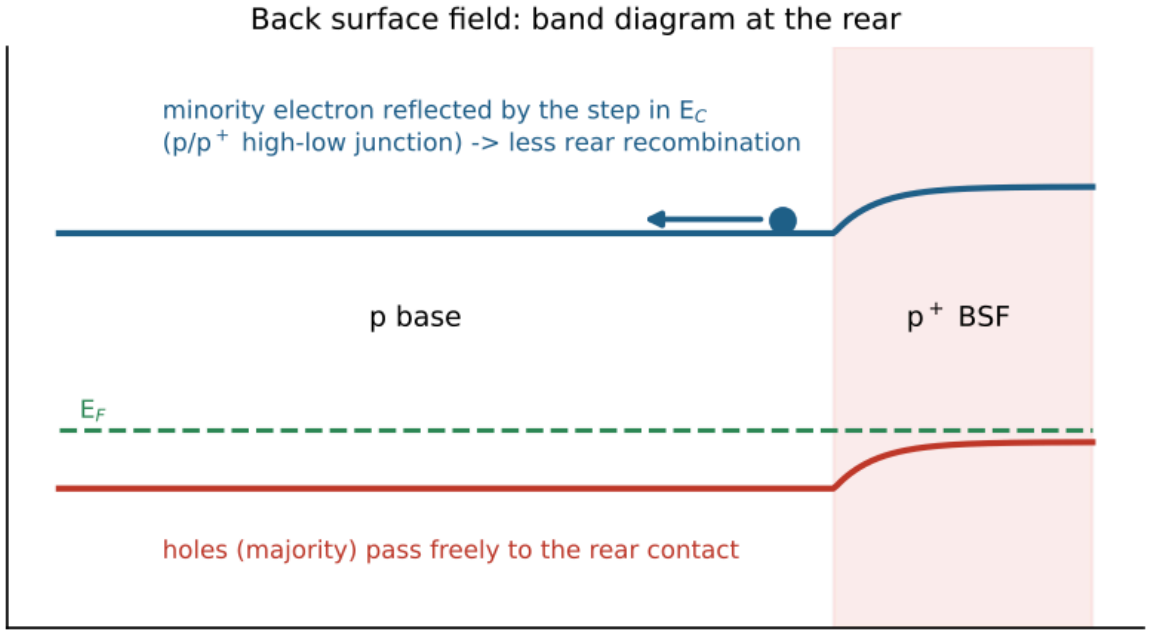
Design rules for a solar cell with respect to loss (2022 3c)

Management area	Design rule
Spectral (bandgap utilisation)	bandgap near 1.1-1.5 eV for one junction; tandem/multi-junction or third-generation concepts for more
Light management (optical)	texture the front (pyramids) to reduce reflection and trap light; single/double-layer ARC; minimise front metal coverage (fine fingers, optimal finger spacing, busbars) or back-contact design; back surface reflector and flat/smooth rear to raise back reflection
Recombination management (electrical)	pure, defect-free base with diffusion length > base thickness; surface passivation (SiO_2, SiN_x, Al_2O_3); back surface field and point contacts at the rear; passivated contacts; heavy doping only under the metal (selective emitter) to keep minority carriers away from the contacts; shallow, lightly doped emitter elsewhere (good blue response)
Ohmic management	deep, heavy doping under contacts (low contact resistance); fine but high, conductive fingers with optimum spacing (resistive loss grows ~ S^2-S^3 with spacing); low-resistivity base; thick busbars or multi-busbar



Each fabrication step targets one of these losses

Why the back surface field is important (2023 3b)



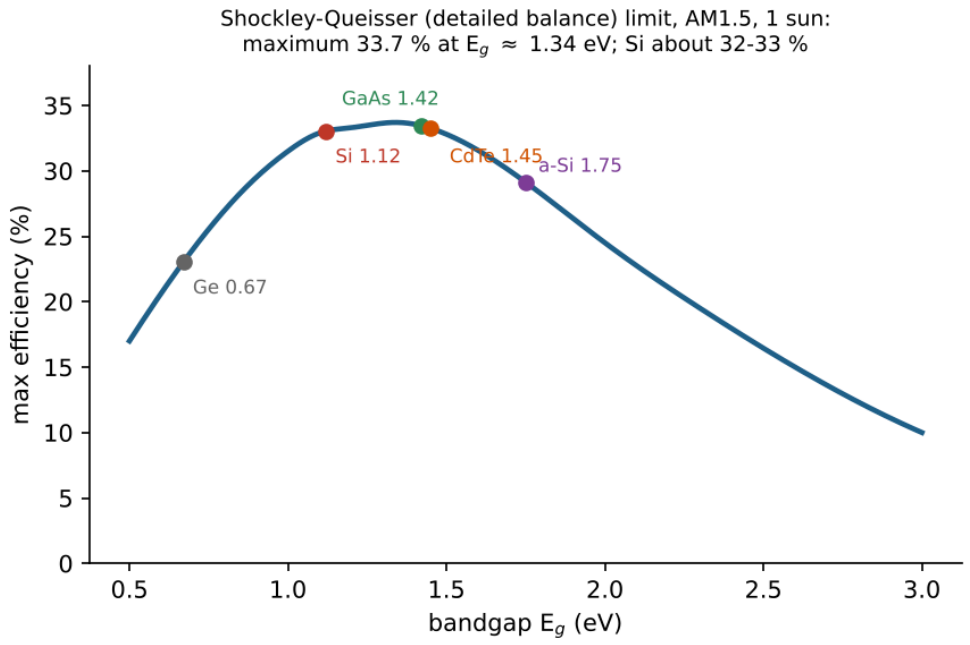
Back surface field: a p/p^+ high-low junction reflects minority electrons

A **back surface field** is a heavily doped p^+ layer at the back of a p-type base (made by alloying Al during co-firing, or boron diffusion). The p/p^+ high-low junction gives a step in the bands with an

electric field that **repels minority electrons** from the highly recombining rear contact, while majority holes pass freely to the contact. Effects: lower rear surface recombination velocity (effective S from ~10^6 to < 200 cm/s) → higher V_{oc} (+20-30 mV) and higher J_{sc} (better red response, carriers from deep in the base are collected); lower contact resistance (ohmic contact); the Al layer also acts as a back reflector and getters impurities. It is essential for thin wafers, where the rear is within a diffusion length of most carriers.

Thermodynamic limit and Shockley-Queisser (detailed balance) limit (2023 3c, 2022 1c, 2019 2c)

- **Thermodynamic limit:** the sun (T_s ≈ 5800 K) is a heat source and the cell (T_c ≈ 300 K) a heat engine. The Carnot limit $1 - T_c/T_s$ = 95%; including the cell's own unavoidable emission (Landsberg) gives about 93%; with infinite junctions it is 86.8% under full concentration and about 68% at one sun. It is the absolute maximum for any solar converter.
- **Shockley-Queisser (1961) detailed-balance limit** for a single p-n junction: assumptions: one bandgap E_g; every photon with $E \geq E_g$ is absorbed and makes exactly one electron-hole pair; carriers thermalise to the band edges; the only recombination is radiative (detailed balance: the cell in equilibrium emits as a blackbody at 300 K, which sets J_0); perfect collection, no resistance losses; one sun AM1.5.
- The efficiency is then limited by: non-absorption ($E_{ph} < E_g$), thermalisation ($E_{ph} > E_g$), and the fact that $V_{oc} < E_g/q$ and FF < 1 due to radiative recombination. A small E_g collects many photons but gives low voltage; a large E_g gives high voltage but few photons, so there is an optimum: **~33.7% at E_g ≈ 1.34 eV** (AM1.5; 31% in the original 1961 paper for a 6000 K blackbody); Si (1.12 eV) ~32-33%, GaAs ~33%. Including Auger recombination for Si, the practical limit is ~29.4%; record Si cells reach 26.8-27.3%.



Shockley-Queisser efficiency limit vs bandgap

Memory trick: SQ (Shockley-Queisser) = 33-1.34: thirty-three per cent at one-point-three-four eV. Two thieves: **too poor photons** (pass through) and **too rich photons** (lose the change as heat).

Remember: Losses: spectral (non-absorption 19%, thermalisation 33%), optical (shading, reflection, parasitic, transmission), electrical (recombination, R_s, R_{sh}). Rules: texture + ARC + few fingers; passivate + BSF + selective emitter; heavy doping under contacts. BSF repels minority carriers. Limits: thermodynamic 86.8% (68% one sun); SQ 33.7% at 1.34 eV.

A4 Solar cell parameters and equivalent circuit: I_{sc}, V_{oc}, FF, efficiency, R_s and R_{sh}, quantum efficiency

Pass set · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q2(b)** [12] Briefly explain external solar cell performance parameters with proper equations.
- **2023 Q2(c)** [12] Briefly explain the effect of series and parallel resistance on the I-V characteristics of a solar cell.
- **2022 Q3(b)** [8] Draw and briefly explain the equivalent circuit of a realistic solar cell which considers all the losses, and also write the modelled equation.
- **2021 Q5(d)** [11] A solar panel 61 cm × 46 cm at STC (standard test conditions (1000 W/m^2, AM1.5, 25 °C)): V_{oc} = 21.5 V, I_{sc} = 3.21 A, V_{MPP} (maximum power point) = 20 V, I_{MPP} = 2.5 A. Calculate the fill factor and efficiency of this panel. (asked in Section B)
- **2021 Q4(a)** [9] Draw an equivalent circuit of an ideal solar cell and a solar cell with series resistance (R_s) and shunt resistance (R_p), mentioning their modelled equations.
- **2021 Q4(b)** [10] Briefly explain the efficiency losses of a solar cell. Also define internal quantum efficiency and external quantum efficiency. (losses part: A5)
- **2020 Q1(a)** [6] A solar cell 12 cm × 12 cm at STC: V_{oc} = 0.8 V, I_{sc} = 3 A, V_{MPP} = 0.75 V, I_{MPP} = 2.5 A. Calculate the fill factor and efficiency.
- **2020 Q1(c)** [12] With the help of mathematical equations, explain the influence of series and shunt resistance on the performance of a solar cell.
- **2019 Q2(b)** [10] Draw and discuss the equivalent circuit of an ideal solar cell and an actual solar cell with R_s and R_p.

LEARN IT FROM ZERO

The I-V curve, from zero

A solar cell is a **current source** (light makes I_L) in parallel with a **diode** (the p-n junction, which leaks current back as voltage rises): $I = I_L - I_0[e^{V/nV_t} - 1]$. Short the terminals: V = 0, I = **I_{sc}**. Leave them open: I = 0, V = **V_{oc}** = $nV_t\ln(I_L/I_0 + 1)$. In between is the **maximum power point** (V_{mp}, I_{mp}).

Fill factor FF (fill factor) **= $V_{mp}I_{mp}/(V_{oc}I_{sc})$**: how "square" the curve is. **Efficiency $\eta = P_{max}/(G \times A)$**, G = 1000 W/m² at STC.

A real cell adds a **series resistance R_s** (contacts, fingers: tilts the curve near V_{oc}) and a **shunt resistance R_{sh}** (leakage paths: tilts the curve near I_{sc}). Good cell: R_s small, R_{sh} large. Both lower FF.

CALCULATOR **Fill factor and efficiency (2020 1(a), 2021 5(d))**

Change any number; every step is recalculated and shown with the numbers put in.

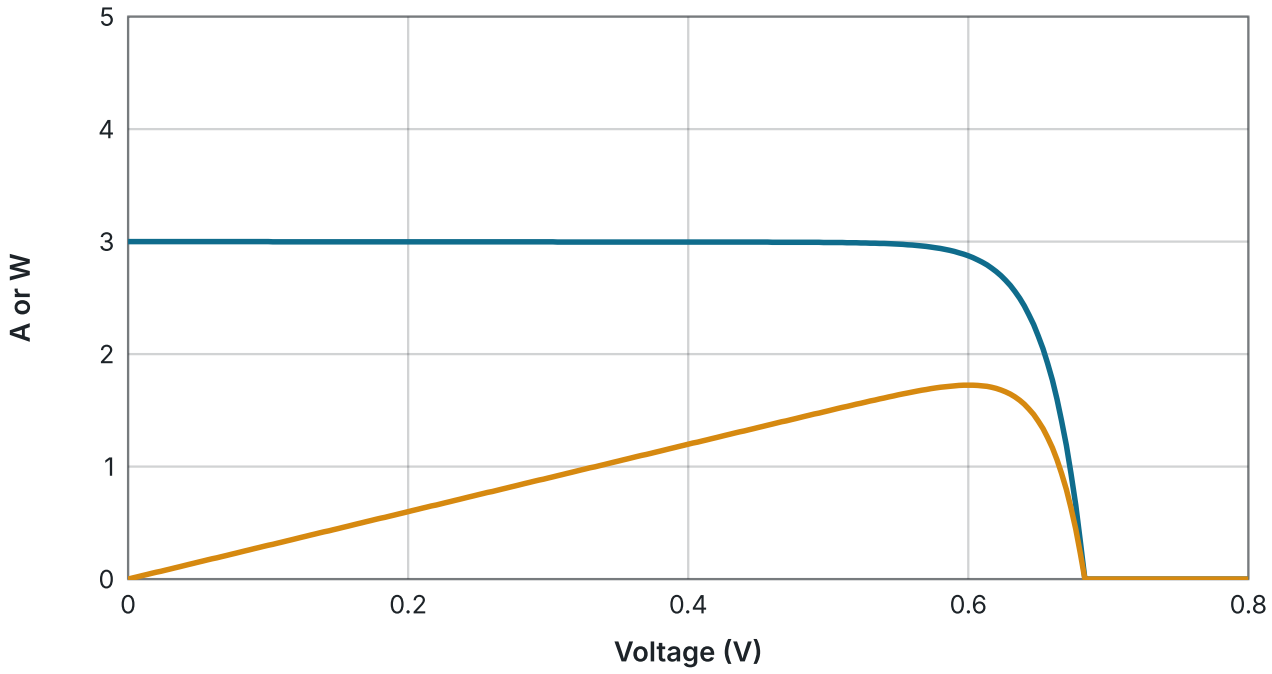
1. Maximum power: $P_{max} = V_{mp}I_{mp}$ = **0.75** × **2.5** = **1.875** W
2. Fill factor: $FF = V_{mp}I_{mp}/(V_{oc}I_{sc})$ = **1.875** / (**0.8** × **3**) = **0.781**

3. Input power: $P_{in} = G \times A =$ **1000** × **12** × **12** / 10^4 = **14.40** W

4. Efficiency: $\eta = P_{max}/P_{in} = 100 \times$ **1.875** / **14.4** = **13.0** %

INTERACTIVE **I-V and P-V curve of a cell: see what R_{sh} and I_0 do**

Move the sliders and watch the curve change.



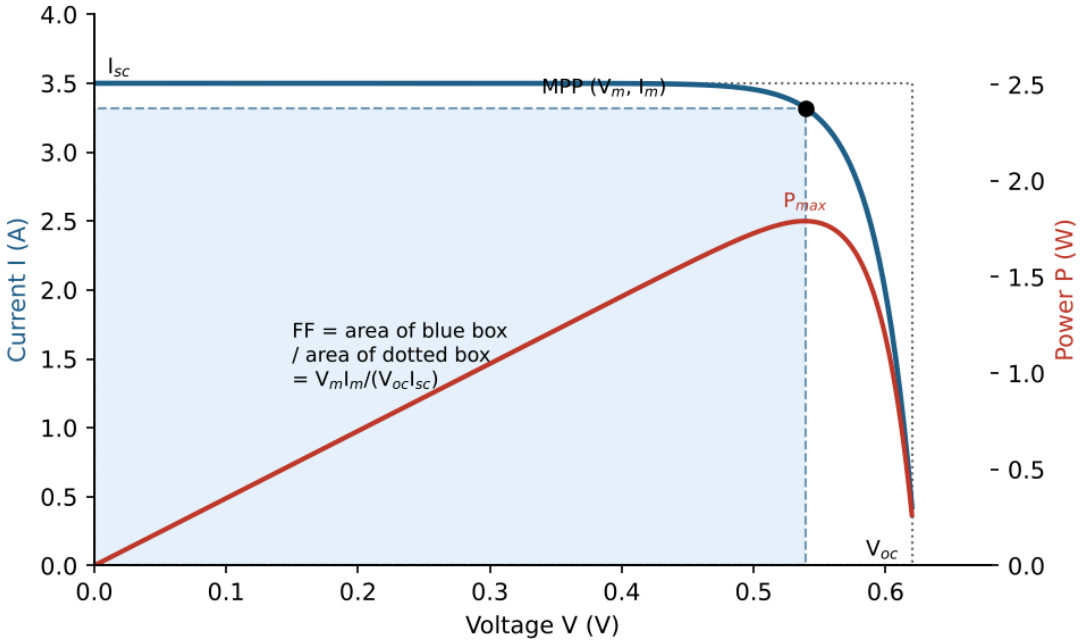
V_{oc} (ignoring R_{sh}): **0.683** V

Drag R_{sh} down to 2 Ω: the flat top tilts and the power peak drops. Raise I_0 (worse recombination): V_{oc} falls. Series resistance is not drawn here.

Before you start: what the data sheet numbers mean

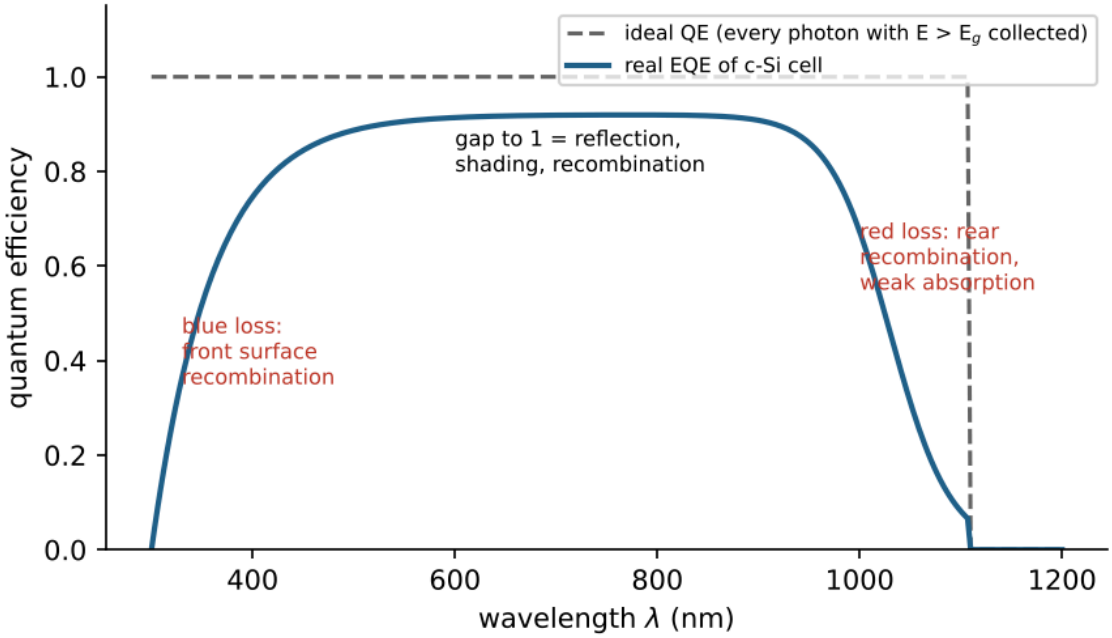
Every cell or module is described by a few numbers measured at **standard test conditions (STC)**: irradiance 1000 W/m^2 with the AM1.5 spectrum (light through 1.5 atmospheres), cell temperature 25 °C (and wind 1 m/s). From the I-V curve we read I_{sc}, V_{oc} and the maximum power point. Source: Rahaman sir Part II ch. 9; Muhit sir L1 (parameters, I-V and P-V curves).

External solar cell performance parameters (2023 2b)



I-V and P-V curves with I_{sc}, V_{oc}, MPP and the fill factor rectangle

Parameter	Definition and equation	Typical c-Si
Short-circuit current I_{sc} (J_{sc} per area)	current when V = 0 (terminals shorted); ideal $J_{sc} = J_{ph} = qG(L_n + W + L_p)$; $\propto$ irradiance and area	35-42 mA/cm^2
Open-circuit voltage V_{oc}	voltage when I = 0; $V_{oc} = (kT/q)\ \ln(J_{ph}/J_0 + 1)$, $J_0 = qn_i^2(D_n/L_nN_A + D_p/L_pN_D)$	0.6-0.72 V
Maximum power point (V_m, I_m)	point where P = VI is largest; $P_{max} = V_mI_m$	$V_m \approx 0.8V_{oc}$
Fill factor FF	$FF = V_mI_m/(V_{oc}I_{sc})$: how "square" the curve is; empirical $FF = [v_{oc} - \ln(v_{oc} + 0.72)]/(v_{oc} + 1)$, $v_{oc} = qV_{oc}/kT$	0.75-0.84
Efficiency η	$\eta = P_{max}/P_{in} = V_{oc}I_{sc}FF/(G \times A)$	17-26%
Quantum efficiency	EQE(λ) = electrons collected/incident photons; IQE (internal quantum efficiency) = electrons collected/absorbed photons; $J_{sc} = q\int$EQE (external quantum efficiency)$\cdot\Phi(\lambda)d\lambda$	EQE > 90% in visible



External quantum efficiency of a c-Si cell and where it is lost

Worked example (teacher's notes): J_{ph} = 350 A/m^2, J_0 = 1.95 × 10^{-9} A/m^2 → V_{oc} = 0.0258 ln(350/1.95 × 10^{-9} + 1) = 0.67 V; v_{oc} = 26.8, FF = 0.84; η = 350 × 0.67 × 0.84/1000 = 19.7%.

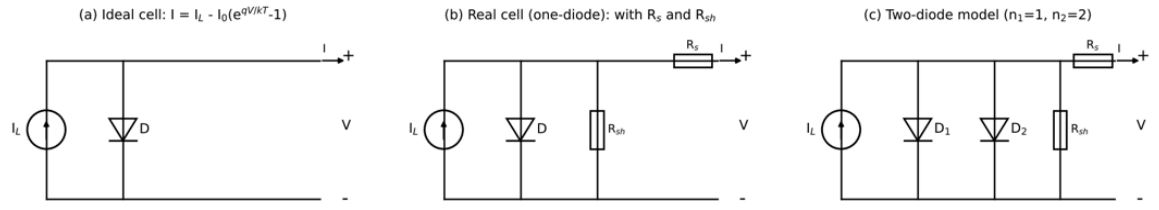
Numerical: fill factor and efficiency of a cell (2020 1a)

Step (write this)	Why / what it means
FF = $V_m I_m/(V_{oc} I_{sc})$ = 0.75 × 2.5/(0.8 × 3) = 1.875/2.4 = **0.781 (78.1%)**	ratio of the two rectangles under the I-V curve
P_{in} = G × A = 1000 × (0.12 × 0.12) = 14.4 W	STC irradiance 1000 W/m^2; convert cm to m
n = P_{max}/P_{in} = 1.875/14.4 = **13.0%**	

Numerical: fill factor and efficiency of a panel (2021 5d)

Step (write this)	Why / what it means
P_{max} = 20 × 2.5 = 50 W; FF = 50/(21.5 × 3.21) = 50/69.0 = **0.724**	module: V adds over 36 series cells, same formula
P_{in} = 1000 × 0.61 × 0.46 = 280.6 W	gross panel area
n = 50/280.6 = **17.8%**	

Equivalent circuits: ideal, with R_s and R_{sh}, two-diode (2022 3b, 2021 4a, 2019 2b)

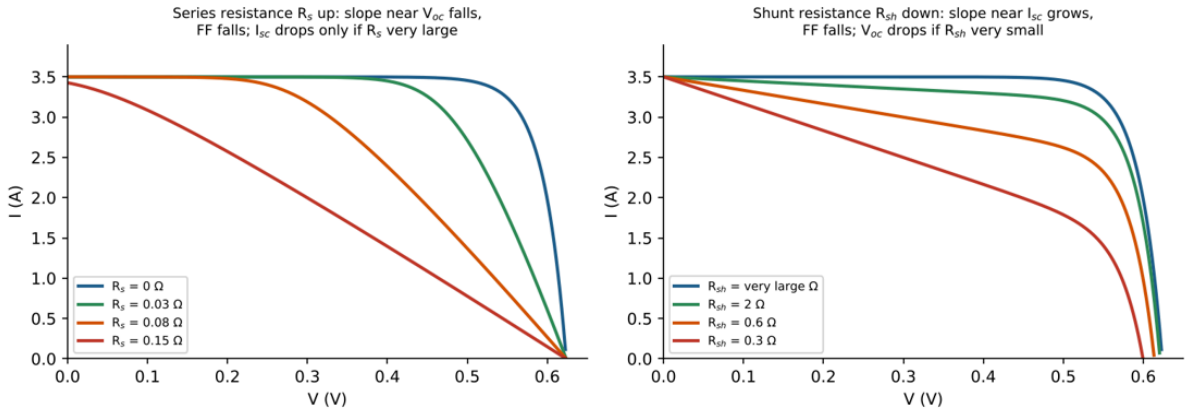


Equivalent circuits of a solar cell

- **Ideal cell:** a current source I_L (photocurrent, ∝ light) in parallel with a diode (the p-n junction). **$I = I_L - I_0[\exp(qV/kT) - 1]$**. At short circuit I = I_L; at open circuit V_{oc} = (kT/q) ln(I_L/I_0 + 1).

- **Real (one-diode) cell:** adds a **series resistance R_s** (bulk of the semiconductor, emitter sheet resistance, fingers and busbars, metal-semiconductor contact) and a **shunt (parallel) resistance $R_{sh} = R_p$** (leakage paths across the junction: edges, crystal defects, pinholes, impurity precipitates). **$I = I_L - I_0[\exp(q(V + IR_s)/nkT) - 1] - (V + IR_s)/R_{sh}$.**
- **Two-diode model (considers all losses):** diode 1 ($n_1 = 1$) for diffusion/recombination in the quasi-neutral regions and diode 2 ($n_2 = 2$) for recombination in the depletion region: $I = I_L - I_{01}[e^{q(V+IR_s)/n_1kT} - 1] - I_{02}[e^{q(V+IR_s)/n_2kT} - 1] - (V + IR_s)/R_{sh}$.

Effect of series and shunt resistance on the I-V curve (2023 2c, 2020 1c)



Rising R_s and falling R_{sh} both reduce the fill factor

- **Series resistance:** the voltage across the junction is $V + IR_s$, so at high current the terminal voltage drops by IR_s. The curve tilts near V_{oc} (slope near $V_{oc} \approx -1/R_s$), **FF falls** and P_{max} falls (loss I^2R_s). V_{oc} is not affected ($I = 0$); I_{sc} drops only when R_s is very large. Ideal $R_s = 0$; keep it below ~0.5 Ωcm^2.
- **Shunt resistance:** part of the photocurrent leaks through R_{sh}, current V/R_{sh}. The curve tilts near I_{sc} (slope $\approx -1/R_{sh}$), **FF falls**; for very low R_{sh} V_{oc} also falls; I_{sc} is not affected. Ideal $R_{sh} = \infty$; low R_{sh} is worst at low light.
- Characteristic resistance $R_{ch} = V_m/I_m$: R_s matters when it approaches R_{ch}; R_{sh} matters when it falls toward R_{ch}. Approximate: $FF_{Rs} \approx FF_0(1 - R_s/R_{ch})$; $FF_{Rsh} \approx FF_0(1 - R_{ch}/R_{sh})$.

Memory trick: **S**eries hurts near **V_{oc}** (both have the letter 'S/V' at the end of the curve); **Sh**unt hurts near **I_{sc}**. Both steal fill factor.

Remember: $FF = V_mI_m/(V_{oc}I_{sc})$; $\eta = V_{oc}I_{sc}FF/(1000 \times A)$. 2020 1a: 0.781, 13.0%. 2021 5d: 0.724, 17.8%. Real cell: $I = I_L - I_0(e^{q(V+IR_s)/nkT} - 1) - (V+IR_s)/R_{sh}$.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2020 1(a) 12 cm × 12 cm cell, $V_{oc} = 0.8$ V, $I_{sc} = 3$ A, $V_{mp} = 0.75$ V, $I_{mp} = 2.5$ A

Fill factor: $FF = V_{mp}I_{mp}/(V_{oc}I_{sc})$

$= (0.75 \times 2.5)/(0.8 \times 3) = 1.875/2.4$

$= \boxed{0.781}$

Input power: $P_{in} = GA$

$= 1000 \times 0.12 \times 0.12$

$= \boxed{\mathbf{14.4\ W}}$

Efficiency: $\eta = P_{max}/P_{in}$

$= 1.875/14.4$

$= \boxed{\mathbf{13.0\%}}$

A8 Multi-junction, perovskite and third-generation cells; generations of solar cells Next to add Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2023 Q4(a)** [10] What is a multijunction solar cell? Explain the working principle of a multijunction solar cell with a proper diagram.
- **2023 Q4(d)** [10] Briefly explain third-generation solar cell technology with its significance.
- **2022 Q4(c)** [7] Explain the challenges for multijunction solar cells.
- **2022 Q2(c)** [8] What is lattice mismatch? Justify the use of Ge, GaAs and GaInP to form triple-junction cells.
- **2022 Q4(d)** [8] What distinguishes perovskite solar cells from traditional silicon-based cells in terms of their construction and materials?
- **2021 Q4(c)** [10] What are the approaches to increase the efficiency of a solar cell? Explain how a multi-junction solar cell increases efficiency compared to a single-junction solar cell.
- **2021 Q3(c)** [10] Same question as 2022 2(c).
- **2021 Q7(a)** [14] What are the properties and characteristics of perovskite solar cells? Write down the classifications of perovskite solar cells. Explain the basic structure of perovskite solar cells. (asked in Section B)
- **2021 Q4(d)** [6] Write a short note on generations of solar cells.
- **2020 Q2(b)** [15] Second part: main challenges in multi-junction solar cells.
- **2020 Q3(a)** [14] Explain the working principle of high-efficiency perovskite solar cells.
- **2019 Q4(b)** [7] Same question as 2022 2(c).

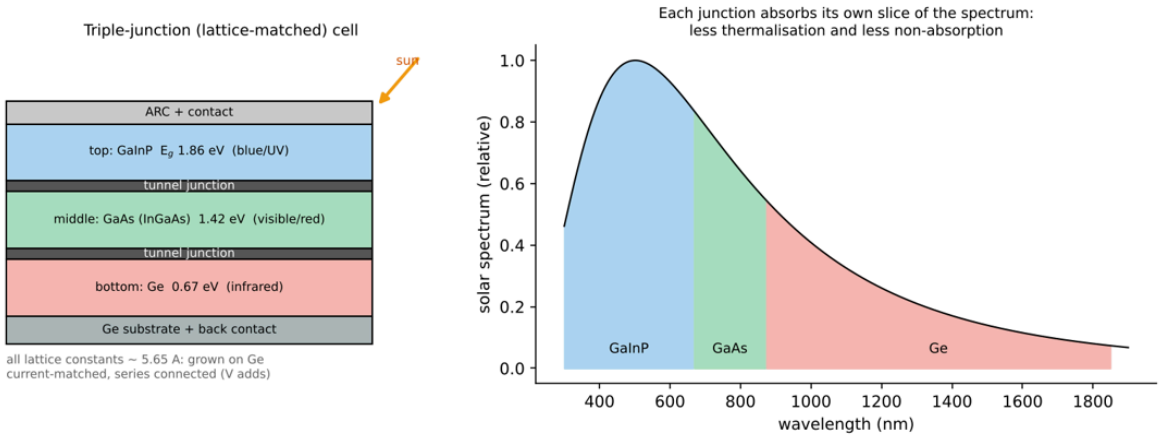
★ REMEMBER IT

- Multi-junction: stack cells with decreasing band gap (top GaInP 1.8 eV, middle GaAs 1.4 eV, bottom Ge 0.67 eV). Each absorbs its own part of the spectrum, cutting thermalization and non-absorption losses. Ge, GaAs and GaInP have almost the same lattice constant (**lattice-matched**), so they grow on each other without defects.
- Challenges: current matching (series stack, the weakest cell limits current), lattice mismatch, tunnel junctions, cost (use with concentrators or in space).
- Generations: 1st c-Si wafers; 2nd thin films (CdTe, CIGS (copper indium gallium diselenide), a-Si); 3rd new concepts beyond the S-Q limit (multi-junction, perovskite, dye-sensitized, organic, quantum dots). Perovskite ABX_3 (e.g. $CH_3NH_3PbI_3$): cheap, solution-processed, tunable gap, > 25% efficiency, but stability and lead are problems.

Before you start: beating the single-junction limit

A single junction loses about half of the sunlight to non-absorption and thermalisation (topic A5). Third-generation concepts try to beat the 33.7% Shockley-Queisser limit, mainly by stacking several junctions, and new materials such as perovskites promise high efficiency at low cost. Source: Rahaman sir Part III ch. 13.6 and 16 (Smets ch. 16).

Multi-junction cell and its working principle; how it increases efficiency (2023 4a, 2021 4c)



Triple-junction GaInP/GaAs/Ge cell and how it splits the spectrum

- A **multi-junction (tandem) cell** is a stack of two or more p-n junctions made of semiconductors with **different bandgaps**, arranged with the **largest bandgap on top**.
- **Working:** sunlight enters the top cell (GaInP, 1.86 eV), which absorbs the high-energy blue/UV photons and transmits the rest; the middle cell (GaAs, 1.42 eV) absorbs visible/red; the bottom cell (Ge, 0.67 eV) absorbs the near-infrared. Each photon is absorbed in a junction whose E_g is just below its energy, so **thermalisation and non-absorption losses are both reduced**.
- In the monolithic (2-terminal) version the sub-cells are connected **in series** by thin, heavily doped **tunnel junctions**; the voltages add (V_{oc} ≈ 2.7 V) while the current is that of the weakest sub-cell, so the cells are designed for **current matching**.
- Efficiency: limit ~49% (3 junctions, 1 sun), 68% for infinite junctions; records 39.5% (1 sun) and 47.6% under concentration; used in space and concentrator PV.
- **Other approaches to increase efficiency (2021 4c):** light trapping and ARC (anti-reflection coating); surface passivation and BSF (back surface field); better material (longer lifetime); low-resistance contacts; concentrators (higher V_{oc}); tandems (perovskite/Si); spectrum conversion; hot-carrier and intermediate-band cells; cooling and tracking at system level.

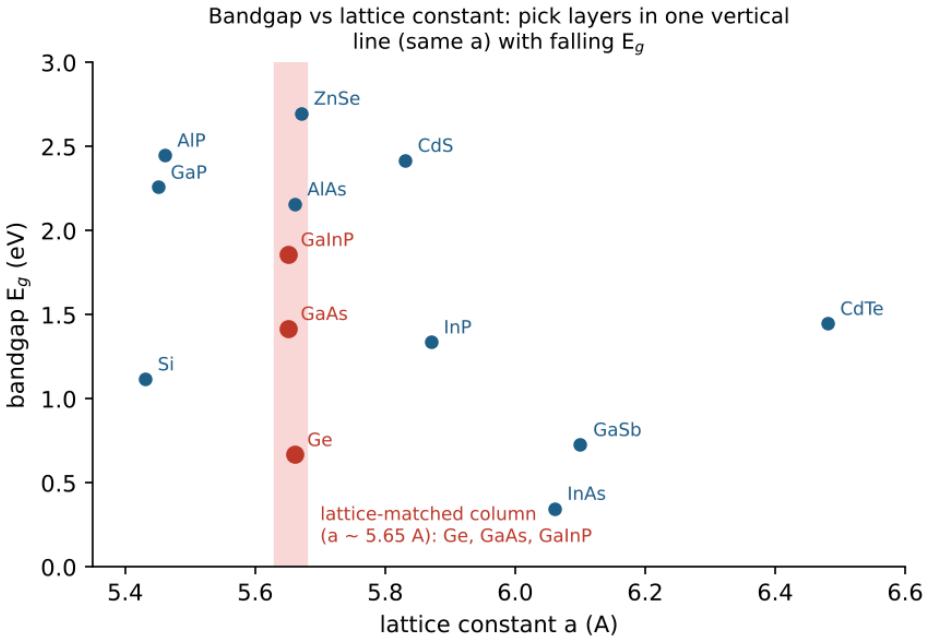
Challenges for multi-junction cells (2022 4c, 2020 2b)

1. **Bandgap matching:** materials with exactly the right gaps (e.g. 1.9/1.4/1.0/0.7 eV) to share the spectrum are needed; not every gap is available.
2. **Current matching:** series sub-cells must produce the same current (lowest one limits); the spectrum changes with time of day, season and air mass, so the match is lost under real conditions.
3. **Lattice matching:** layers grown on each other must have nearly the same lattice constant, otherwise dislocations form (recombination centres). Metamorphic (graded) buffers or wafer bonding are needed for mismatched layers.
4. **Tunnel junctions:** must be very thin, transparent and low-resistance, even at high (concentrated) currents.
5. Thermal expansion mismatch, complex epitaxial growth (MOCVD), many layers (tens), high cost: viable mainly with concentrators or in space.

Lattice mismatch; why Ge, GaAs and GaInP form the triple-junction cell (2022 2c, 2021 3c, 2019 4b)

Lattice mismatch is the difference between the lattice constants of two crystalline materials grown on each other: $\mathbf{f = (a_{layer} - a_{substrate})/a_{substrate}}$. If f is more than ~0.1-1%, the growing layer is

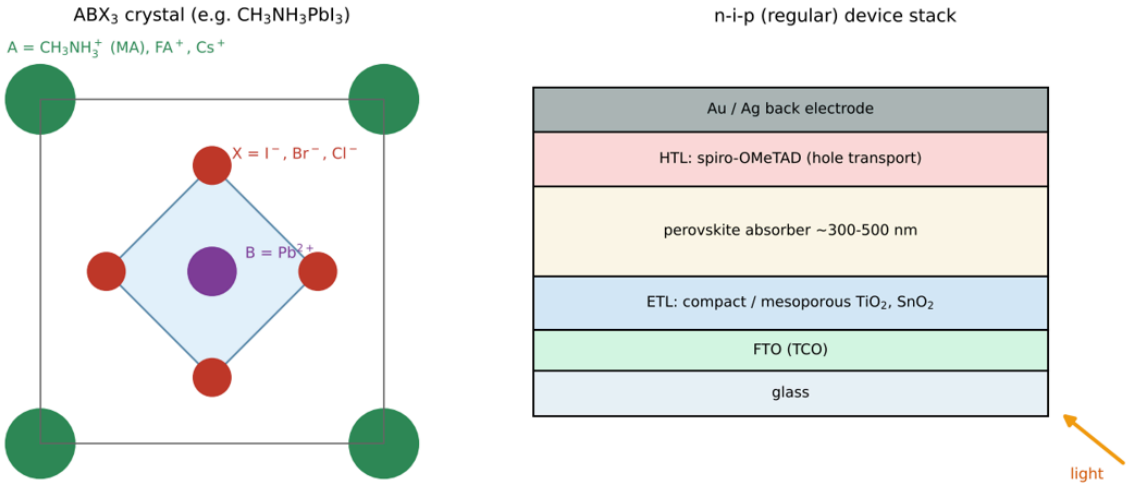
strained and, beyond a critical thickness, **misfit dislocations** form; these act as recombination centres and shorten lifetimes, lowering V_{oc} and efficiency.



Bandgap vs lattice constant of III-V and group-IV semiconductors

- **Lattice match:** Ge (a = 5.658 Å), GaAs (5.653 Å) and $Ga_{0.5}In_{0.5}P$ (5.653 Å) have nearly identical lattice constants (mismatch < 0.1%; adding 1% In to GaAs makes it exact), so they can be grown epitaxially on a Ge wafer without dislocations.
- **Bandgap sequence:** GaInP 1.86 eV (top), GaAs 1.42 eV (middle), Ge 0.67 eV (bottom) fall in the right order and divide the solar spectrum reasonably (about 670 nm and 870 nm cut-offs).
- **Practical:** Ge is a cheap, strong substrate that is also the active bottom cell; all three are direct (except Ge) high-quality materials with established MOCVD (metal-organic chemical vapour deposition) growth; the combination gives > 30% at one sun and ~41% under concentration. (Current is slightly limited by the top/middle cells; Ge produces excess current.)

Perovskite solar cells: properties, classification, structure, working, vs silicon (2020 3a, 2021 7a, 2022 4d)



ABX_3 perovskite crystal and an n-i-p device stack

Perovskite means any material with the crystal structure of $CaTiO_3$, formula **ABX_3**: A = a large monovalent cation (methylammonium $CH_3NH_3^+$ (MA), formamidinium (FA), Cs^+), B = a divalent metal (Pb^{2+}, Sn^{2+}), X = halide (I^-, Br^-, Cl^-). The typical absorber is $MAPbI_3$.

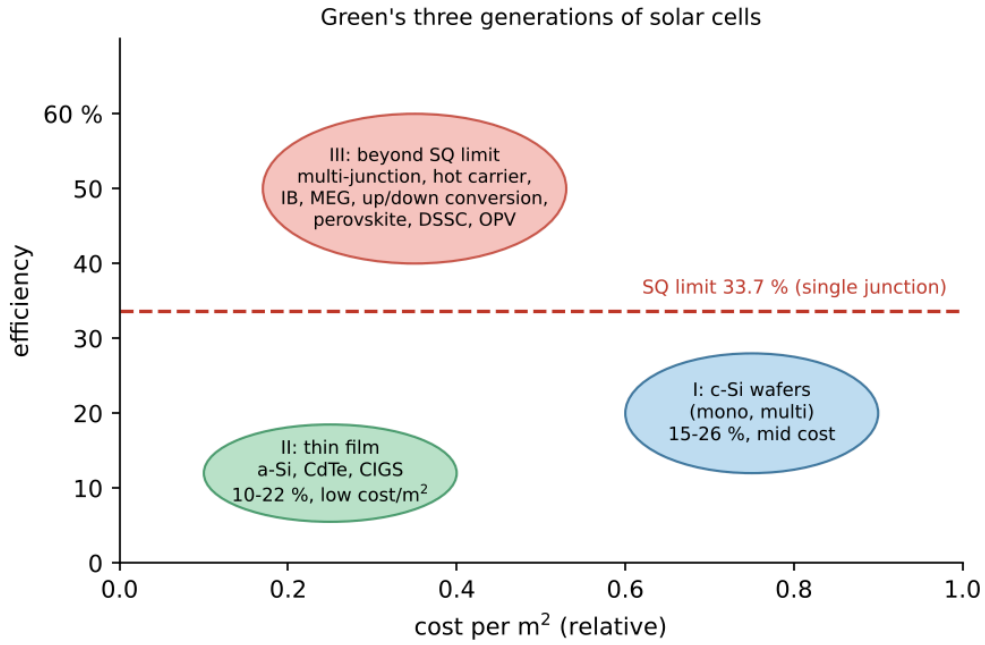
Properties and characteristics: direct bandgap ~1.5-1.6 eV, tunable 1.2-2.3 eV by changing the halide/cation (ideal for tandems); very high absorption coefficient (~10^5 cm^{-1}, 300-500 nm absorbs all light); long carrier diffusion length (> 1 µm) and lifetime; low exciton binding energy (free carriers at room temperature); defect tolerance; ambipolar transport; solution processing at low temperature (spin coating, printing) on glass or flexible foil; rapid efficiency rise from 3.8% (2009) to > 26% (single) and > 33% on Si tandem. Weaknesses: degradation by moisture, oxygen, heat, UV and ion migration; hysteresis; toxic lead; scaling up.

Classification: (a) by structure: **mesoporous** (mesoporous TiO_2 scaffold, from dye cells) and **planar**, which is either **n-i-p (regular)**: glass/FTO/ETL/perovskite/HTL/Au, or **p-i-n (inverted)**: glass/ITO/HTL (PEDOT:PSS, PTAA)/perovskite/ETL (C_{60}, BCP)/Ag; (b) by composition: organic-inorganic hybrid ($MAPbI_3$, $FAPbI_3$, mixed cation), all-inorganic ($CsPbI_3$), lead-free (Sn-based), 2D/3D layered; (c) single junction vs tandem (perovskite/Si, perovskite/CIGS, all-perovskite).

Basic structure and working (n-i-p): light enters through glass and FTO (fluorine-doped tin oxide); the compact/mesoporous TiO_2 (or SnO_2) **electron transport layer** blocks holes; the perovskite absorbs light and creates free electrons and holes (p-i-n type, drift + diffusion); electrons go to the ETL (electron transport layer) and FTO, holes to the **hole transport layer** (spiro-OMeTAD) and the Au/Ag back electrode. The selective contacts with suitable energy levels separate the carriers.

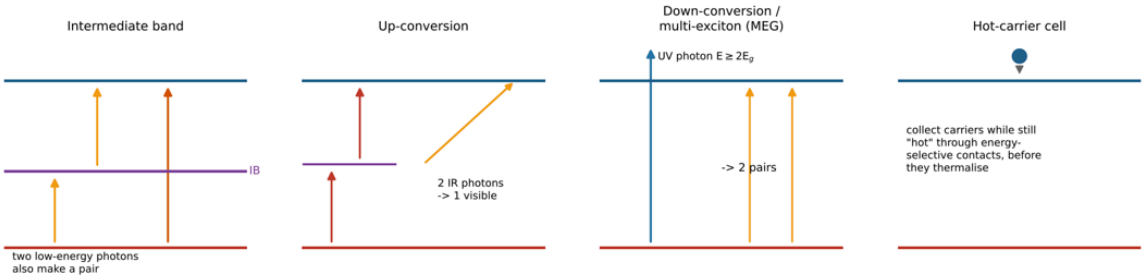
	Perovskite cell	**Traditional c-Si cell**
Absorber	ABX_3 hybrid organic-inorganic halide, polycrystalline thin film 0.3-0.5 µm	single/multi-crystalline Si wafer 150-200 µm
Bandgap	direct, tunable 1.2-2.3 eV	indirect, fixed 1.12 eV
Junction	p-i-n/n-i-p with selective transport layers (ETL/HTL)	p-n homojunction made by diffusion
Processing	solution/vapour at < 150 °C, flexible substrates, low energy	purification + crystal growth at > 1400 °C, energy intensive
Contacts	TCO (transparent conducting oxide) front + ETL/HTL + Au/Ag/C	Ag fingers, Al rear (screen printed)
Efficiency / stability	> 26% lab, short lifetime so far, Pb content	26.8% lab, 22-24% commercial, 25-30 year lifetime
Cost potential	very low materials cost, tandem partner for Si	mature, cheap at scale

Generations and third-generation technology (2023 4d, 2021 4d)



Three generations of solar cells by efficiency and cost

- **First generation:** wafer-based crystalline silicon (mono and multi), p-n homojunction; 15-26%; mature, > 95% of the market; high material and energy cost.
- **Second generation:** thin films (a-Si:H, CdTe, CIGS, thin III-V) on cheap substrates: less material and energy, lower cost per m^2, but usually lower efficiency (10-23%).
- **Third generation:** aims for high efficiency at low cost, beyond the SQ (Shockley-Queisser) limit: **multi-junction** cells, **spectral up- and down-conversion**, **multi-exciton generation** in quantum dots, **intermediate-band** cells, **hot-carrier** cells, **concentrator** cells, and emerging materials (**perovskite, dye-sensitised, organic, quantum dot**).



Third-generation concepts

Significance: they attack the two big spectral losses: intermediate band and up-conversion use sub-gap photons (non-absorption); down-conversion, MEG (multiple exciton generation) and hot-carrier cells use the excess energy of blue photons (thermalisation); multi-junction does both. Theoretical limits: up to ~68% (1 sun) and 86.8% (concentration). They also open low-cost, flexible, lightweight, semi-transparent and building-integrated PV.

> **Memory trick:** Multi-junction = **"BIG gap on TOP"**. Its three matches: **B**and, **C**urrent, **L**attice ("Bangladesh Cricket League"). Ge-GaAs-GaInP all sit at 5.65 Å.

Remember: MJ: stacked gaps 1.86/1.42/0.67 eV, series + tunnel junctions, cuts thermalisation and non-absorption; challenges: bandgap, current, lattice matching, tunnel junctions, cost. Perovskite ABX_3: tunable direct gap, solution processed, > 26%; unstable, Pb. Generations: I wafer Si, II thin film, III beyond SQ.

A1 Semiconductor basics: bands, doping, Fermi level, direct/indirect gap, drift and diffusion, Poisson and continuity equations

Pass set · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2023 Q1(a)** [10] What is the purpose of doping in a semiconductor? Explain different types of doping with band diagram levelling the Fermi level energy.
- **2023 Q1(c)** [15] Write the physical significance of the Poisson equation, continuity equation, minority carrier diffusion length, density of states and Fermi level energy, including their equations.
- **2022 Q1(a)** [11] Define Fermi level in a semiconductor. Briefly explain the physical significance of the Poisson and continuity equations together for semiconductor material.
- **2021 Q1(a)** [10] Same question as 2023 1(a).
- **2021 Q1(b)** [15] Briefly explain direct and indirect band gap semiconductors. Describe the SRH (Shockley-Read-Hall (trap-assisted recombination)) and Auger recombination process mechanisms with their band diagrams. (recombination part: A2)
- **2020 Q2(a)** [11] Define doping. Discuss the p-n junction's working procedure of a solar cell. (p-n part: topic A3)
- **2020 Q1(b)** [12] Explain the basic two transport mechanisms in a semiconductor device.
- **2019 Q1(a)** [5] Why is the Fermi level of an n-type semiconductor near the lowest conduction band energy, and for a p-type near the highest valence band energy? What is a direct and indirect bandgap semiconductor?
- **2019 Q1(b)** [10] What are the four main ideas represented in quantum theory? Explain surface recombination, and mention its significance in a solar cell.

LEARN IT FROM ZERO

Semiconductors, from zero

In a solid, electrons can only have energies inside allowed **bands**. The **valence band** is full of electrons bound in bonds; the **conduction band** above it is where free electrons move. The empty gap between them is the **band gap E_g** (silicon 1.12 eV). Light with photon energy above E_g can lift an electron across, leaving a **hole** (a missing electron that behaves like a positive charge).

Doping adds a few impurity atoms per million: phosphorus (5 outer electrons) gives a spare electron, making **n-type**; boron (3 outer electrons) leaves a hole, making **p-type**. The **Fermi level** is the energy with a 50% chance of being filled: it moves up towards the conduction band in n-type (many electrons) and down towards the valence band in p-type (many holes).

Carriers move in two ways: **drift** (pushed by an electric field) and **diffusion** (spreading from high to low concentration). The **Poisson equation** links charge to the electric field; the **continuity equation** is the carrier bookkeeping: change = generation − recombination ± flow.

★ REMEMBER IT

- **n-type: P for Plus an electron** (phosphorus, group V, donor). **p-type: B for Bye electron** (boron, group III, acceptor).

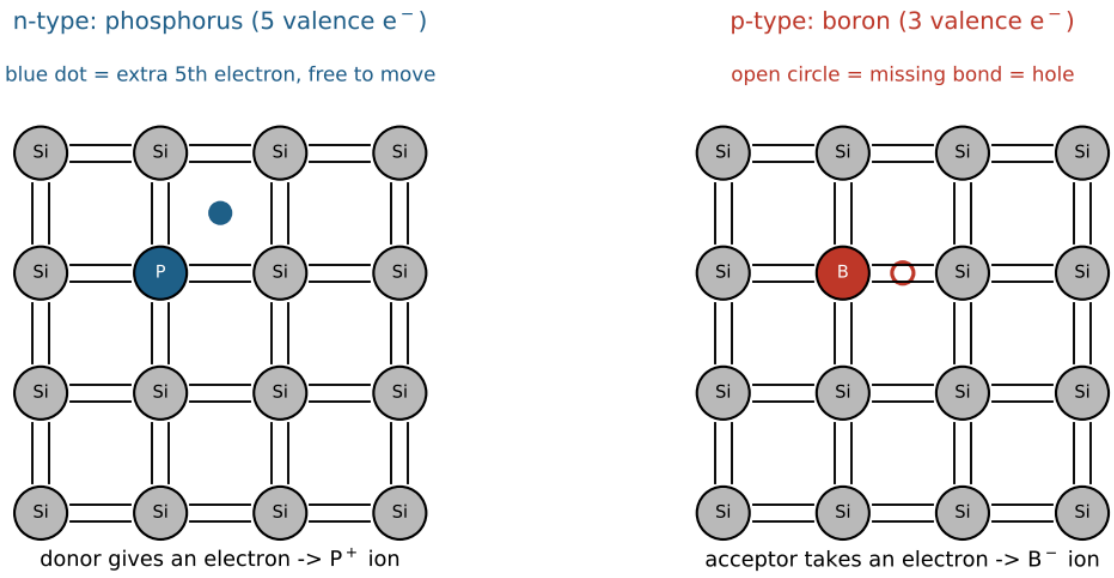
- Direct gap (GaAs): electron jumps straight up, absorbs light strongly, thin cells. Indirect gap (Si): needs a phonon (lattice vibration) too, weak absorption, thick wafers.

Before you start: electrons in a crystal

In a single atom an electron can only have certain energies. When billions of Si atoms bond into a crystal, these levels spread into **bands**. The **valence band** (VB) holds the electrons that sit in the covalent bonds; the **conduction band** (CB) holds electrons that are free to move. Between them is a forbidden gap, the **bandgap E_g** (Si 1.12 eV, GaAs 1.42 eV). When a bond breaks (by heat or a photon) we get a free electron in the CB and a missing electron, a **hole**, in the VB; the hole behaves like a positive carrier. Source: Rahaman sir's Part II ch. 6 (follows Smets et al. ch. 6). 1 eV = 1.602×10^{-19} J; kT = 0.0258 eV at 300 K.

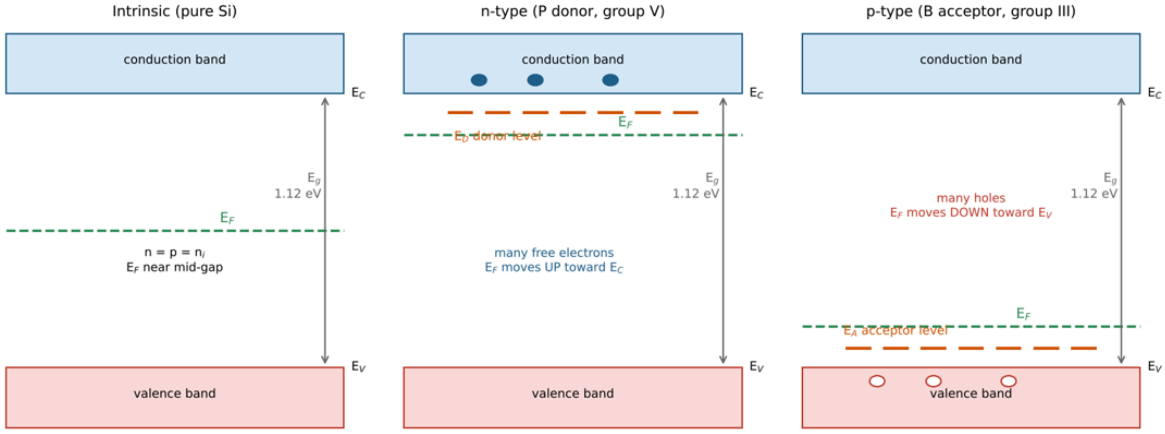
Purpose of doping; types of doping with band diagrams and Fermi level (2023 1a, 2021 1a, 2020 2a)

Doping is the deliberate addition of a tiny amount of impurity atoms (about 1 in 10^6-10^8 Si atoms) to a pure (intrinsic) semiconductor to control its type and number of charge carriers. **Purpose:** pure Si has only $n_i = 1.5 \times 10^{10}$ cm^{-3} carriers at 300 K, so it conducts badly and has no internal field. Doping (i) raises conductivity by many orders of magnitude, (ii) makes one carrier the majority (electrons or holes), and (iii) allows a **p-n junction** with a built-in electric field, which is what separates the light-generated carriers in a solar cell.



Doping in the bond model: phosphorus donates an electron, boron accepts one

	Intrinsic	n-type	p-type
Dopant	none	group V donor: P, As, Sb (5 valence e^-)	group III acceptor: B, Al, Ga (3 valence e^-)
What happens	thermal bond breaking only	4 electrons bond, the 5th is loosely held (0.045 eV) and becomes free; P^+ ion left	one bond is incomplete = hole; a neighbouring electron fills it; B^- ion left
Carriers	$n = p = n_i$	$n \approx N_D$ (majority), $p = n_i^2/N_D$	$p \approx N_A$ (majority), $n = n_i^2/N_A$
New level in the gap	none	donor level E_D just below E_C	acceptor level E_A just above E_V
Fermi level	near mid-gap, E_{Fi}	moves up toward E_C: $E_C - E_F = kT \ln(N_C/N_D)$	moves down toward E_V: $E_F - E_V = kT \ln(N_V/N_A)$



Band diagrams: the Fermi level moves toward the band of the majority carrier

There is also **compensated** doping (both donors and acceptors; the larger one wins: $n \approx N_D - N_A$) and **degenerate** (very heavy) doping, n^+/p^+, where E_F reaches or enters the band (used for emitters, BSF (back surface field) and contacts).

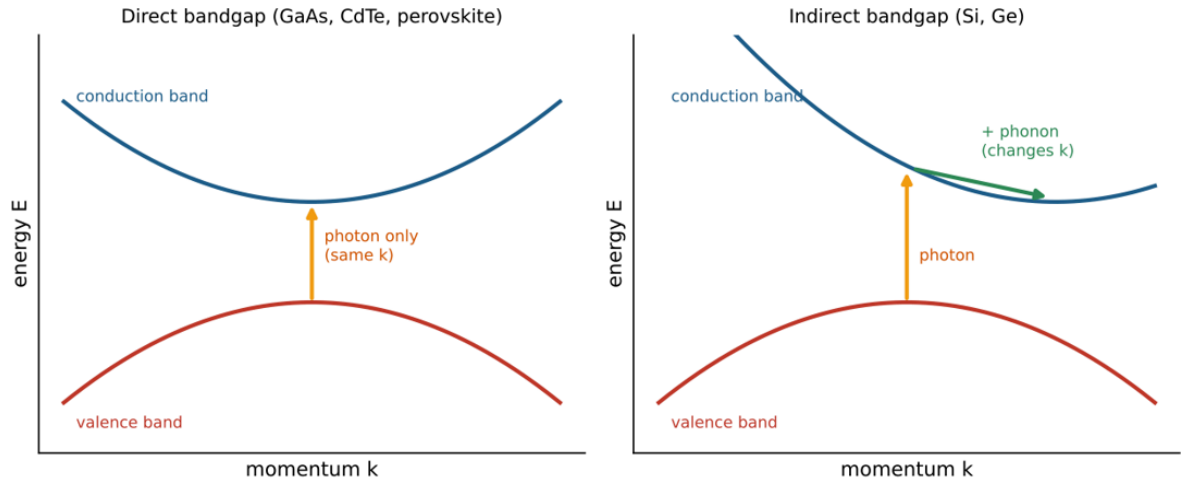
> **Side note:** Why does E_F move toward E_C in n-type (2019 1a)? The Fermi level is the energy where a state is occupied with probability ½. In n-type material there are many more electrons in the CB, so the occupation probability near E_C must be higher; from $n = N_C e^{-(E_C - E_F)/kT}$, a large n needs a small $E_C - E_F$. In p-type there are few electrons (many holes), so E_F must sit close to E_V.

Fermi level, density of states, Poisson and continuity equations, diffusion length (2023 1c, 2022 1a)

Quantity and equation	Physical significance
Fermi level E_F; Fermi-Dirac $f(E) = 1/[1 + e^{(E-E_F)/kT}]$	The energy at which the probability that a state is filled is exactly 50% (at T = 0 K all states below E_F are full, above are empty). Its position tells the doping type and carrier density; in equilibrium E_F is flat through a whole device (no net current).
Density of states $g_C(E) = 4\pi(2m_n^*/h^2)^{3/2}\sqrt{(E - E_C)}$	The number of allowed electron states per unit energy per unit volume. Multiplying by f(E) and integrating gives the carrier density: $n = \int g_C f\, dE = N_C e^{-(E_C-E_F)/kT}$, with $N_C = 3.22 \times 10^{19}$, $N_V = 1.83 \times 10^{19}$ cm^{-3} for Si.
Poisson equation $d^2\psi/dx^2 = -dE/dx = -\rho/\varepsilon = -(q/\varepsilon)(p - n + N_D^+ - N_A^-)$	Links the space charge to the electric field and potential. It is used to find the built-in field, potential (V_{bi}) and width of the depletion region of a junction; wherever the net charge is zero (quasi-neutral region) there is no field.
Continuity equations $\partial n/\partial t = (1/q)\partial J_n/\partial x + G_n - R_n$; $\partial p/\partial t = -(1/q)\partial J_p/\partial x + G_p - R_p$	Carrier bookkeeping (conservation of charge): the change of carriers in a slice = what flows in minus out (drift + diffusion current) + generation − recombination. Solving them with the current equations gives the dark current J_0 and the photocurrent of the cell.
Minority-carrier diffusion length $L_n = \sqrt{(D_n\tau_n)}$, $L_p = \sqrt{(D_p\tau_p)}$	The average distance a minority carrier diffuses before it recombines. Only carriers generated within about L of the junction are collected, so L should be longer than the absorber thickness. Example: $\mu_n = 1250$ cm^2/Vs, $\tau = 1$ µs → $L = \sqrt{(0.0258 \times 1250 \times 10^{-6})} = 57$ µm.

Memory trick: F-D-P-C-L: Fermi = who fills; Density of states = how many seats; Poisson = charge makes field; Continuity = in − out + made − lost; Length = how far a minority carrier lives.

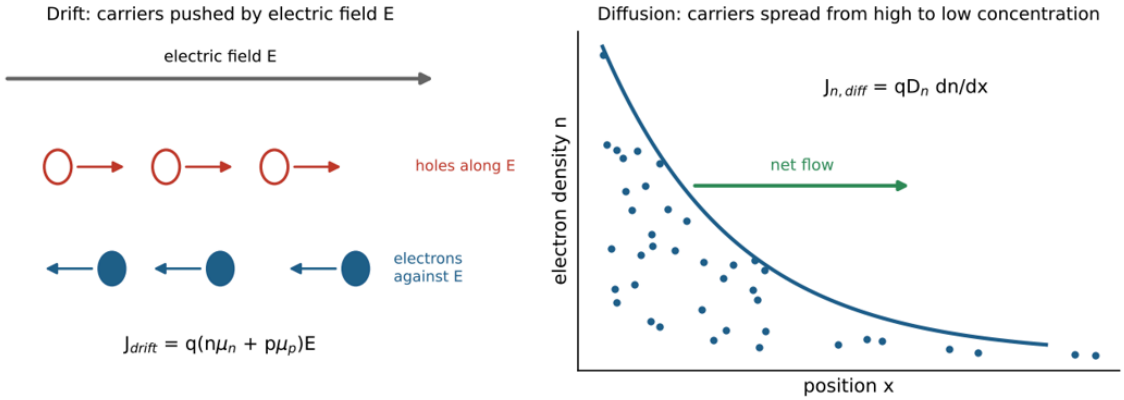
Direct and indirect bandgap semiconductors (2019 1a, 2021 1b)



E-k diagrams: a direct gap needs only a photon; an indirect gap also needs a phonon

	Direct bandgap	Indirect bandgap
Band edges	CB minimum and VB maximum at the **same** crystal momentum k	CB minimum and VB maximum at **different** k
Absorption/emission	photon alone (momentum conserved); strong absorption, $\alpha \sim 10^4$-10^5 cm^{-1}, a few µm absorbs all light	needs a photon plus a **phonon** (lattice vibration) to supply momentum: weak absorption, 100-200 µm needed
Main recombination	radiative (used in LEDs, lasers)	SRH and Auger; long lifetime
Examples	GaAs, InP, CdTe, CIGS (copper indium gallium diselenide), perovskite	Si, Ge, GaP
PV consequence	thin-film cells possible	thick wafers + light trapping

Two transport mechanisms: drift and diffusion (2020 1b)



Drift is driven by an electric field; diffusion by a concentration gradient

- **Drift** is motion of charged carriers caused by an electric field E: holes move along E, electrons against it, with drift velocity $v = \mu E$ (µ = mobility, cm^2/Vs). $J_{drift} = q(n\mu_n + p\mu_p)E$. In a solar cell drift happens in the depletion region, where the built-in field separates electrons and holes.
- **Diffusion** is the spreading of carriers from high to low concentration by random thermal motion; it does not need charge or a field. $J_{n,diff} = qD_n dn/dx$, $J_{p,diff} = -qD_p dp/dx$. In the quasi-neutral p and n regions, light-made minority carriers reach the junction by diffusion.
- They are linked by the **Einstein relation** $D/\mu = kT/q$ (0.0258 V at 300 K), e.g. $\mu_n = 1360$ cm^2/Vs → $D_n = 35$ cm^2/s. Total current $J = q(n\mu_n + p\mu_p)E + q(D_n dn/dx - D_p dp/dx)$. Diffusion devices (c-Si) rely mainly on diffusion; drift devices (a-Si p-i-n) on drift.

Four main ideas of quantum theory (2019 1b, first part)

1. **Energy is quantised:** energy is exchanged in discrete packets; light comes as photons with $E = h\nu = hc/\lambda$ (Planck, Einstein photoelectric effect).
2. **Wave-particle duality:** light and electrons behave both as waves and particles; an electron has a de Broglie wavelength $\lambda = h/p$.
3. **Uncertainty principle:** position and momentum (or energy and time) cannot both be known exactly: $\Delta x \Delta p \geq h/4\pi$; so electrons are described by probabilities (wave function, Schrödinger equation).
4. **Pauli exclusion and allowed states:** electrons occupy only allowed energy levels, no two in the same quantum state; in a crystal the levels form bands separated by gaps, which is the basis of the semiconductor band model.

Remember: Doping: P (V) → n, E_F up; B (III) → p, E_F down; $n \cdot p = n_i^2$. Drift = field, diffusion = gradient, $D/\mu = kT/q$. Direct gap: photon only (GaAs); indirect: photon + phonon (Si).

A7 Thin-film cells: III-V, CdTe, CIGS, thin-film silicon (a-Si:H), materials criteria

Next to add | Very likely · 87% | Medium

PAST-PAPER QUESTIONS

- **2023 Q4(b)** [10] Explain the operation of a CdTe solar cell using its layer structure.
- **2023 Q4(c)** [5] Why is III-V technology for solar cells called thin-film solar cell?
- **2022 Q4(b)** [10] Explain the operation of a CdTe solar cell using its layer structure.
- **2022 Q4(a)** [10] Why TCO (transparent conducting oxide) is used; why III-V material is considered for thin-film technology.
- **2022 Q2(b)** [7] What are the various types of materials used in the construction of thin-film solar cells and how do they contribute to the overall efficiency?
- **2021 Q2(a)** [12] Mention the names of some III-V materials used in thin-film solar cell production. Also discuss the design of thin-film silicon solar cells.
- **2021 Q3(d)** [5] What are the criteria for materials to be used in solar cells?
- **2019 Q4(a)** [3] Mention the names of some III-V materials used in thin-film solar cell production.
- **2019 Q4(d)** [15] Discuss the design of thin-film silicon solar cells.
- **2019 Q4(c)** [10] Why is hydrogen used in thin-film silicon alloys? Why is a thin-film silicon solar cell called a drift device?

★ REMEMBER IT

- Thin-film materials: **CdTe** (glass/TCO/CdS window/CdTe absorber/back contact; E_g 1.45 eV), **CIGS** (copper indium gallium diselenide) ($Cu(In,Ga)Se_2$), **a-Si:H** (hydrogenated amorphous silicon), **III-V** (GaAs, InP, GaInP, AlGaAs: direct gap, absorb in a few µm, so a thin film works).
- Hydrogen in a-Si:H passivates dangling bonds (defect density from 10^{21} to 10^{16}). a-Si is a **drift** device (p-i-n): diffusion lengths are tiny, so the built-in field across the thick intrinsic layer has to sweep carriers out.

Before you start: why thin films

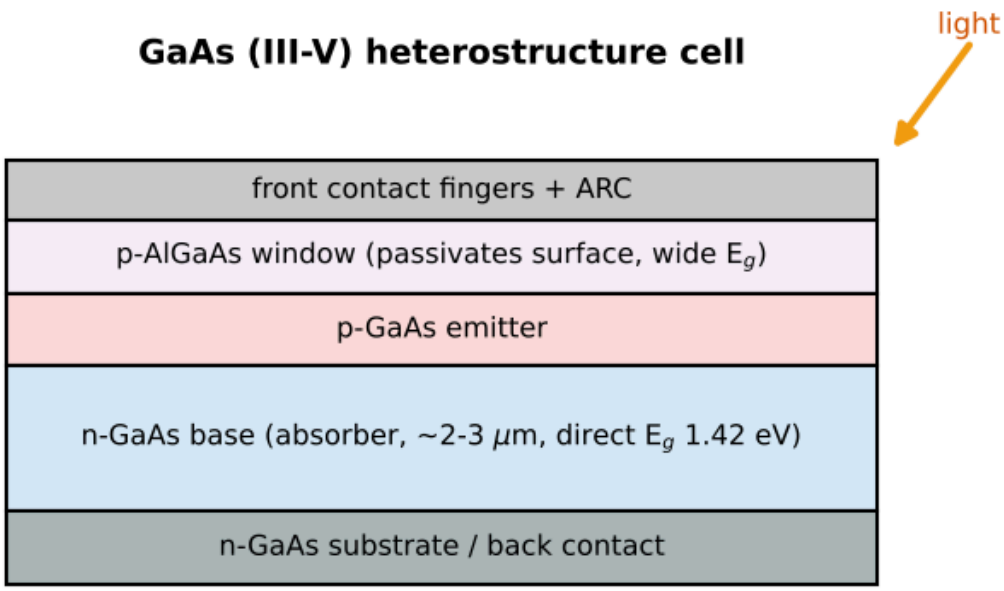
Silicon absorbs weakly (indirect gap), so wafers are 150-200 µm thick. Direct-gap materials absorb sunlight in 1-3 µm, so a cell can be a thin film deposited on cheap glass or foil: less material, less energy, large-area processing. The price is more complex materials and interfaces. Source: Rahaman sir Part III ch. 13 (Smets ch. 13).

Materials used in thin-film cells and how they contribute (2022 2b); criteria for solar cell materials (2021 3d)

Family	Materials	Contribution
III-V compounds	GaAs, GaP, InP, InAs, GaInAs, GaInP, AlGaAs, AlGaInP, AlGaInAs	direct gap, highest efficiency (29% single, 47% multi-junction), radiation hard
II-VI	CdTe (+ CdS buffer)	E_g 1.45 eV ideal; cheap, fast deposition; 22%
I-III-VI (chalcopyrite)	$CuInSe_2$, $CuGaSe_2$, $Cu(In,Ga)Se_2$ (CIGS), CIGSS; kesterite CZTS (I_2-II-IV-VI_4)	tunable E_g 1.0-1.7 eV, very high absorption; 23%; CZTS (copper zinc tin sulfide) uses abundant elements
Thin-film silicon (IV-IV)	a-Si:H, nc-Si:H (hydrogenated nanocrystalline silicon), a-SiGe:H, a-SiC:H, nc-SiO_x:H	abundant, non-toxic, low temperature, tandems; 10-14%
Organic / hybrid	polymers + fullerenes (OPV), dye-sensitised TiO_2 (DSSC), perovskites	cheap, flexible, solution-processed; perovskite 26%
Supporting layers	TCO (ITO, FTO (fluorine-doped tin oxide), AZO (aluminium-doped zinc oxide)), buffers (CdS, ZnO), Mo, back reflectors	front/back contact, junction formation, light trapping

Criteria for a good solar cell material: (1) bandgap 1.1-1.7 eV (close to the SQ (Shockley-Queisser) optimum 1.34 eV); (2) high absorption coefficient (preferably direct gap) so a thin layer suffices; (3) long minority-carrier diffusion length / lifetime (high purity, few defects, high mobility); (4) ability to be doped n and p or to form a good heterojunction; (5) stability under light, heat and moisture; (6) abundant, cheap, non-toxic elements; (7) easy, low-energy, large-area processing; (8) good contacts and passivation possible.

III-V technology: names, why it is thin film (2023 4c, 2022 4a, 2021 2a, 2019 4a)

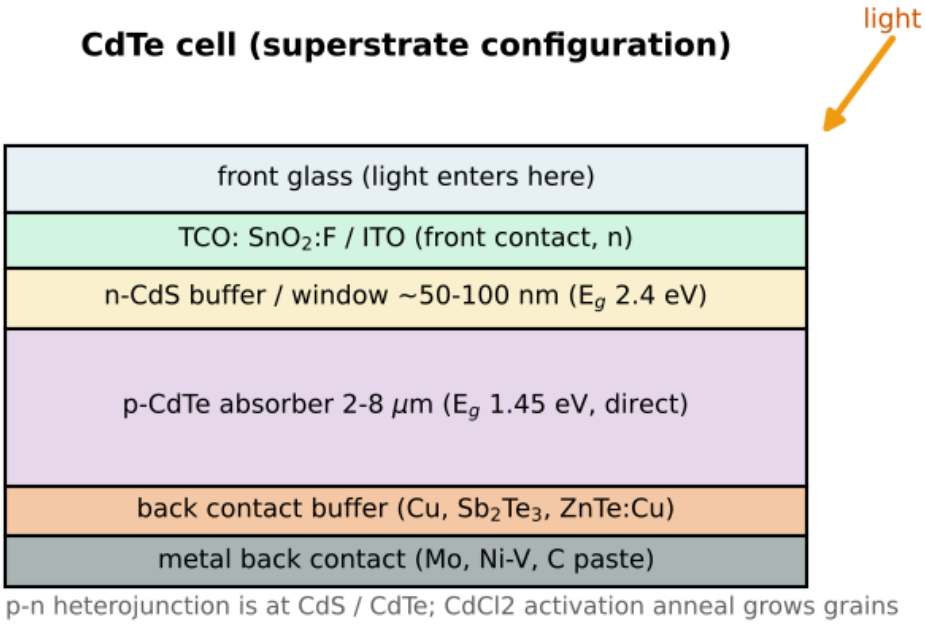


GaAs heterostructure cell with an AlGaAs window

- **Materials:** GaAs, InP, GaP, InAs, GaSb, AlAs, and alloys GaInP, AlGaAs, GaInAs, AlGaInP, AlGaInAs (elements from groups III: Al, Ga, In; and V: N, P, As, Sb).

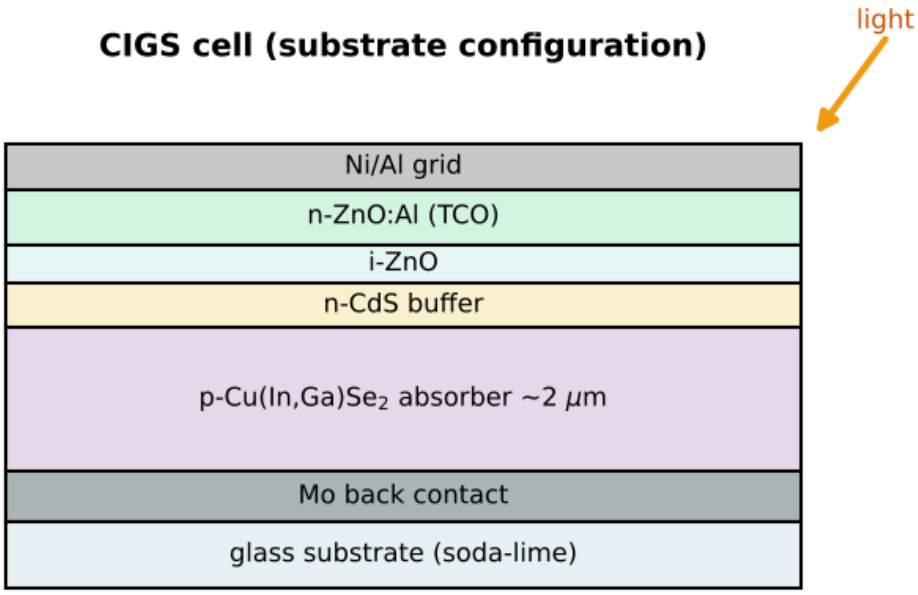
- **Why called thin film:** they have a direct bandgap with absorption coefficients ~10^4-10^5 cm^{-1}, so the active absorber is only 1-3 µm thick (vs 180 µm for Si). The layers are grown epitaxially as thin films (MOCVD, MBE (molecular beam epitaxy)) on a substrate (GaAs or Ge) which can even be removed (epitaxial lift-off) to make thin, flexible cells.
- **Why considered:** highest efficiencies (GaAs 29.1% single junction, record 47.6% multi-junction under concentration); bandgap and lattice constant tunable by alloying (multi-junction stacks); high radiation resistance and low temperature coefficient (space satellites); excellent for concentrator PV. Drawbacks: expensive substrates and growth, scarce In/Ga, toxic As.

CdTe solar cell: layer structure and operation (2023 4b, 2022 4b)



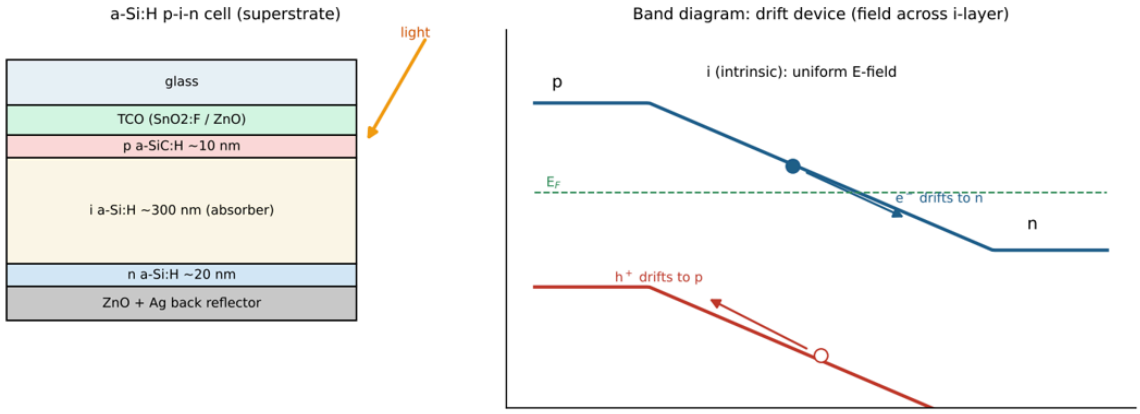
CdTe cell in superstrate configuration

1. Light enters through the **front glass** (superstrate: layers are deposited on the glass that faces the sun).
2. **TCO** (SnO_2:F or ITO (indium tin oxide), n-type) is the transparent front contact.
3. **n-CdS buffer/window** (50-100 nm, E_g 2.4 eV): transparent to most light and forms the **n-p heterojunction** with p-CdTe. (Thin to reduce blue absorption; now often replaced by ZnO/MgZnO.)
4. **p-CdTe absorber** (2-8 µm, E_g 1.45 eV direct, zinc-blende structure): absorbs almost all light above E_g; deposited by close-spaced sublimation or vapour transport; a **$CdCl_2$ activation anneal** (~400 °C) grows grains, passivates grain boundaries and makes it p-type.
5. **Back contact buffer** (Cu, ZnTe:Cu, Sb_2Te_3): CdTe has a very high work function (~5.7 eV), so a heavily doped p^+ layer is needed for an ohmic back contact; then **metal back contact** (Ni-V, Mo, carbon paste).
6. **Operation:** photons absorbed in CdTe create pairs; the built-in field at the CdS/CdTe junction separates them: electrons cross to CdS and the TCO (front, −), holes go to the back contact (+). Efficiency up to 22% (cell), 19% (module, First Solar); low cost, good in hot climates. Issues: Cd toxicity (recycled), Te scarcity.



For comparison: CIGS cell in substrate configuration

Thin-film silicon: design, hydrogen, drift device (2021 2a, 2019 4d, 4c)



Thin-film a-Si:H p-i-n cell and its band diagram

- **Why hydrogen is used:** amorphous silicon has no long-range order, so many Si atoms have **dangling bonds** (~10^{19}-10^{20} cm^{-3} in pure a-Si), which act as deep traps and pin the Fermi level; the material is useless. Depositing from silane with hydrogen (PECVD, SiH_4 + H_2) gives **a-Si:H** with ~10% H that bonds to and **passivates the dangling bonds**, cutting defect density to ~10^{15}-10^{16} cm^{-3} so it can be doped and used. Hydrogen also widens the gap (1.7-1.8 eV) and high H dilution produces nano-crystalline nc-Si:H. Weakly bonded H causes the **Staebler-Wronski effect** (light-induced degradation, ~10-30% relative).
- **Why a drift device:** even in a-Si:H the diffusion length is very short (~0.1-1 µm), so carriers cannot reach a junction by diffusion. A **p-i-n** structure puts the absorber (i layer) inside the electric field between the thin p and n layers; carriers are swept out by **drift** as soon as they are generated. Hence it is called a drift device (c-Si is a diffusion device).
- **Design:** (1) superstrate p-i-n on glass/TCO (or substrate n-i-p on steel/plastic); p-layer on the light side (holes have lower mobility so they travel the shorter path); (2) thin, wide-gap p window (a-SiC:H, ~10 nm) to reduce absorption losses; (3) i-layer 200-300 nm: thick enough to absorb, thin enough to keep a strong field and limit degradation; (4) textured TCO and ZnO/Ag back reflector for light trapping; (5) **multi-junction**: a-Si:H/nc-Si:H (“micromorph”) or a-Si/a-SiGe/nc-

Si triple junction to use more of the spectrum and reduce degradation; (6) laser scribing (P1, P2, P3) to connect cells monolithically into a module.

Remember: Thin film = direct gap, 1-3 μm absorber. III-V: GaAs, InP, GaInP, AlGaAs... best but costly. CdTe: glass / TCO / n-CdS / p-CdTe / back contact, 1.45 eV. a-Si:H: H fixes dangling bonds; short L → p-i-n drift device.

A2 Generation and recombination: radiative, SRH, Auger, surface; diffusion length; photocurrent numerical

Pass set · Very likely · 87% · Medium

PAST-PAPER QUESTIONS

- **2023 Q1(b)** [10] Explain the recombination process through SRH (Shockley-Read-Hall (trap-assisted recombination)) and Auger recombination with proper diagrams.
- **2022 Q1(b)** [11] Briefly explain the role of carrier generation and recombination in maintaining dynamic equilibrium. Define minority carrier diffusion length with physical significance.
- **2021 Q1(b)** [15] Direct and indirect band gap; SRH and Auger recombination with band diagrams.
- **2021 Q1(c)** [10] The total absorption in a d = 300 μm thick c-Si wafer for light of wavelength 500 nm. Incident photons reflected from the surface are 0.38 and absorption coefficient $\alpha = 1.11 \times 10^4$ cm^{-1}. The incident irradiance is 1000 W/m^2. Calculate the photocurrent density.
- **2019 Q2(a)** [10] Explain Auger recombination in detail.
- **2019 Q1(b)** [10] Surface recombination and its significance in a solar cell (second part).

★ REMEMBER IT

- Recombination = an electron falls back into a hole and the energy is lost. **Radiative**: gives out a photon (direct-gap materials, LEDs). **SRH** (Shockley-Read-Hall): via a trap level from a defect or impurity, energy goes to heat (dominant in Si). **Auger**: energy is given to a third carrier which loses it as heat (high doping, high injection). **Surface**: dangling bonds at the surface act as traps (fix with passivation).
- Diffusion length $L = \sqrt{(D\tau)}$: how far a minority carrier travels before it recombines. It must be longer than the cell thickness for good collection.

CALCULATOR **Photocurrent from absorption (Lambert-Beer, 2021 1(c))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Incident photon flux: $\Phi_0 = P\lambda/(hc)$ = **1000** × **500** × 10^{-9} / (6.626×10^{-34} × 2.998×10^{8}) = **2.517 × 10^{21}** $1/m^2s$
2. αd: **11100** × **300** × 10^{-4} = **333.00**
3. Absorbed fraction of entering light: $1 - e^{-\alpha d}$ = 1 - exp(-**333**) = **1.0000**
4. Absorbed flux: $(1 - R)\Phi_0(1 - e^{-\alpha d})$ = (1 - **0.38**) × **2.517 × 10^{21}** × **1** = **1.561 × 10^{21}** $1/m^2s$
5. Photocurrent density: $J_{ph} = qA$ = 1.602×10^{-19} × **1.561 × 10^{21}** = **250.0** A/m^2
6. Photocurrent density: **250** / 10 = **25.0** mA/cm^2

Try λ = 1000 nm with α = 64 /cm: about 85% of that infrared light passes straight through a 300 μm wafer.

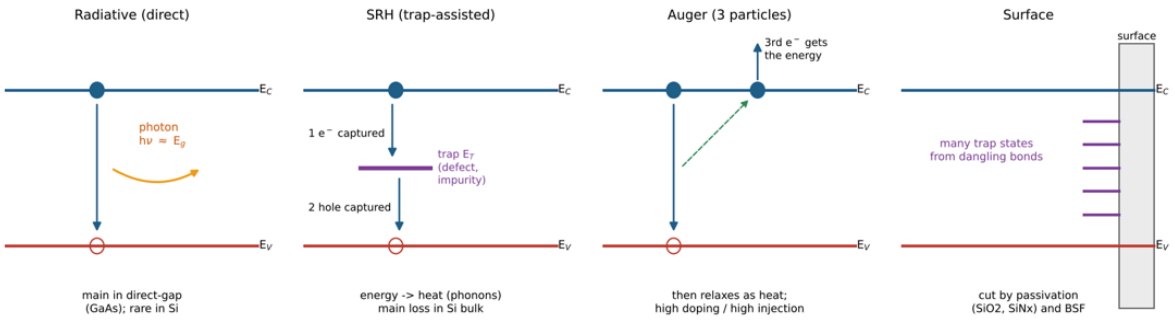
Before you start: making and losing carriers

A solar cell works only if light **generates** electron-hole pairs faster than they **recombine**, and if the pairs reach the junction before they are lost. Generation needs a photon with energy ≥ E_g. Recombination is the reverse: the electron falls back into a hole and the energy is given out as a photon, as heat, or to another carrier. Source: Rahaman sir's Part II ch. 7 (Smets ch. 7).

Role of generation and recombination in dynamic equilibrium; diffusion length (2022 1b)

- At any temperature above 0 K, bonds are continuously broken thermally (**generation**) and electrons continuously fall back (**recombination**). In thermal equilibrium the two rates are equal (G = R), so carrier numbers stay constant and $\mathbf{np = n_i^2}$ (mass-action law). This is a *dynamic* equilibrium: both processes keep going.
- If there are extra carriers ($np > n_i^2$, e.g. under light or forward bias), recombination exceeds generation and drives the system back. If there are too few ($np < n_i^2$, e.g. reverse bias), generation wins.
- Under illumination the extra (excess) carriers Δn live for a **lifetime τ** ($R = \Delta n/\tau$). During that time they diffuse an average distance $L = \sqrt{(D\tau)}$, the **minority-carrier diffusion length**. Physical significance: carriers generated farther than about L from the junction recombine before collection; so high-quality material (long τ, long L > wafer thickness) gives high current and voltage.

Recombination mechanisms: radiative, SRH, Auger, surface (2023 1b, 2021 1b, 2019 2a, 2019 1b)



Recombination paths in band diagrams

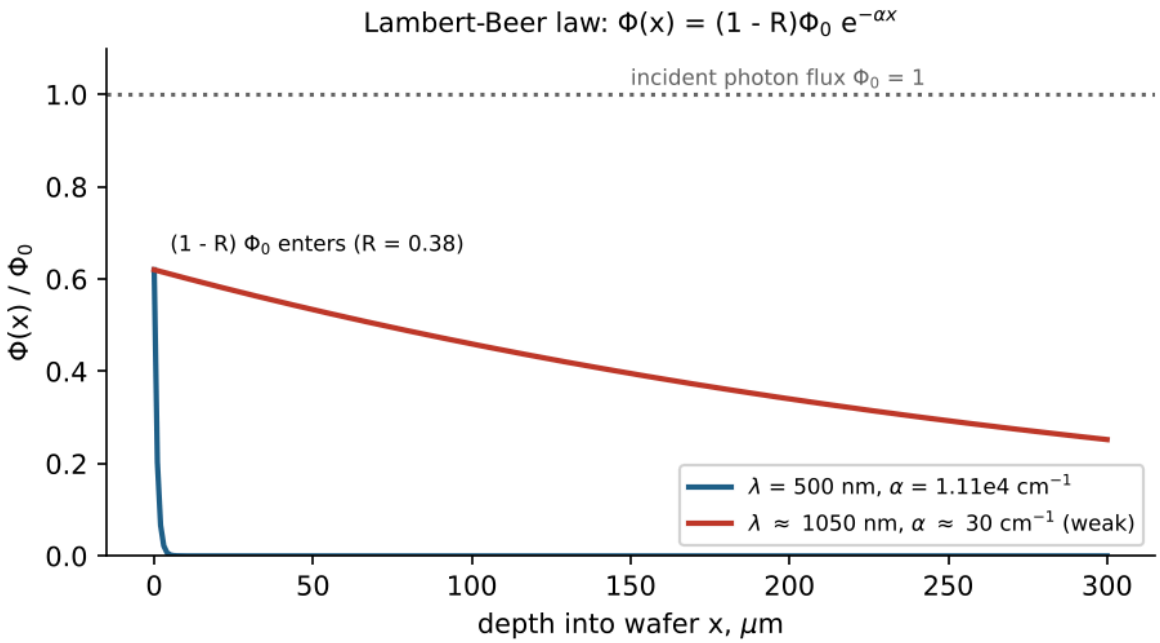
Mechanism	How it works	Where important
Radiative (band-to-band, direct)	a CB electron drops directly into a VB hole and emits a photon of energy ≈ E_g; rate $R = B(np - n_i^2)$	direct-gap materials (GaAs, perovskite); unavoidable; very small in Si
Shockley-Read-Hall (SRH), trap-assisted	defects or impurities (Fe, Au, dislocations, grain boundaries) create an energy level E_T inside the gap. Step 1: the trap captures a CB electron; step 2: the trap captures a VB hole (or emits the electron to the VB). Energy is released in small steps as **phonons (heat)**. Four processes: electron capture, electron emission, hole capture, hole emission. Most effective when E_T is near mid-gap.	dominant in Si and most real cells; controlled by material purity
Auger	three-particle process: an electron and a hole recombine and give their energy (and momentum) to a **third carrier** (an electron in the CB, eeh, or a hole in the VB, ehh). The third carrier is kicked high into the band and then relaxes back by emitting phonons (heat). Rate $\propto n^2p$ or np^2	heavily doped regions (emitter, BSF (back surface field)), high injection, concentrators; indirect semiconductors; sets the intrinsic limit of Si (~29.4%)
Surface	the crystal ends at the surface; unsatisfied **dangling bonds** create many trap states throughout the gap, so SRH recombination is very fast at the surface (surface recombination velocity S, cm/s)	front and back surfaces, contacts; reduced by passivation (SiO_2, SiN_x, Al_2O_3, a-Si:H (hydrogenated amorphous silicon)), BSF, small contact area

Significance of surface recombination in a solar cell: blue light is absorbed close to the front surface and long-wavelength carriers reach the back surface, so high S reduces the blue and red response (EQE), lowers J_{sc}, raises the dark current J_0 and so lowers V_{oc}. Thin wafers make surfaces even more important. Hence high-efficiency cells (PERC, PERL (passivated emitter, rear locally diffused), HIT (heterojunction with intrinsic thin layer)) are mainly about passivating surfaces.

> **Memory trick: R-S-A-S**: **R**adiative = light out; **S**RH = Stepping via a trap; **A**uger = A third carrier takes it; **S**urface = dangling bonds.

Numerical: photocurrent density from absorption (2021 1c)

Given: d = 300 µm c-Si, λ = 500 nm, R = 0.38, $\alpha = 1.11 \times 10^4$ cm^{-1} = 1.11×10^6 m^{-1}, irradiance P = 1000 W/m^2. **Find:** J_{ph}. This is Example 7.1 in the teacher's Part II notes (Smets).



Lambert-Beer decay: 500 nm light is absorbed in the first few micrometres

Step (write this)	Why / what it means
$\Phi_0 = P\lambda/(hc) = 1000 \times 500 \times 10^{-9}/(6.626 \times 10^{-34} \times 2.998 \times 10^8) =$ **2.52×10^{21} $m^{-2}s^{-1}$**	photons per second per m^2: power divided by the energy of one photon hc/λ (= 3.97 × 10^{-19} J = 2.48 eV)
Entering flux $= (1 - R)\Phi_0 = 0.62 \times 2.52 \times 10^{21} = 1.56 \times 10^{21}$	38% is reflected at the front surface
$\Phi(d) = (1 - R)\Phi_0 e^{-\alpha d}$; $\alpha d = 1.11 \times 10^6 \times 300 \times 10^{-6} = 333 \rightarrow e^{-333} \approx 0$	Lambert-Beer law: nothing reaches the back; 500 nm light dies within ~4/α = 3.6 μm
Absorbed $A = (1 - R)\Phi_0 - \Phi(d) = 1.56 \times 10^{21}$ $m^{-2}s^{-1}$	all photons that enter are absorbed
$J_{ph} = qA = 1.602 \times 10^{-19} \times 1.56 \times 10^{21} =$ **250 A/m^2 = 25 mA/cm^2**	assume each photon makes one collected pair (η_g = 1)

Side note: The teacher's notes round Φ_0 to 2.5 × 10^{21} and get 248 A/m^2; both are correct. If you use h = 6.625 × 10^{-34} you get the same to 3 figures.

Remember: np = n_i^2 in equilibrium (G = R). L = √(Dτ). $\Phi_0 = P\lambda/hc$, $\Phi(x) = (1-R)\Phi_0 e^{-\alpha x}$, J = q × absorbed flux = 250 A/m^2.

A3 Junctions: p-n junction solar cell, p-i-n, metal-semiconductor, diode equation; junction numericals Pass set Very likely · 100% Hard

PAST-PAPER QUESTIONS

- **2023 Q2(a)** [11] Explain the working principle of a p-i-n junction solar cell using a schematic diagram.
- **2022 Q3(a)** [10] Explain the working principle of a p-n junction solar cell using a schematic diagram.
- **2022 Q3(d)** [8] A c-Si wafer is doped with 10^{16} acceptor atoms/cm^3. A 1 µm thick emitter layer is formed at the surface with a uniform 10^{18} donors/cm^3. Assume a step p-n junction, all dopants ionized, $n_i = 1.5 \times 10^{10}$ cm^{-3} at 300 K. Calculate the built-in voltage, width of the depletion region and maximum electric field at the metallurgical junction.
- **2021 Q3(b)** [12] Same wafer as 2022 3(d). Calculate the electron and hole concentrations in the p- and n-type quasi-neutral regions at thermal equilibrium.
- **2020 Q2(a)** [11] Define doping. Discuss the p-n junction's working procedure of a solar cell.
- **2020 Q2(c)** [4] Write the diode equation and draw the I-V curve of an ideal diode.
- **2020 Q3(b)** [10] Si cell at 300 K: A = 1.5 cm^2, $N_A = 5 \times 10^{17}$, $N_D = 8 \times 10^{17}$ cm^{-3}, $D_n = 22$, $D_p = 10$ cm^2/s, $\tau_n = 3 \times 10^{-7}$ s, $\tau_p = 5 \times 10^{-7}$ s, generation rate $G_L = 10^{22}$ cm^{-3}s^{-1}. Calculate the photocurrent. The cell is reverse biased with 2 V.
- **2019 Q1(d)** [10] Explain the metal-semiconductor junction with energy band diagrams under different bias conditions.

UNDERSTAND **Read this to understand; you do not need to write it**

Put p-type and n-type side by side. Electrons diffuse into the p side and holes into the n side, leaving fixed charged ions behind. These ions create an electric field (the **depletion region**, width W) and a **built-in voltage V_{bi}**. When light makes an electron-hole pair near the junction, the field sweeps the electron to the n side and the hole to the p side: that separation is the solar cell's current. Pairs made within one diffusion length of the junction (L_n on the p side, L_p on the n side) are collected.

CALCULATOR **p-n junction: carriers, V_{bi}, W and photocurrent (2021 3(b), 2022 3(d), 2020 3(b))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Minority electrons in p side: $n_{p0} = n_i^2/N_A$ = **1.500 × 10^{10}** × **1.500 × 10^{10}** / **5.000 × 10^{17}** = **450.000** 1/cm^3
2. Minority holes in n side: $p_{n0} = n_i^2/N_D$ = **1.500 × 10^{10}** × **1.500 × 10^{10}** / **8.000 × 10^{17}** = **281.250** 1/cm^3
3. Built-in voltage: $V_{bi} = (kT/q)\ln(N_A N_D/n_i^2)$ = 0.0258 × ln(**5.000 × 10^{17}** × **8.000 × 10^{17}** / (**1.500 × 10^{10}** × **1.500 × 10^{10}**)) = **0.906** V
4. Depletion width: $W = \sqrt{[(2\varepsilon/q)(V_{bi} + V_R)(1/N_A + 1/N_D)]}$ = √(2 × 11.7 × 8.854×10^{-14} / 1.602×10^{-19} × (**0.9059** + **2**) × (1 / **5.000 × 10^{17}** + 1 / **8.000 × 10^{17}**)) = **1.105 × 10^{-5}** cm

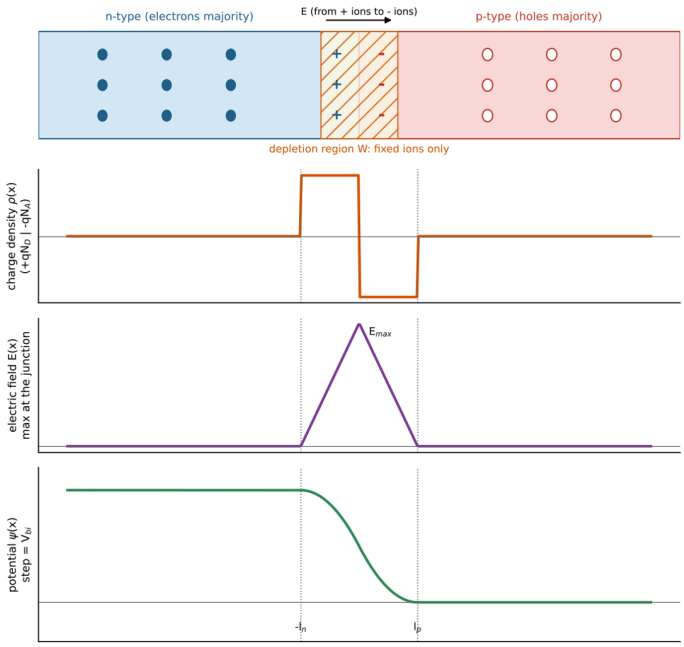
5. L_n: $L = \sqrt{(D\tau)} = \sqrt{(\mathbf{22} \times \mathbf{3.000 \times 10^{-7}})} = \mathbf{0.003}$ cm
6. L_p: $\sqrt{(\mathbf{10} \times \mathbf{5.000 \times 10^{-7}})} = \mathbf{0.002}$ cm
7. Photocurrent: $I_L = qAG_L(L_n + W + L_p) = 1.602\times10^{-19} \times \mathbf{1.5} \times \mathbf{1.000 \times 10^{22}} \times (\mathbf{0.002569} + \mathbf{1.105 \times 10^{-5}} + \mathbf{0.002236}) = \mathbf{11.57}$ A

Silicon $\varepsilon = 11.7\varepsilon_0$, $kT/q = 0.0258$ V at 300 K. For 2021 3(b) set $N_A = 10^{16}$, $N_D = 10^{18}$.

Before you start: the junction is the heart of the cell

Light makes electron-hole pairs anywhere in the semiconductor, but without a built-in electric field they would just recombine. Joining n-type and p-type material creates a **p-n junction** with a permanent field that pushes electrons one way and holes the other: this is the "semi-permeable membrane" in Rahaman sir's Part I figure. Source: Part I sec. 3 and Part II ch. 8 (Smets ch. 8). Constants: $q = 1.602 \times 10^{-19}$ C, $\varepsilon_0 = 8.854 \times 10^{-14}$ F/cm, $\varepsilon_r(\text{Si}) = 11.7$.

Working principle of a p-n junction solar cell (2022 3a, 2020 2a)

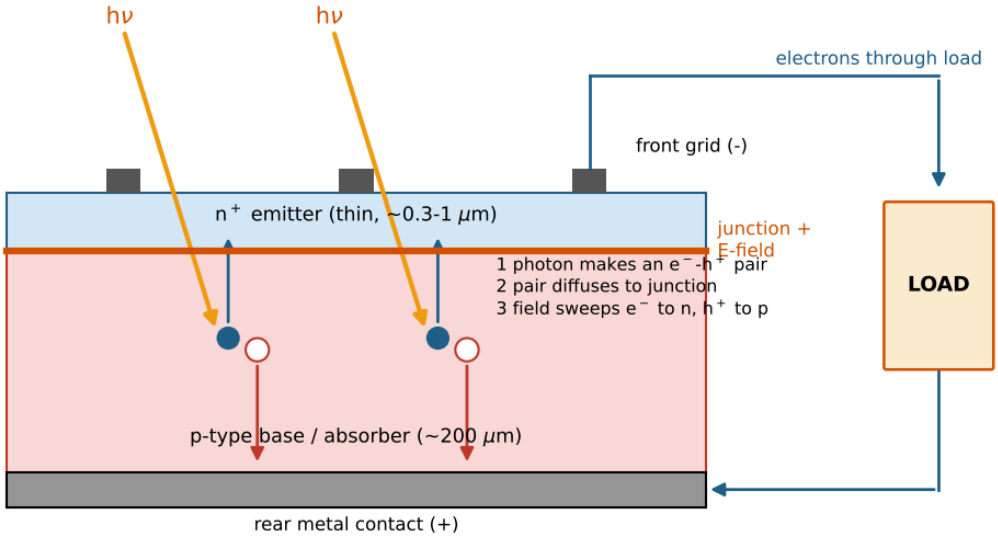


p-n junction: space charge, electric field and potential across the depletion region

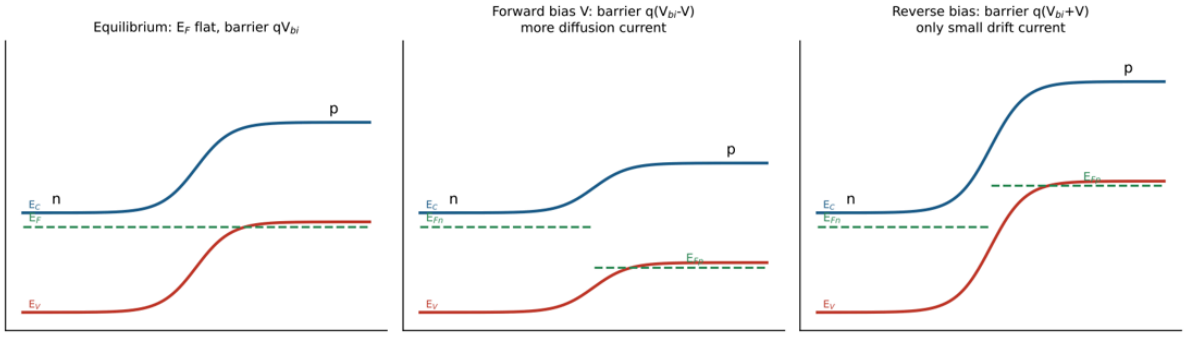
1. **Junction formation:** when n-type and p-type Si are joined, electrons diffuse from n to p and holes from p to n (concentration gradient). They recombine near the junction, leaving uncovered fixed ions: P^+ on the n side and B^- on the p side. This is the **depletion (space-charge) region**, W ≈ 0.3 μm.
2. The ions create a **built-in electric field** (from n to p) and potential V_{bi} ≈ 0.7-0.9 V. The field causes a drift current that exactly balances diffusion in equilibrium; E_F is flat.
3. **Absorption:** sunlight photons with $h\nu \geq E_g$ are absorbed in the emitter, depletion region and base, creating electron-hole pairs.
4. **Diffusion to the junction:** minority carriers created within a diffusion length (L_n in the p base, L_p in the n emitter) diffuse to the edge of the depletion region.
5. **Separation:** the field sweeps electrons into the n side and holes into the p side. Electrons pile up in n, holes in p: the n side becomes negative, p side positive, a **photovoltage** appears (forward-

biasing the junction).

6. **Collection:** the front grid (n side, −) and rear contact (p side, +) collect the carriers; if a load is connected, electrons flow through it and do work, then recombine with holes at the rear contact. Current $I = I_L - I_0(e^{qV/kT} - 1)$.

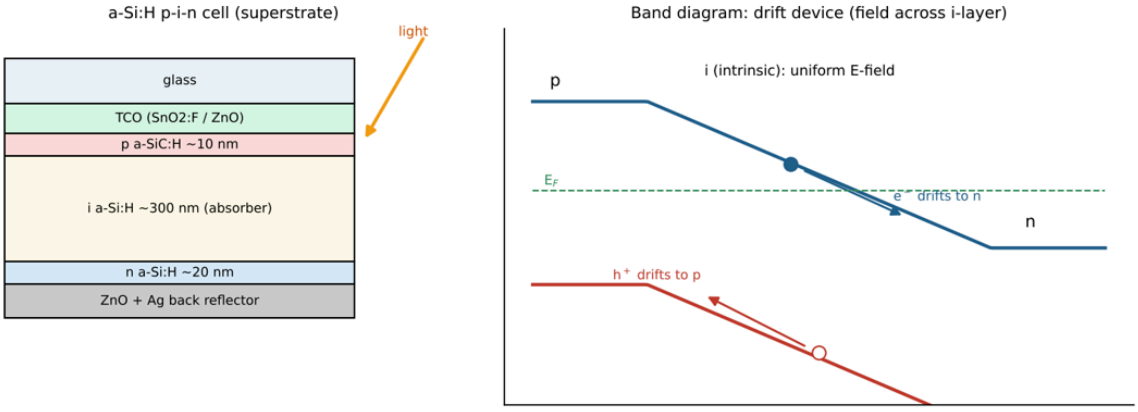


Working of an n^+-p silicon solar cell



Band diagram of the p-n junction in equilibrium, forward and reverse bias

Working principle of a p-i-n junction solar cell (2023 2a)

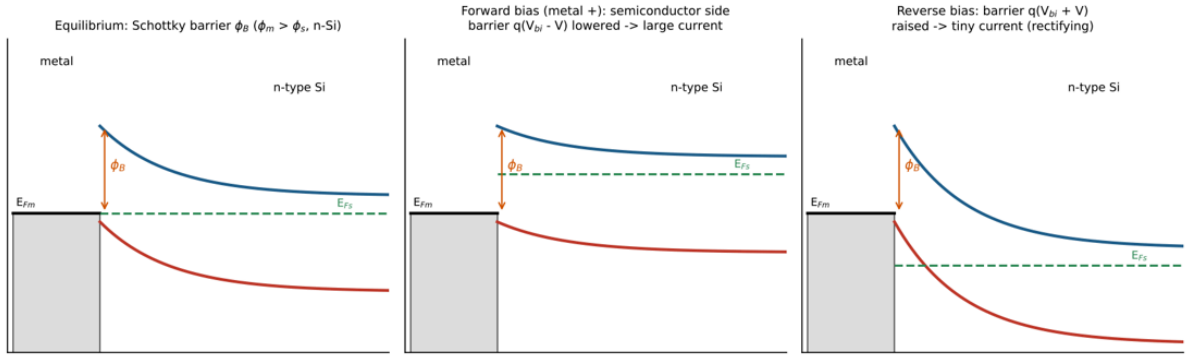


a-Si:H (hydrogenated amorphous silicon) p-i-n cell: layer stack and band diagram

- A thick **intrinsic (undoped) absorber layer i** (~300 nm for a-Si:H) is placed between very thin p (~10 nm) and n (~20 nm) layers. Light enters through glass/TCO and the p layer (p is on the light side because holes are the slower carriers and must travel less).

- The p and n layers set up the built-in voltage, but because the i layer has almost no charge, the **electric field extends uniformly across the whole i layer** (like a capacitor).
- Photons absorbed in the i layer create pairs; the field immediately **drifts** electrons to n and holes to p. Collection does not depend on diffusion, which is why a p-i-n cell is a **drift device**; this is essential for materials with a very short diffusion length (a-Si:H, L < 1 µm) and is also used in perovskite cells.
- Thickness trade-off: a thicker i layer absorbs more light but weakens the field (more recombination, more light-induced degradation).

Metal-semiconductor junction with band diagrams under bias (2019 1d)



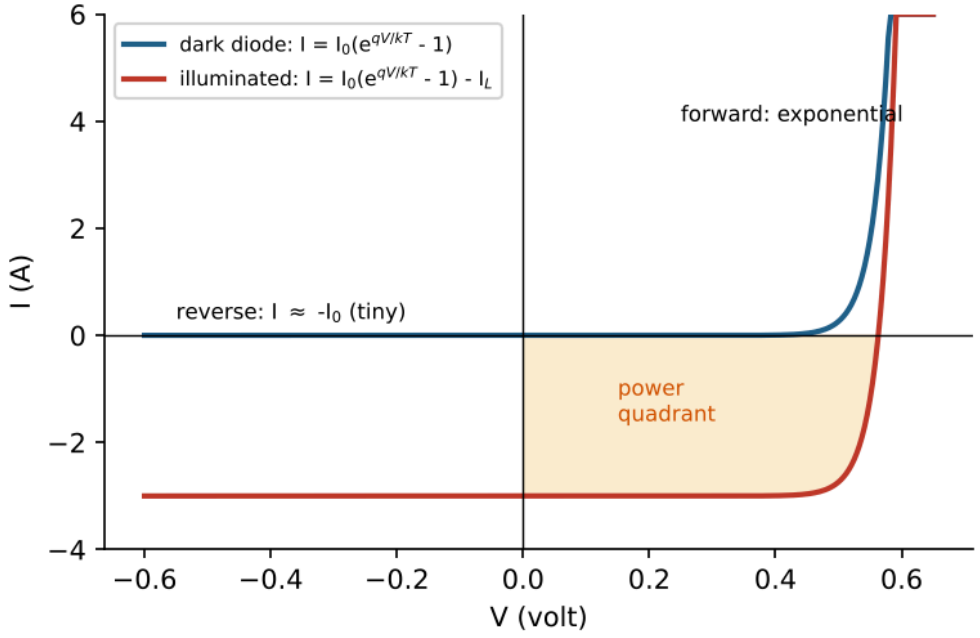
Metal / n-type semiconductor Schottky contact in equilibrium, forward and reverse bias

- Work function $\varphi = E_{vac} - E_F$; electron affinity $\chi = E_{vac} - E_C$. When metal and semiconductor touch, electrons flow until the Fermi levels line up; the vacuum level stays continuous.
- **Schottky (rectifying) contact:** metal/n-type with $\varphi_m > \varphi_s$ (or metal/p-type with $\varphi_m < \varphi_s$). Electrons leave the semiconductor, the bands bend up, a barrier $\varphi_B = \varphi_m - \chi$ forms. **Forward bias** (metal +): the semiconductor-side barrier falls to $q(V_{bi} - V)$ and electrons flow easily into the metal, current rises exponentially. **Reverse bias**: the barrier rises to $q(V_{bi} + V)$; only a tiny current over φ_B flows. It behaves like a diode.
- **Ohmic contact:** metal/n-type with $\varphi_m < \varphi_s$ (or metal/p with $\varphi_m > \varphi_s$), or a very heavily doped semiconductor (thin barrier, tunnelling). Current flows equally both ways with low resistance: needed for solar cell contacts (hence n^{++} under the front fingers, p^+ Al-BSF at the back).

Diode equation and ideal I-V curve (2020 2c)

$$I = I_0[\exp(qV/nkT) - 1]; \quad \text{illuminated cell: } I = I_0[\exp(qV/nkT) - 1] - I_L$$

I_0 = saturation (dark) current, n = ideality factor (1 ideal), kT/q = 25.8 mV. Forward bias: current rises exponentially; reverse bias: $I \approx -I_0$ (tiny) until breakdown. Light shifts the curve down by I_L into the 4th (power) quadrant.



Ideal diode I-V curve, dark and illuminated

Numerical: built-in voltage, depletion width and maximum field (2022 3d)

Given: $N_A = 10^{16}$ cm^{-3} (p wafer), $N_D = 10^{18}$ cm^{-3} (1 µm emitter), $n_i = 1.5 \times 10^{10}$ cm^{-3}, T = 300 K. Teacher's Part II example (Smets Ex. 8.1).

Step (write this)	Why / what it means
$V_{bi} = (kT/q)\ \ln(N_A N_D/n_i^2) = 0.0258\ \ln(10^{34}/2.25 \times 10^{20}) = 0.0258 \times 31.43 =$ **0.81 V**	the potential step that balances diffusion; grows with doping
$W = \sqrt{[(2\varepsilon_r\varepsilon_0/q)V_{bi}(1/N_A + 1/N_D)]} = \sqrt{[(2 \times 11.7 \times 8.854 \times 10^{-14}/1.602 \times 10^{-19}) \times 0.81 \times 1.01 \times 10^{-16}]}$	use cm units throughout: ε in F/cm, N in cm^{-3}
$W = 3.25 \times 10^{-5}$ cm = **0.325 µm** ($l_p = 0.322$ µm, $l_n = 0.003$ µm)	the region spreads mostly into the lightly doped p side; only 0.1% of a 300 µm wafer
$E_{max} = \sqrt{[(2q/\varepsilon_r\varepsilon_0)V_{bi}N_A N_D/(N_A + N_D)]} =$ **5.0×10^4 V/cm**	peak field at the metallurgical junction (x = 0); check: $E_{max} \approx 2V_{bi}/W = 4.98 \times 10^4$ V/cm

Numerical: carrier concentrations in the quasi-neutral regions (2021 3b)

Step (write this)	Why / what it means
p-region: $p_{p0} \approx N_A =$ **10^{16} cm^{-3}**; $n_{p0} = n_i^2/N_A = (1.5 \times 10^{10})^2/10^{16} =$ **2.25×10^4 cm^{-3}**	all acceptors ionised; mass-action law $np = n_i^2$ gives the minority density
n-region: $n_{n0} \approx N_D =$ **10^{18} cm^{-3}**; $p_{n0} = n_i^2/N_D =$ **2.25×10^2 cm^{-3}**	14-16 orders of magnitude between majority and minority
Fermi levels ($E_C = 0$): n side $E_F = -kT\ \ln(N_C/n) = -0.0258\ \ln(3.22 \times 10^{19}/10^{18}) = -0.09$ eV; p side $-0.0258\ \ln(3.22 \times 10^{19}/2.25 \times 10^4) = -0.90$ eV	extra marks: E_F sits 0.09 eV below E_C in the emitter and 0.22 eV above E_V in the base

Numerical: photocurrent of a reverse-biased cell (2020 3b)

Given: A = 1.5 cm^2, $N_A = 5 \times 10^{17}$, $N_D = 8 \times 10^{17}$ cm^{-3}, $D_n = 22$, $D_p = 10$ cm^2/s, $\tau_n = 3 \times 10^{-7}$ s, $\tau_p = 5 \times 10^{-7}$ s, $G_L = 10^{22}$ $cm^{-3}s^{-1}$, $V_R = 2$ V. Method: $J_{ph} = qG(L_n + W + L_p)$ (Part II eq. for J_{ph}), the long-diode result (Neamen).

Step (write this)	Why / what it means
$L_n = \sqrt{(D_n\tau_n)} = \sqrt{(22 \times 3 \times 10^{-7})} = 2.57 \times 10^{-3}$ cm = 25.7 μm	electrons generated within L_n of the junction in the p side are collected
$L_p = \sqrt{(10 \times 5 \times 10^{-7})} = 2.24 \times 10^{-3}$ cm = 22.4 μm	same for holes in the n side
$V_{bi} = 0.0258 \ln(5 \times 10^{17} \times 8 \times 10^{17}/2.25 \times 10^{20})$ = 0.906 V	needed for W
$W = \sqrt{[(2\varepsilon/q)(V_{bi} + V_R)(1/N_A + 1/N_D)]} = \sqrt{[1.293 \times 10^7 \times 2.906 \times 3.25 \times 10^{-18}]} = 1.11 \times 10^{-5}$ cm = 0.11 μm	reverse bias adds to V_{bi} and widens W, but W is still tiny compared with L
$I_L = qAG_L(L_n + W + L_p) = 1.602 \times 10^{-19} \times 1.5 \times 10^{22} \times 4.82 \times 10^{-3}$ = **11.6 A** (J_L = 7.7 A/cm^2)	under reverse bias the diode term is only $-I_0$ (negligible), so the current is just the photocurrent

Side note: The given generation rate 10^{22} $cm^{-3}s^{-1}$ is about 100 times one sun, so the current is very large; that is what the data give. With $G_L = 10^{21}$ (the usual textbook value) I_L = 1.16 A.

Remember: Junction: diffusion → ions → field → separation. $V_{bi} = kT/q \ln(N_A N_D/n_i^2) = 0.81$ V; W = 0.325 μm; $E_{max} = 5 \times 10^4$ V/cm. p-i-n = drift device. Schottky if bands bend into a barrier; ohmic if not.

Section B: PV systems

B5 PV system types: standalone (SHS), grid-connected, with storage, hybrid; net metering; grid interfacing; standalone applications; water pumping; SHS installation and testing

Pass set · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q7(a)** [14] Discuss the working principle of on-grid and off-grid solar PV systems with proper diagrams.
- **2023 Q7(b)** [14] What is net metering? How does it work? Explain with proper illustration.
- **2023 Q8(a)** [8] Provide examples of real-life applications of standalone PV systems and explain how they meet specific energy needs.
- **2022 Q7(a)** [12] Discuss the working principle of an on-grid solar PV system with storage.
- **2022 Q5(a)** [10] What are the key components of a solar home system and explain the role of each component in converting sunlight into usable electric energy.
- **2022 Q7(b)** [10] What is net metering? How does it work?
- **2021 Q8(a)** [8] How does a grid-connected PV system work? Write the conditions for grid interfacing of an on-grid PV system.
- **2021 Q7(b)** [8] What is net metering? Write down the eligibility conditions for a customer to be considered for a net metering connection in Bangladesh.
- **2020 Q5(c)** [6] Write down the components of an on-grid PV system.
- **2019 Q7(a)** [10] Discuss the working principle of on-grid and off-grid solar systems with necessary diagrams.
- **2019 Q7(c)** [6] What are the conditions for grid interfacing of an on-grid PV system?
- **2019 Q7(b)** [11] What is net metering? How does it work?
- **2019 Q7(d)** [8] Write down the eligibility conditions for a customer to be considered for a net metering connection in Bangladesh.
- **2019 Q5(d)** [9] Explain the working principle of a PV-powered water pumping system.
- **2019 Q6(b)** [6] Write the merits and demerits of a PV pumping system when it is directly connected to the PV array or to a battery.
- **2019 Q8(d)** [10] Discuss the testing, installation and replacing procedures of a solar home system with different components.

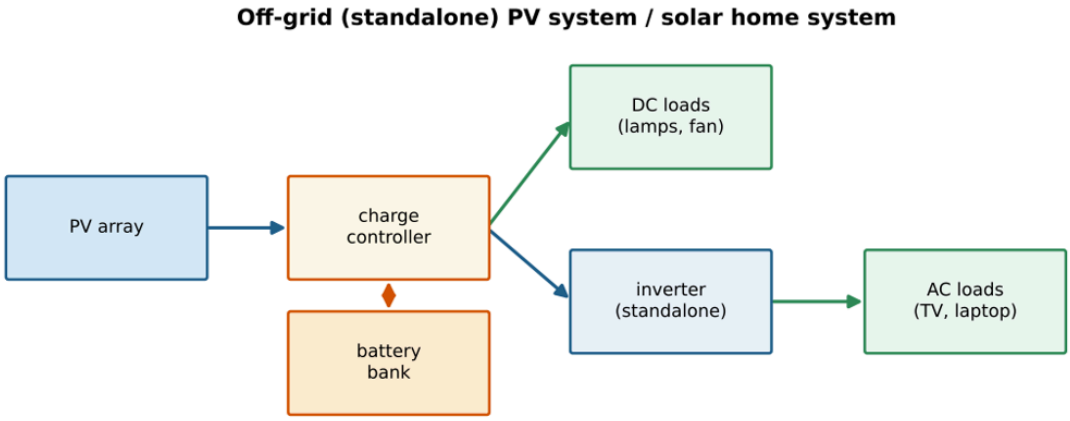
★ REMEMBER IT

- **Off-grid (standalone)**: PV → charge controller → battery → inverter → loads. **On-grid**: PV → grid-tie inverter (MPPT + anti-islanding) → bidirectional meter → grid. **Hybrid / with storage**: both, battery backs up during outages.
- Grid interfacing conditions: same **voltage, frequency, phase sequence and phase angle** as the grid, low THD (total harmonic distortion), power factor near 1, anti-islanding (disconnect when the grid fails, to protect line workers).
- **Net metering**: one meter runs both ways; you pay only for (imported − exported) energy over the billing period. Bangladesh (2018 guideline): consumers with grid connection, system up to 70% of sanctioned load, on own premises, with an approved inverter.

Before you start: how the parts fit together

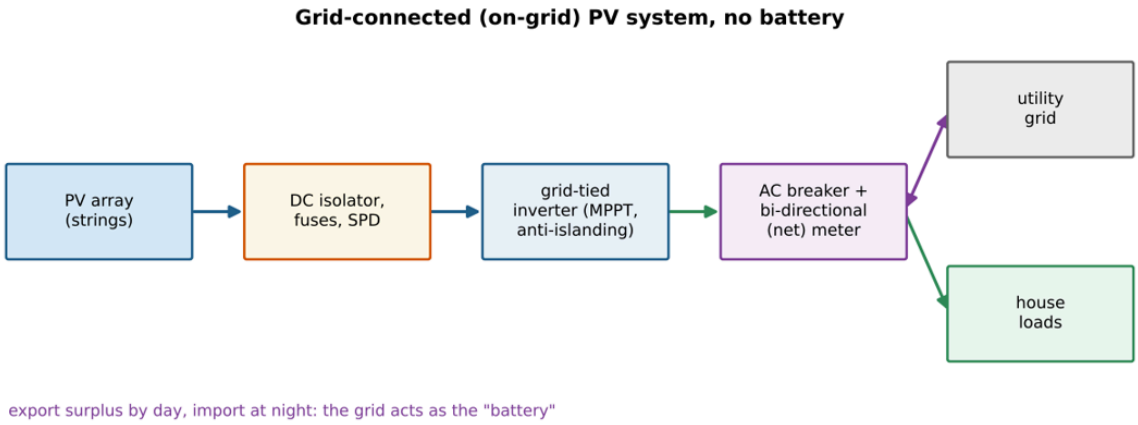
A PV system can stand alone (off-grid), feed the utility grid (on-grid), do both with a battery, or work with another generator (hybrid). Source: Muhit sir L9 (system types), L10 (net metering, feed-in tariffs); Messenger ch. 3-4 and 7-8 for grid interfacing and pumping; Bangladesh Net Metering Guidelines 2018 (SREDA).

Off-grid (standalone) and on-grid systems: working with diagrams (2023 7a, 2019 7a)



Off-grid (standalone) solar PV system with battery

Off-grid system: works without the utility grid, for remote houses, islands, telecom, street lights. Components: PV array, charge controller, battery bank, inverter (for AC loads), DC and AC loads, fuses and switches. **Working:** in the day the array produces DC; the charge controller feeds DC loads and charges the battery, protecting it against overcharge; the inverter converts DC to AC for AC loads; at night or in cloudy weather the battery supplies the loads; the controller disconnects the load at LVD (low-voltage disconnect). Simplest variants (L9): (i) PV + DC load only (day-time loads, e.g. a fan or pump), (ii) PV + control circuitry + DC load, (iii) PV + controller + battery + DC load, (iv) PV + controller + battery + inverter + AC/DC load. The system must be sized for the worst month and days of autonomy.



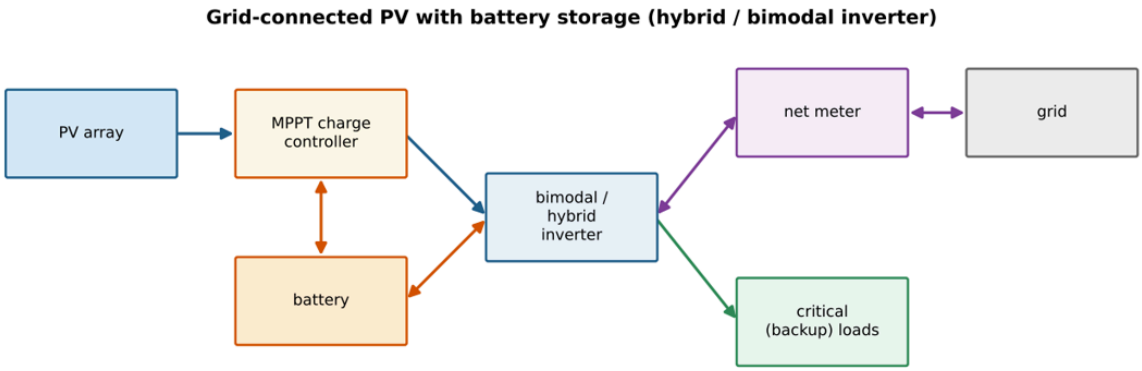
On-grid (grid-tied) solar PV system without storage

On-grid system: the PV array feeds a **grid-tied inverter**, which tracks the MPP (maximum power point), converts DC to grid-synchronised AC and feeds the house distribution board; a bi-directional (net) meter connects the house to the utility. **Working:** when PV production exceeds the house demand, the surplus is exported to the grid; when PV is less (evening, night) the deficit is imported

from the grid. The grid acts as an infinite “battery”, so no storage is required; cheaper, ~95% inverter efficiency, but it shuts down during a blackout (anti-islanding). Large solar parks use the same principle with step-up transformers.

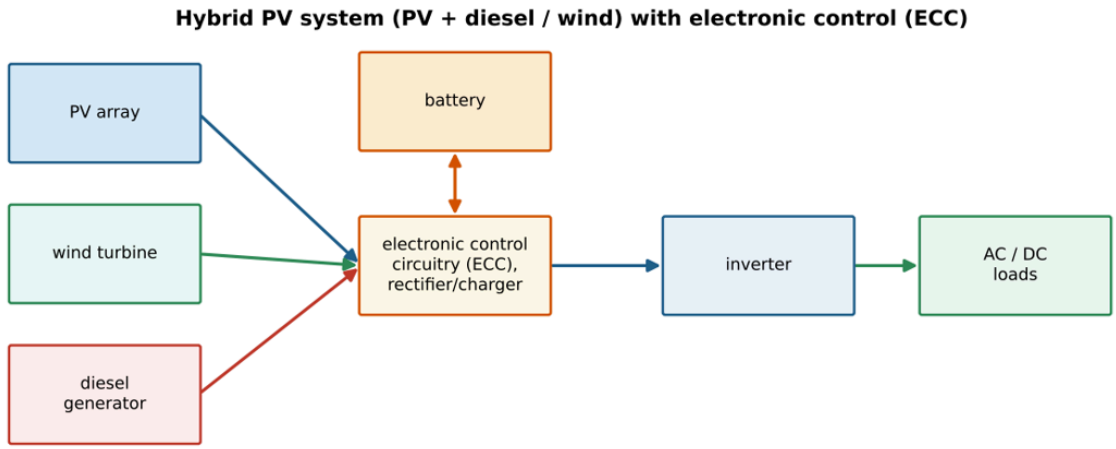
Components of an on-grid PV system (2020 5c): PV array (modules, mounting structure); DC combiner box with string fuses, surge protection, DC isolator; grid-tied inverter (MPPT, synchronisation, anti-islanding); AC distribution board with AC breaker/isolator; bi-directional (net) energy meter; earthing and lightning protection; cables; monitoring system; (for large plants) step-up transformer and switchgear.

On-grid PV system with storage (2022 7a)



Grid-connected PV with battery storage (hybrid / bimodal inverter)

A battery bank is added via a **bimodal (hybrid) inverter** or a battery inverter on the AC side. **Working:** (1) in the day PV supplies the loads first; (2) surplus PV charges the battery; (3) when the battery is full, further surplus is exported to the grid; (4) in the evening and at night the battery supplies the loads (increasing self-consumption), and the grid covers what the battery cannot; (5) during a grid outage the inverter disconnects from the grid (anti-islanding) and runs in island mode, supplying **critical loads** from the battery and PV (backup power). The control can also shift energy to peak-tariff hours, shave peaks and reduce grid instability from high PV penetration. Costlier (battery, maintenance) but more reliable.



Hybrid PV-diesel system with electronic control circuitry (L9)

How a grid-connected system works and the conditions for grid interfacing (2021 8a, 2019 7c)

Working as above: PV DC → grid-tied inverter (MPPT + DC/AC + synchronisation) → house loads and the grid via a meter. The inverter is a current source synchronised to the grid voltage; it can only push power when the grid is present.

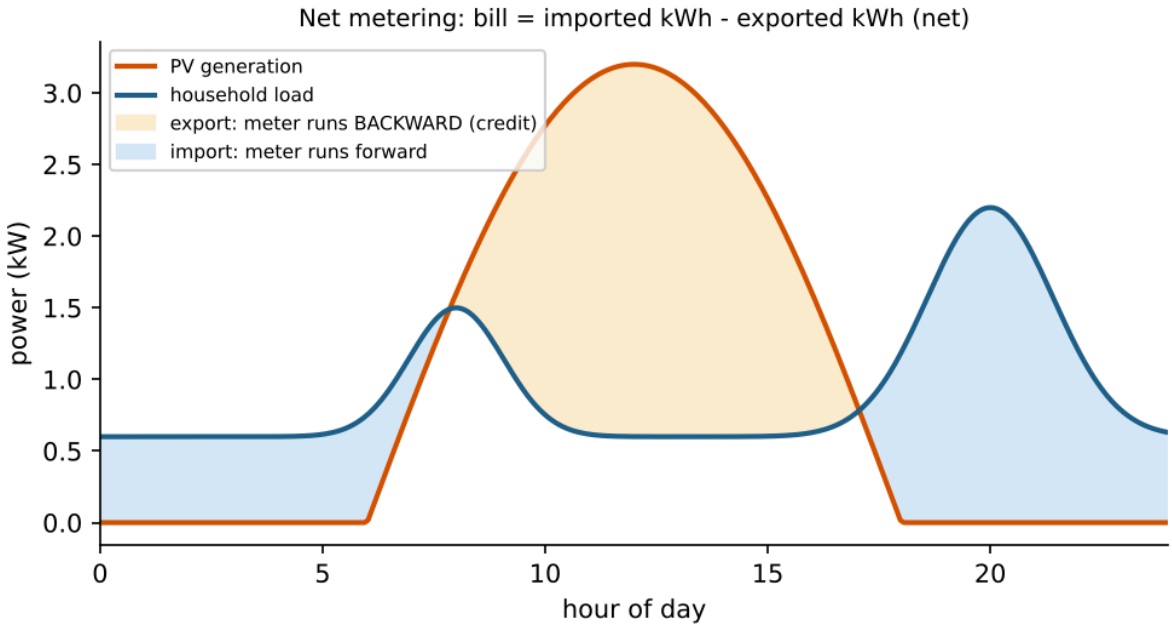
Conditions (requirements) for interfacing with the grid (IEEE 929/1547, UL 1741; Messenger ch. 8):

1. **Voltage:** output voltage must match the grid voltage (230 V ± 10%); the inverter must disconnect if the grid voltage goes outside the limits (e.g. < 88% or > 110%) within the specified clearing time.
2. **Frequency:** same frequency (50 Hz in Bangladesh); disconnect if outside the band (about 49.5-50.5 Hz).
3. **Synchronisation:** same phase sequence and phase angle at the moment of connection, so no inrush current.
4. **Power quality:** current THD < 5% (each odd harmonic limited), power factor > 0.85-0.9, DC current injection < 0.5-1% of rated current, flicker limits.
5. **Anti-islanding:** the inverter must detect loss of the utility grid and stop exporting within 2 s, so line workers are safe and equipment is not damaged.
6. **Reconnection:** after a trip, reconnect only when grid voltage and frequency have been normal for about 5 minutes.
7. **Protection and safety:** lockable visible AC disconnect switch accessible to the utility, earthing, over-current and surge protection, fault ride-through as required.
8. **Metering and approval:** bi-directional/net meter, certified (tested) inverter, interconnection agreement with the utility.

Key components of a solar home system and their roles (2022 5a)

Component	Role in converting sunlight into usable energy
PV module (20-130 Wp) on a mounting frame	converts sunlight into DC electricity (photovoltaic effect); faces south, tilted ~23°
Charge controller	regulates the PV current into the battery, prevents overcharge and deep discharge, blocks reverse current, controls loads
Battery (12 V deep-cycle lead-acid or Li-ion)	stores the daytime energy for use at night and on cloudy days
Inverter (if AC loads)	converts 12 V DC into 230 V, 50 Hz AC
Loads	efficient DC LED lamps, DC fans, TV, mobile charger
Wiring, switches, fuses, earthing	carry current with low loss and protect people and equipment

Net metering: what it is and how it works (2023 7b, 2022 7b, 2019 7b, 2021 7b)



Net metering: the meter runs forward when importing and backward when exporting

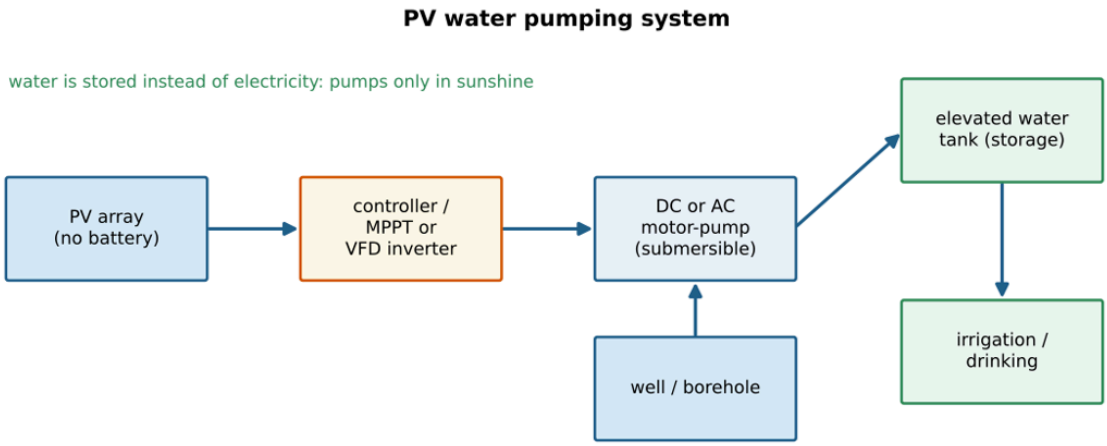
- **Net metering** is a billing arrangement in which a consumer with a grid-connected rooftop PV system (a **prosumer**) is credited for the electricity exported to the grid, and pays only for the **net energy** = energy imported from the grid − energy exported to the grid, over the billing period.
- **How it works:** a **bi-directional meter** (or an old analogue meter that runs backwards) records import and export. In the day, when PV production exceeds consumption, the surplus flows to the grid and the meter runs backwards (export register counts up); at night the house draws from the grid and the meter runs forward. At the end of the month the utility bills the net import; if export exceeds import, the surplus is carried forward as credit (in Bangladesh to the next bill, settled annually at a set rate).
- Attractive when the LCOE (levelized cost of electricity) of the PV system is lower than the grid tariff. Related schemes: **feed-in tariff** (gross: all PV energy sold at a fixed price, all consumption bought; net: only the surplus sold) and **self-consumption**.

Eligibility for a net metering connection in Bangladesh (Net Metering Guidelines 2018, SREDA (Sustainable and Renewable Energy Development Authority (Bangladesh))/Power Division): (1) the customer must be a consumer of the distribution utility (BPDB, DPDC, DESCO, REB/PBS, WZPDCL, NESCO) with a **three-phase** (LT or above) connection; (2) the renewable energy system must be installed on the customer's own premises (rooftop/land) and owned or leased by the consumer; (3) system capacity may not exceed **70% of the sanctioned load** of the connection (and is limited by the distribution transformer capacity); (4) no outstanding bills; (5) equipment must meet the technical standards: grid-tied inverter with anti-islanding, synchronisation and power quality requirements, protection devices and a **bi-directional energy meter** approved by the utility; (6) application, inspection and a net metering agreement signed with the utility. Applies to residential, commercial and industrial consumers; the system is mainly solar PV.

Real-life standalone applications (2023 8a)

Application	How it meets the energy need
Solar home systems (Bangladesh: > 6 million)	20-130 Wp + battery give lights, fan, TV, phone charging to off-grid rural homes; replace kerosene
Solar irrigation pumps	array drives a DC/AC pump in the day, water is the storage; no diesel
Solar street lights	module charges a battery in the day; dusk-to-dawn LED with controller
Telecom towers, remote monitoring	reliable unattended power with batteries and days of autonomy
Health centres (vaccine refrigerators)	solar direct-drive fridges keep vaccines cold without the grid
Highway signs, navigation buoys, railway signals, calculators, satellites	small, maintenance-free power where cables are impossible
Solar mini-grids for villages/islands; water purification; e-rickshaw charging	community-scale supply

PV-powered water pumping (2019 5d, 2019 6b)



Directly coupled PV water pumping system

Working principle: the PV array supplies DC power to a pump motor (DC brushless, or AC through a VFD (variable frequency drive)/pump inverter) driving a submersible or surface pump. The pump lifts water from a well/river into a storage tank or field. Flow depends on irradiance: more sun, more water, which matches irrigation need. A **pump controller/linear current booster** (MPPT) matches the motor to the array, giving extra current at low light to start the motor and protecting against dry running. Water stored in an elevated tank is the energy storage (cheaper than batteries); hydraulic energy $E = \rho g H V$.

	Directly connected to the array	Connected through a battery
Merits	simple, cheap, high reliability, no battery cost or maintenance, water tank is the storage, efficient (no battery losses), long life	pumping at night or on demand, constant head and flow, motor runs at its design voltage, can start at low light, other loads can share the battery
Demerits	pumps only during sunshine, flow varies with irradiance, motor may stall at low light (needs a controller/MPPT), operating point not always at MPP, larger tank needed	battery adds cost, losses (~20%), maintenance and replacement every 3-5 years, charge controller needed, lower overall efficiency, more complex

Testing, installation and replacement of a solar home system (2019 8d)

- **Installation:** site survey (shade-free, south facing, sturdy roof); fix the mounting structure and module at ~23° tilt, facing south; place the battery indoors in a ventilated, protected place near the controller; mount the charge controller near the battery; wiring **in the order battery → controller first, then the module, then the loads** (so the controller detects the system voltage), with correct polarity, sized cables, fuses and switches; install loads and earthing.
- **Testing and commissioning:** measure module V_{oc} and I_{sc} in sunlight; check battery voltage and specific gravity; check controller indicators (charging LED), LVD and HVD (high-voltage disconnect) operation; check every load and switch; check polarity, connections and earthing; train the user.
- **Maintenance and replacement:** clean the module regularly, check for shading; top up distilled water in flooded batteries, clean terminals, check specific gravity; replace the battery when its capacity falls below ~70-80% (typically 3-5 years) with the same type and voltage, disconnecting the module first, then the loads, then the battery; replace a faulty controller with one of equal or higher rating; replace damaged modules with modules of the same voltage and similar current; replace lamps with efficient LEDs; record and dispose of old batteries through recycling.

L9 problem: array 100 V × 50 A = 5000 W DC; 20% losses → AC power to the grid = 0.8 × 5000 = **4000 W**; AC current = 4000/240 = **16.7 A**.

Remember: Off-grid: PV → controller → battery → inverter → loads. On-grid: PV → grid-tied inverter → loads + net meter + grid, no battery. With storage: hybrid inverter, backup. Grid conditions: V, f, phase, THD < 5%, DC injection, anti-islanding 2 s, 5 min reconnect, disconnect switch. Net metering: pay import − export; Bangladesh: 3-phase, ≤ 70% sanctioned load, own premises, no dues, bi-directional meter.

B1 PV cell, module, panel and array; series and parallel; effects of irradiance, temperature, area, orientation and tilt

Pass set · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q5(a)** [12] What are the effects of radiation and PV module temperature on PV cell performance? Explain with necessary figures.
- **2023 Q5(b)** [6] Write short notes on (i) PV module, (ii) PV array.
- **2022 Q8(b)** [10] In what ways does temperature affect solar cell efficiency? What strategies and technologies can be employed to reduce temperature-related challenges in PV systems?
- **2022 Q5(c)** [7] The maximum rated power of a crystalline solar cell is 75 Wp. Calculate the maximum power output at 45, 55 and 65 °C cell temperature. Decrease in maximum power per °C rise (from STC (standard test conditions (1000 W/m^2, AM1.5, 25 °C)) 25 °C) is −0.45 %/°C.
- **2022 Q7(c)** [13] Discuss the effect of PV module temperature, orientation and tilt angle on the performance of a solar PV system.
- **2022 Q5(b)** [10] How does the surface area of a solar cell impact its output current, and what key factors should be considered in optimizing the design of a solar cell for efficient energy conversion?
- **2021 Q3(a)** [8] "Performance of a solar cell depends on irradiance." Justify the statement. (asked in Section A)
- **2021 Q6(d)** [7] How does heat affect solar panel efficiency? Describe.
- **2021 Q5(a)** [6] Write short notes on (i) PV cell, (ii) PV module, (iii) PV array.
- **2021 Q5(c)** [8] Explain the working principle of photovoltaic panels in series and parallel connection. Why are series and parallel connections necessary in PV systems?
- **2020 Q5(a)** [8] "Performance of a solar cell depends on irradiance." Justify the statement.
- **2019 Q8(a)** [10] Same question as 2022 7(c).
- **2019 Q5(a)** [4] Write short notes on (i) PV module and (ii) PV array.
- **2019 Q5(b)** [12] Same question as 2021 5(c).

★ REMEMBER IT

- Irradiance up → I_{sc} up in proportion, V_{oc} up only slightly (log). Temperature up → V_{oc} down (about −2.3 mV/°C per Si cell), I_{sc} up very slightly, power down about 0.4–0.5%/°C.
- **Series adds voltage, parallel adds current.** Cell (0.5–0.6 V) → module (36 or 60/72 cells in series) → panel → string → array.

CALCULATOR **Power at a hotter cell temperature (2022 5(c))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Power at T: $P = P_{STC}[1 + \gamma(T - 25)]$ = **75** × (1 + **(-0.45)** / 100 × (**45** - 25)) = **68.25** W
2. Loss: 100 × (1 - **68.25** / **75**) = **9.0** %

Try 55 and 65 °C: 64.9 W and 61.5 W.

CALCULATOR **Array from modules: series and parallel**

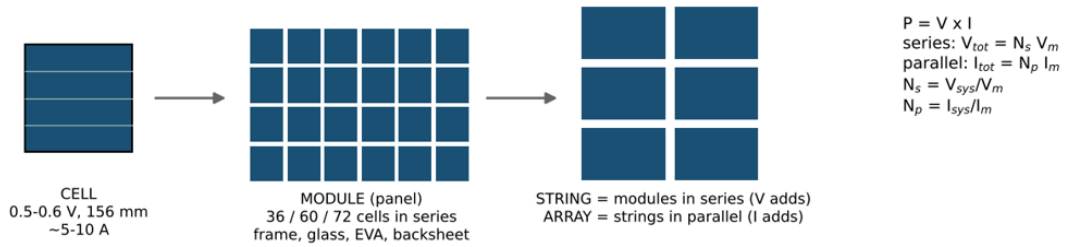
Change any number; every step is recalculated and shown with the numbers put in.

1. Array voltage: $V = N_s V_m$ = **18** × **2** = **36.0** V
2. Array current: $I = N_p I_m$ = **5.56** × **3** = **16.68** A
3. Array power: **36** × **16.68** = **600** W

Before you start: from one cell to a power plant

One silicon cell gives only about 0.5-0.6 V and a few amperes (about 4-6 W for a 156 mm cell). To charge a 12 V battery or feed a house we connect cells in **series** (to add voltage) and in **parallel** (to add current). How much power comes out also depends on the light, the temperature, the area and the angle of the panel. Source: Muhit sir L2 (factors), L3 (cell, module, array), L4 (temperature); Messenger ch. 3-4.

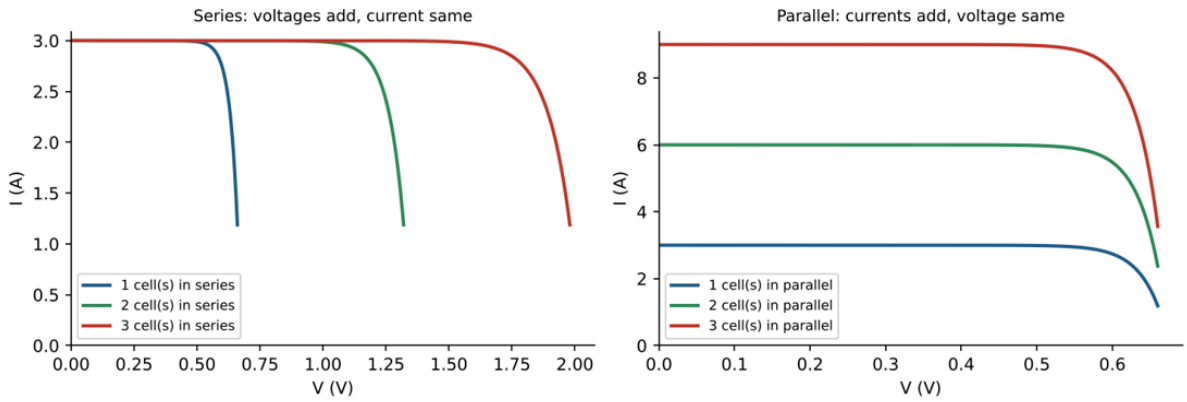
Short notes: PV cell, PV module, PV array (2023 5b, 2021 5a, 2019 5a)



Cell, module (panel) and array

- **PV cell:** the basic semiconductor p-n junction device that converts sunlight directly into DC electricity. A c-Si cell (156-210 mm square, ~180 µm thick) gives V_{oc} ≈ 0.6-0.7 V, V_m ≈ 0.5 V and J_{sc} ≈ 35-40 mA/cm^2, i.e. 4-8 W. It is fragile and its voltage is too low for any practical load.
- **PV module:** a number of cells (typically 36, 60, 72, or 120/144 half-cells) connected **in series**, sealed between tempered glass and a back sheet with EVA encapsulant, in an aluminium frame with a junction box and bypass diodes. It is the smallest unit sold and rated in **watt-peak (Wp)** at STC: e.g. a 36-cell module, V_m ≈ 17-18 V, used to charge a 12 V battery; the encapsulation protects the cells from moisture, dust and impact for 25 years. A **panel** is one or more modules mounted together.
- **PV array:** a group of modules connected in series (a **string**, to reach the system voltage) and strings in parallel (to reach the required current and power), mounted on a common structure with wiring, fuses and disconnects: the complete DC generator of a PV system, from a few hundred watts (SHS) to megawatts (power plants).

Series and parallel connection: working principle and why both are needed (2021 5c, 2019 5b)

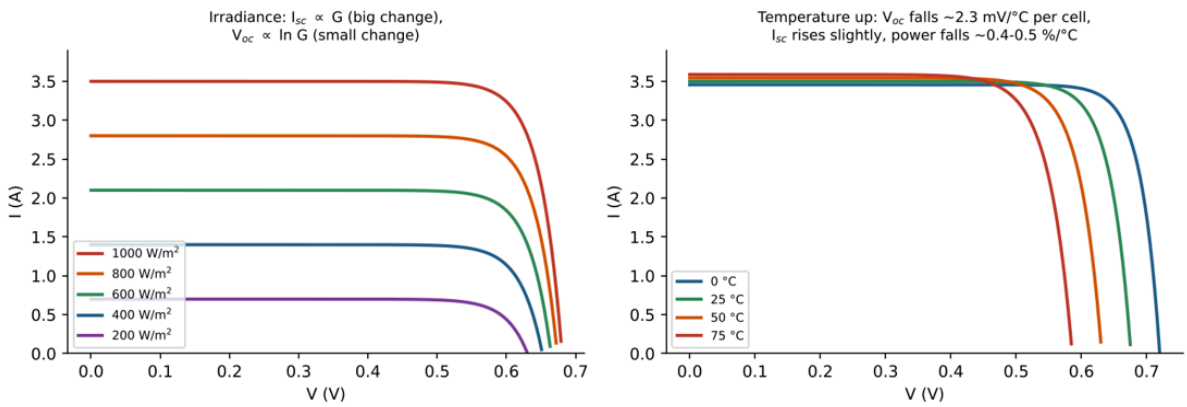


I-V curves of identical cells or modules in series and in parallel

- **Series connection:** the positive terminal of one cell/module is joined to the negative terminal of the next. The **same current** flows through all, and the **voltages add**: $V_{oc,total} = N_s V_{oc}$, $V_{m,total} = N_s V_m$, I_{sc} unchanged. The I-V curve stretches along the voltage axis. Because current is common, the weakest (shaded) cell limits the whole string.
- **Parallel connection:** all positive terminals are joined together and all negative terminals together. The **voltage is the same**, the **currents add**: $I_{sc,total} = N_p I_{sc}$, V_{oc} unchanged. The I-V curve stretches along the current axis. A weak module only reduces its own share.
- **Series-parallel (array):** $P = (N_s V_m)(N_p I_m)$.
- **Why necessary:** a single cell gives ~0.5 V and a few amperes, but loads and batteries need 12/24/48 V DC or inverter inputs of 200-1000 V, and kilowatts of power. Series connection raises voltage to charge the battery or match the inverter window (and higher voltage means lower current, so smaller wire and lower I^2R loss); parallel connection raises current and power; together they give any voltage and power rating with standard modules, and parallel strings add redundancy.

Teacher's problems (L3): (1) a new cell has V_{oc} = 0.90 V, V_m = 0.80 V at STC, −1 mV/°C; for a 12 V battery at 60 °C: $V_m(60\,°C) = 0.80 - 0.001 \times 35 = 0.765$ V; a 12 V battery needs about 14.5 V for charging, so $N_s = 14.5/0.765 = 18.95$ → **19 cells in series** (20 if 15 V is taken). (2) Modules of V_m = 16 V, I_m = 7 A for 16 V and 42 A: $N_s = 16/16 = 1$, $N_p = 42/7 =$ **6 modules in parallel**.

Performance depends on irradiance; effect of radiation and temperature (2023 5a, 2021 3a, 2020 5a)



I-V curves for different irradiance (left) and cell temperature (right)

- **Irradiance:** the photocurrent is proportional to the number of photons: **$I_{sc} \propto G$** ($I_{sc}(G) = I_{sc,STC} \times G/1000$). Halving the irradiance halves the current. $V_{oc} = (kT/q)\ln(I_L/I_0)$ rises only **logarithmically** (about +60 mV per tenfold increase at the cell). So the power, $P = V_m I_m$, is nearly proportional to irradiance; e.g. 1000 W/m^2 gives 100% power, 500 W/m^2 about 48%. Output therefore rises from morning to noon and falls in the afternoon; clouds, dust, air mass and angle all lower G. At very low light the shunt resistance makes efficiency fall. This justifies the statement: the cell output follows the irradiance.
- **Temperature:** as T rises, the bandgap shrinks and I_0 rises very fast, so **V_{oc} falls** (−2.0 to −2.3 mV/°C per Si cell, about −0.3%/°C); I_{sc} rises slightly (+0.05%/°C) because more long-wavelength photons are absorbed. Net effect: **power falls about 0.4-0.5%/°C** above 25 °C for c-Si (−0.2%/°C for a-Si/CdTe). Cells in the sun run 20-35 °C above ambient: $T_{cell} = T_{amb} + (NOCT - 20)G/800$, with NOCT (nominal operating cell temperature) ≈ 45 °C.

Teacher's problems (L2): $P = \eta GA$ with A = 100 cm^2 = 0.01 m^2: η = 30, 24, 19, 16, 12% gives 3.0, 2.4, 1.9, 1.6, 1.2 W. At 40 °C with 0.7 V and −2.3 mV/°C: V = 0.7 − 0.0023 × 15 = 0.6655 V.

Numerical: power at higher cell temperatures (2022 5c)

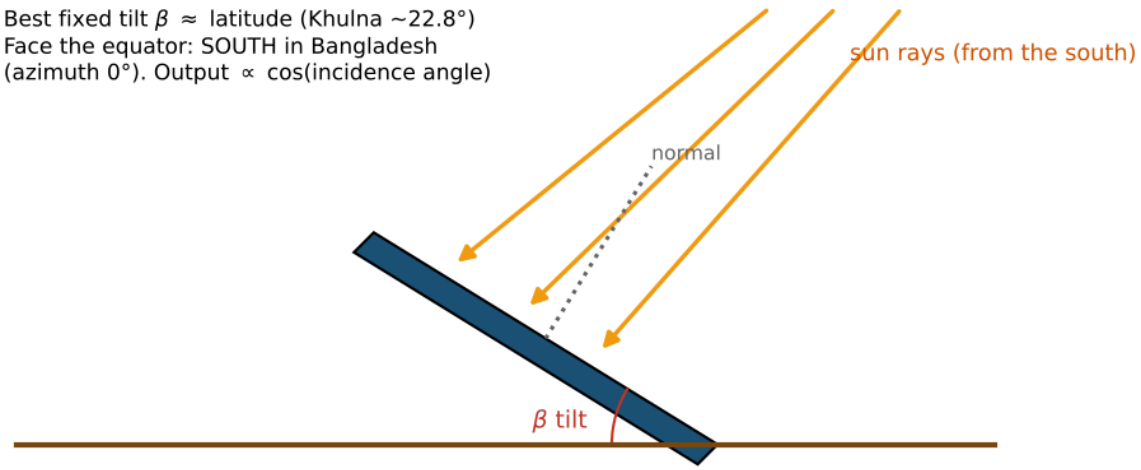
Step (write this)	Why / what it means
$P(T) = P_{STC}[1 + \gamma(T - 25)]$, γ = -0.45%/°C = -0.0045 /°C	linear temperature coefficient of maximum power
45 °C: 75[1 - 0.0045 × 20] = 75 × 0.910 = **68.25 W**	9% loss
55 °C: 75[1 - 0.0045 × 30] = 75 × 0.865 = **64.9 W**	13.5% loss
65 °C: 75[1 - 0.0045 × 40] = 75 × 0.820 = **61.5 W**	18% loss: a hot roof module loses almost a fifth

How heat affects efficiency and how to reduce temperature problems (2022 8b, 2021 6d)

Heat raises the intrinsic carrier density and dark current, so V_{oc}, FF (fill factor) and efficiency all fall (above); in addition high temperature speeds up ageing: EVA browning and delamination, solder fatigue from thermal cycling, potential-induced degradation, and hot spots. A module at 65 °C produces ~18% less than at 25 °C.

Strategies and technologies: (1) **passive cooling**: mount with an air gap of 10-15 cm behind the modules for ventilation, open racks instead of flush roofs, light-coloured surfaces; (2) **active cooling**: water spray or film, forced air, heat pipes, phase change materials behind the module; (3) **PV-thermal (PV/T) hybrid collectors**: remove heat as useful hot water/air; (4) choose **low temperature coefficient technologies**: HIT (heterojunction with intrinsic thin layer)/heterojunction (−0.26%/°C), CdTe, a-Si, or modules with low NOCT; (5) **bifacial and glass-glass** modules, reflective/IR-rejecting coatings; (6) design: use the cell temperature (not 25 °C) when sizing voltage strings, cover temperature drops with MPPT (maximum power point tracking (tracker)), keep cleaning to avoid hot spots; (7) cooler sites, higher elevation, avoid placing near hot walls.

Effect of temperature, orientation and tilt angle (2022 7c, 2019 8a)



Tilt and orientation: the panel should face the equator, tilted near the latitude

- **Temperature:** as above: about −0.45%/°C power for c-Si.
- **Orientation (azimuth):** a fixed panel gets most energy when it faces the equator: **due south in the northern hemisphere** (Bangladesh), due north in the southern hemisphere. Facing east or west shifts production to morning or evening and cuts annual yield by ~10-20%.
- **Tilt angle:** output is largest when sunlight hits the panel perpendicularly; at angle of incidence θ, irradiance on the panel $G = G_b\cos\theta$ + diffuse + reflected. For the best annual yield the tilt β ≈ **latitude** (Khulna ≈ 22.8°, so ~23°); latitude + 15° favours winter, latitude − 15° favours summer. Too flat a panel also collects dust and water. **Tracking** (single-axis +15-25%, dual-axis +30-40%) keeps θ small all day.

Surface area and its impact on current; design optimisation (2022 5b)

Area: the photocurrent is generated per unit area, so $\mathbf{I = J_{sc} \times A}$: doubling the area doubles the current (and power), while the **voltage does not change** with area (it depends on the material and junction). Example (L1): J = 35 mA/cm^2, A = 100 cm^2 → I = 3.5 A. Larger area, however, means larger current in thin fingers and emitter, so series-resistance (I^2R) loss grows unless the grid is redesigned; and more chance of defects/shunts.

Key design factors for efficient conversion: bandgap and absorber material; anti-reflection coating and texturing; minimal shading by metal grid (finger width/spacing optimisation, busbars, back contacts); low series and high shunt resistance; surface passivation and BSF (back surface field); absorber thickness vs diffusion length; light trapping; cell temperature and NOCT; matching cells (same current) when they are put in a module; and cost per watt.

Remember: Cell 0.5 V; module = cells in series (36 for 12 V); array = modules in series strings and parallel. Series: V adds, I same; parallel: I adds, V same. $I_{sc} \propto G$ and area; V_{oc} falls ~2.3 mV/°C; power −0.45%/°C: 75 Wp → 68.3, 64.9, 61.5 W. Face south, tilt ≈ latitude.

B7 Economics and ecology: payback time, energy payback time, LCOE, life-cycle cost, types of cost, carbon footprint, recycling Pass set

Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2023 Q7(c)** [7] What are the differences between payback time and energy payback time?
- **2023 Q8(c)** [7] Does a solar panel reduce carbon footprint and how does production of solar energy contribute to carbon footprint?
- **2022 Q5(d)** [8] Why is energy payback time important in PV system design? What factors influence the energy payback time of a PV system?
- **2022 Q8(c)** [7] Same question as 2023 8(c).
- **2022 Q8(d)** [8] Can solar modules be reused after reaching the end of their life cycle, and if so, how? What environmental impact is associated with improper disposal of solar modules?
- **2021 Q8(d)** [10] How does the concept of levelized cost of electricity work? Describe.
- **2021 Q6(a)** [8] What is the carbon footprint of making solar panels? Explain the environmental impact of PV systems.
- **2020 Q6(b)** [5] Why is energy payback time important in PV system design?
- **2020 Q6(c)** [13] Compare the life-cycle cost of a highway construction warning sign that is PV powered vs a gasoline generator. 24 h/day operation, load 3 kWh/day, 20-year life. PV: 750 W array at $4/W, $900 of batteries replaced every 5 years, $300 charge controller, maintenance $100/year. Generator: 500 W for $250, about 545 gallons of gasoline per year, maintenance about $1500/year, replaced every 5 years. Inflation 3%, discount rate 10%.
- **2019 Q6(c)** [6] Discuss the different types of cost of a PV power generation system.

★ REMEMBER IT

- **Payback time** = money: years until savings repay the cost. **Energy payback time** = energy: years until the module generates the energy used to make it (1–3 years for c-Si; depends on manufacturing energy, irradiance, efficiency, BOS (balance of system)).
- **LCOE (levelized cost of electricity)** = (total lifetime cost)/(total lifetime energy), both discounted: Tk or $ per kWh. Costs: capital (modules, BOS, installation), O&M, replacement (batteries, inverter), fuel (none for PV), salvage.
- Carbon footprint: PV emits nothing while running; its footprint (about 20–50 g CO_2/kWh) comes from making the modules (energy-hungry silicon purification). Recycling recovers glass, aluminium, silicon and silver; improper disposal leaks lead and cadmium.

CALCULATOR **Simple LCOE and payback**

Change any number; every step is recalculated and shown with the numbers put in.

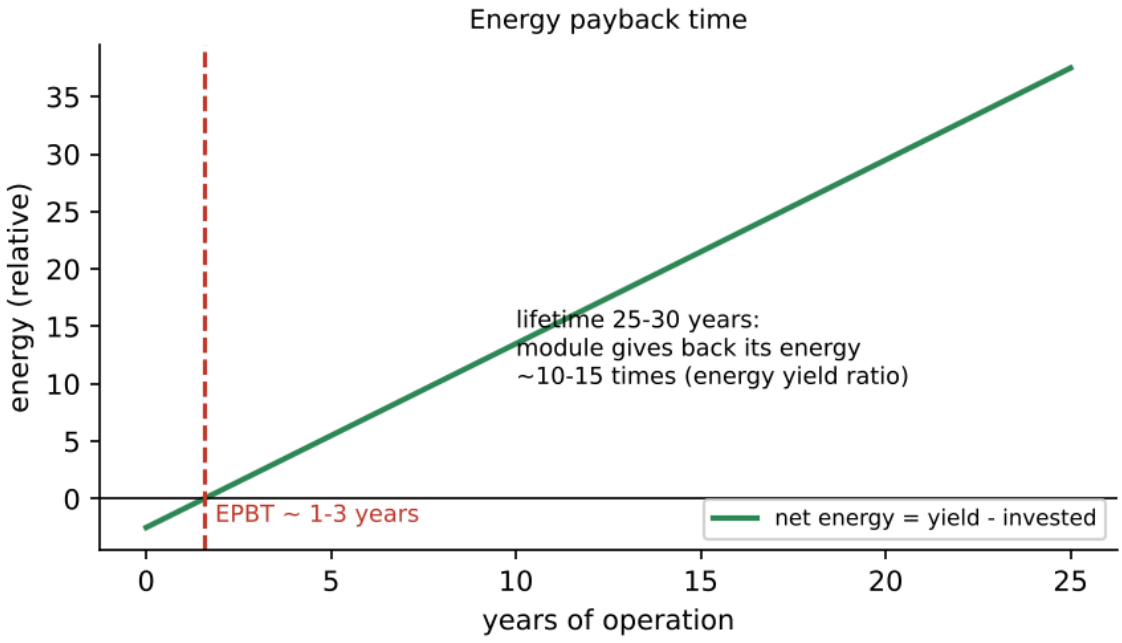
1. Capital recovery factor: CRF = $i(1 + i)^n/[(1 + i)^n - 1]$ = (**8** / 100) × pow(1 + **8** / 100, **20**) / (pow(1 + **8** / 100, **20**) - 1) = **0.1019**
2. LCOE: LCOE = (C × CRF + O&M)/E = (**100000** × **0.1019** + **1000**) / **1100** = **10.17** Tk/kWh
3. Simple payback: payback = C/(annual saving) = **100000** / (**1100** × **9** - **1000**) = **11.2** years

Before you start: is it worth it, in money and in energy

A PV system costs a lot at the start and almost nothing to run, while a generator is cheap to buy and expensive to run. To compare them fairly, all future costs are converted to today's money (present worth) using a discount rate. The same thinking in energy terms gives the energy payback time. Source: Muhit sir L10 (economics), L11 (ecology); Messenger ch. 5 (life-cycle cost, Example 5.2).

Payback time vs energy payback time (2023 7c)

	Payback time (financial)	Energy payback time (EPBT)
Definition	time needed to recover the money invested in the system from the savings or income it produces	time the system needs to generate as much energy as was used to make, transport, install and recycle it
Formula	PBT (payback time) = initial cost/annual net savings (simple)	EPBT (energy payback time) = $E_{invested}$ (total primary energy over the life cycle)/$E_{generated\ per\ year}$
Unit and basis	years; money (tariff, subsidies, interest, prices)	years; energy (kWh or MJ primary energy)
Depends on	initial cost, grid electricity price, feed-in tariff, irradiance, financing	technology and production energy (Si purification is the largest), irradiance, system components (BOS)
Typical	5-10 years for rooftop PV	1-3 years for c-Si (< 1 year for thin film in sunny places)
Tells	economic attractiveness for the investor	environmental (net energy) benefit, sustainability



Energy payback: cumulative net energy becomes positive after 1-3 years

Why EPBT matters in PV design and what influences it (2022 5d, 2020 6b)

Importance: a renewable energy system must give back much more energy than it took to make, otherwise it does not really reduce fossil fuel use or emissions. EPBT (and the related **energy yield ratio** = lifetime energy output/total energy invested, 10-15 for modules) measures this net energy

benefit; it is used to compare technologies and designs, to estimate CO_2 savings and the carbon footprint, and to choose low-energy materials and processes. If the system lasts 25-30 years and EPBT is 2 years, it produces ~12-15 times the energy invested.

Factors influencing EPBT: (1) the PV technology and the energy for production: polysilicon purification and crystal growth dominate for c-Si; thin films use far less material; wafer thickness and kerf loss; (2) cell/module efficiency (more output per m^2 of material); (3) solar irradiance at the site, orientation, tilt, tracking (sunnier sites shorten EPBT); (4) balance-of-system components: frames (aluminium), mounting structure, inverter, cables, batteries; (5) performance ratio, degradation rate and lifetime; (6) the energy mix used in the factory and transport distance; (7) recycling at end of life.

Levelized cost of electricity (2021 8d)

$$LCOE = \Sigma_{t=1..n} (I_t + M_t + F_t)/(1 + r)^t \div \Sigma_{t=1..n} E_t/(1 + r)^t$$

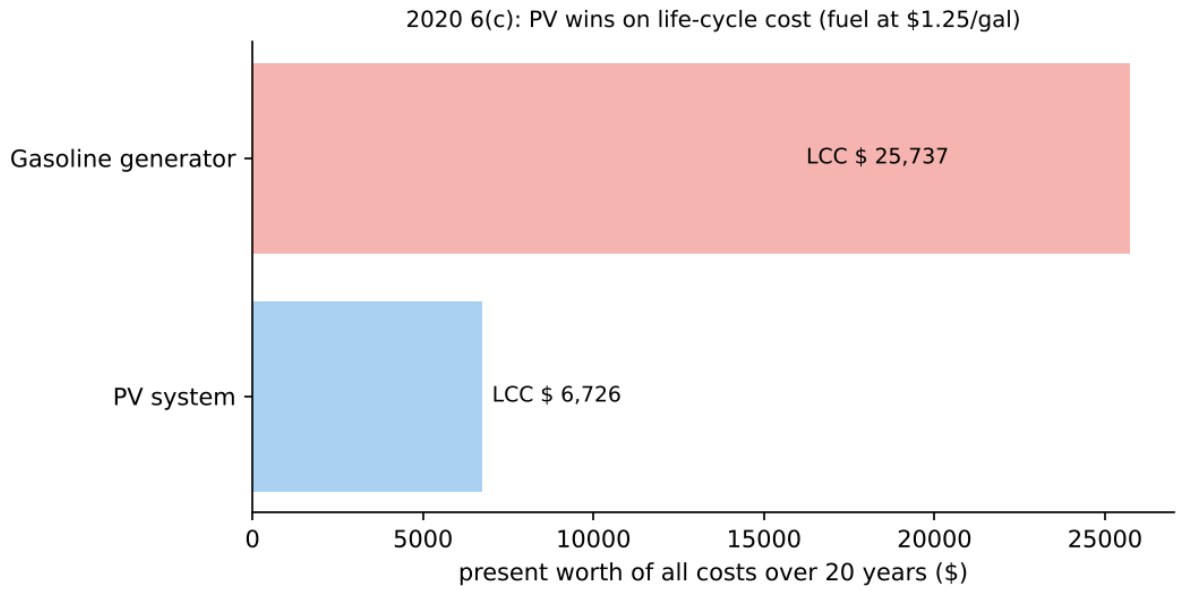
The **LCOE** is the cost per kWh of electricity produced by a plant over its whole life: the present value of all lifetime costs divided by the present value of all electricity produced. I_t = investment expenditures in year t, M_t = operation and maintenance, F_t = fuel (zero for PV), E_t = electricity yield in year t (falls slightly with degradation), r = discount rate, n = lifetime. **How it works:** it spreads the upfront capital cost over the energy produced over the lifetime, like averaging, while the discount rate makes future money and energy worth less than today's; so a cheap system with expensive fuel and an expensive system with no fuel can be compared on one number ($/kWh or Tk/kWh). Uses: compare technologies (wind, PV, gas, coal), set the minimum selling price (price must exceed LCOE to make a profit), assess **grid parity** (PV LCOE = cost of other generation) and **socket parity** (PV LCOE = retail price the consumer pays, reached first). For PV the LCOE is dominated by investment, so it falls with lower module prices, higher irradiance, longer life and lower discount rate.

Types of cost of a PV generation system (2019 6c)

- **Initial (capital) cost:** modules, inverter, batteries, charge controller, mounting structure, cables and protection (BOS), design, land, transport, installation and commissioning; usually 70-90% of the life-cycle cost of PV.
- **Operation and maintenance (O&M) cost:** cleaning, inspection, monitoring, insurance, security, annual repairs (~1-2% of capital per year).
- **Replacement cost:** battery (every 3-7 years), inverter (every 10-15 years), controller.
- **Fuel cost:** zero for PV (but present for generators in hybrid systems).
- **Financing cost:** interest on loans; taxes, permits, grid connection fees.
- **Salvage (end-of-life) value or decommissioning/recycling cost.**
- Combined as the **life-cycle cost**: LCC (life-cycle cost) = $C_{initial}$ + PW(maintenance) + PW(replacements) + PW(fuel) − PW(salvage); annualised LCC = LCC/P_a; unit cost = ALCC (annualised life-cycle cost)/annual kWh.

Numerical: life-cycle cost, PV vs gasoline generator (2020 6c)

Method of Messenger Example 5.2. x = (1 + i)/(1 + d) = 1.03/1.10 = 0.93636. Present worth of a single future cost: $P_r = x^n$. Present worth of an annual recurring cost starting now: $P_a = (1 - x^n)/(1 - x)$ = 11.50; starting after one year (PV maintenance): $P_{a1} = xP_a$ = 10.76 (n = 20).



Life-cycle cost breakdown: PV is capital-heavy, the generator is fuel and maintenance heavy

PV system	Initial ($)	Present worth ($)
Array 750 W × $4/W	3000	3000
Charge controller	300	300
Batteries	900	900
Batteries at 5 yr (x^5 = 0.720)	900	648
Batteries at 10 yr (x^{10} = 0.518)	900	466
Batteries at 15 yr (x^{15} = 0.373)	900	336
Maintenance $100/yr × P_{a1}	100	1076
LCC		**6,726**

Gasoline generator	Initial/annual ($)	Present worth ($)
Generator 500 W	250	250
Replacement at 5, 10, 15 yr	250 each	180 + 130 + 93 = 403
Fuel 545 gal/yr × $1.25/gal = $681/yr × P_a	681	7,831
Maintenance $1500/yr × P_a	1500	17,243
LCC		**25,727**

Step (write this)	Why / what it means
`PV LCC =` **`$6,726`**`; generator LCC =` **`$25,727`** `(at $3/gal: $36,691)`	the PV system costs more at the start (4200 vs 250) but less than a third over 20 years
`Annualised:` $ALCC_{PV}$ `= 6726/11.50 = $585/yr;` $ALCC_{gen}$ `= 25727/11.50 = $2,238/yr`	divide by P_a
`Unit cost: energy = 3 kWh × 365 = 1095 kWh/yr → PV $0.53/kWh; generator $2.04/kWh`	⇒ the PV-powered sign is far cheaper

Side note: The paper does not give the gasoline price. $1.25/gal is the value used in Messenger's same example (300 gal for $375); state your assumption. The conclusion (PV much cheaper) holds for any realistic fuel price.

Carbon footprint and environmental impact of PV (2023 8c, 2022 8c, 2021 6a)

- **Does PV reduce the carbon footprint? Yes.** In operation a PV system emits no CO_2, noise or pollutants. Over its life cycle (cradle-to-grave LCA (life-cycle assessment)) c-Si PV emits about **20-50 g CO_2-eq/kWh** (thin film 10-25), against ~400-500 for natural gas and 800-1000 for coal; every kWh of PV replacing grid electricity in Bangladesh (~0.6 kg CO_2/kWh) saves more than 0.55 kg.
- **How production contributes:** almost all PV emissions occur before operation: quartz reduction in arc furnaces ($SiO_2 + 2C \rightarrow Si + 2CO_2$, plus electricity), Siemens purification and crystal growth at > 1000 °C (the most energy-intensive steps, often powered by coal electricity), wafer sawing, cell processing, aluminium frames and glass, inverters and batteries, transport and installation. The carbon footprint depends on the factory's electricity mix and is paid back in ~1-3 years (EPBT).
- **Other environmental impacts:** toxic and hazardous chemicals in manufacturing (HF, HCl, $SiCl_4$ by-product, silane, NF_3, Cd and Te in CdTe, Pb in solders and perovskites) need strict legislation and treatment; water use for cleaning and processing; land use for ground-mounted plants (can be combined with agriculture: agrivoltaics); battery lead and acid in off-grid systems; end-of-life waste (about 78 million tonnes of panels expected by 2050); visual impact and glare. Positive: no air pollution, saves fossil fuels, quiet, decentralised supply.

Reuse and recycling of modules at end of life; impact of improper disposal (2022 8d)

Yes. After 25-30 years modules still give ~80% of their power. (1) **Reuse/repurposing:** tested and graded modules are resold for less demanding uses (SHS, pumps, off-grid lighting, second-life markets). (2) **Recycling** (about 90-95% of the mass can be recovered): remove the frame (aluminium, ~15% of weight) and junction box (copper cables); separate the glass (~75%) from the laminate by mechanical crushing, thermal treatment (burning the EVA at ~500 °C) or chemical delamination; recover silicon wafers/cells, then chemically etch to recover silver, copper and high-purity silicon; for CdTe, First Solar's process recovers glass and ~95% of the Cd and Te for new modules. EU WEEE directive makes producers responsible; Bangladesh has SHS (solar home system) battery recycling programmes. (3) Repair: replacing junction boxes or bypass diodes.

Impact of improper disposal: modules dumped in landfills or burned can leach **lead** (solder), **cadmium** (CdTe), selenium, tin and antimony into soil and groundwater; burning the EVA and back sheet releases toxic fluorinated gases and dioxins; broken glass and aluminium waste land; valuable materials (silver, silicon, copper, indium, tellurium, aluminium, glass) are lost, so more mining and energy are needed; it undermines the clean image and the circular economy of solar energy.

> **Memory trick: PBT = money back, EPBT = energy back**. LCOE = all costs today ÷ all kWh today. LCC: $x = (1+i)/(1+d)$, $P_r = x^n$, $P_a = (1 - x^n)/(1 - x)$.

> **Remember:** PV LCC $6,726 vs generator $25,727 (20 yr, i 3%, d 10%). EPBT 1-3 yr, depends on technology, irradiance, BOS. PV 20-50 g CO_2/kWh vs coal ~900; emissions are from Si purification. Recycle: frame, glass, Si, Ag, Cu; dumping leaks Pb and Cd.

B2 Shading and mismatch; hot spots; bypass and blocking diodes Pass set

Very likely · 93% Easy

PAST-PAPER QUESTIONS

- **2023 Q5(c)** [6] Why are bypass and blocking diodes used in a solar panel?
- **2023 Q5(d)** [11] What are the different types of mismatches in solar power systems?
- **2022 Q6(a)** [10] How does mismatch in current affect the performance of series-connected PV modules, and what strategies or technologies are employed to mitigate current mismatch issues to optimize overall efficiency?
- **2021 Q5(b)** [10] Describe the PV system's shading effect. Give a justifiable explanation of how the PV system performs when there is shading.
- **2020 Q5(b)** [16] Explain shading of a PV system. With proper justification, explain the performance of a PV system during shading.

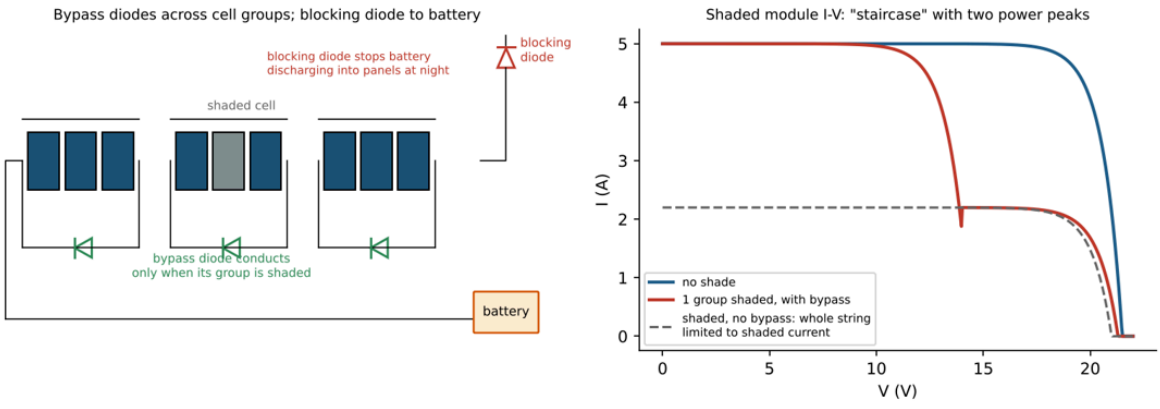
★ REMEMBER IT

- Cells in series must carry the same current. One shaded cell makes less current, so it limits the whole string and is forced into **reverse bias**: it dissipates power as heat (**hot spot**, can crack or burn the module).
- **Bypass diode**: across a group of cells inside the module (usually one per 18–24 cells); lets current flow around the shaded group. **Blocking diode**: in series with each string; stops reverse current from the battery or other strings at night or under shade.
- Mismatch types: current mismatch (shading, soiling, different I_{SC}), voltage mismatch (different V_{OC}, temperature), and degradation mismatch. Fixes: bypass diodes, sorting modules, string inverters/micro-inverters, power optimizers, module-level MPPT (maximum power point tracking (tracker)).

Before you start: one weak cell spoils the string

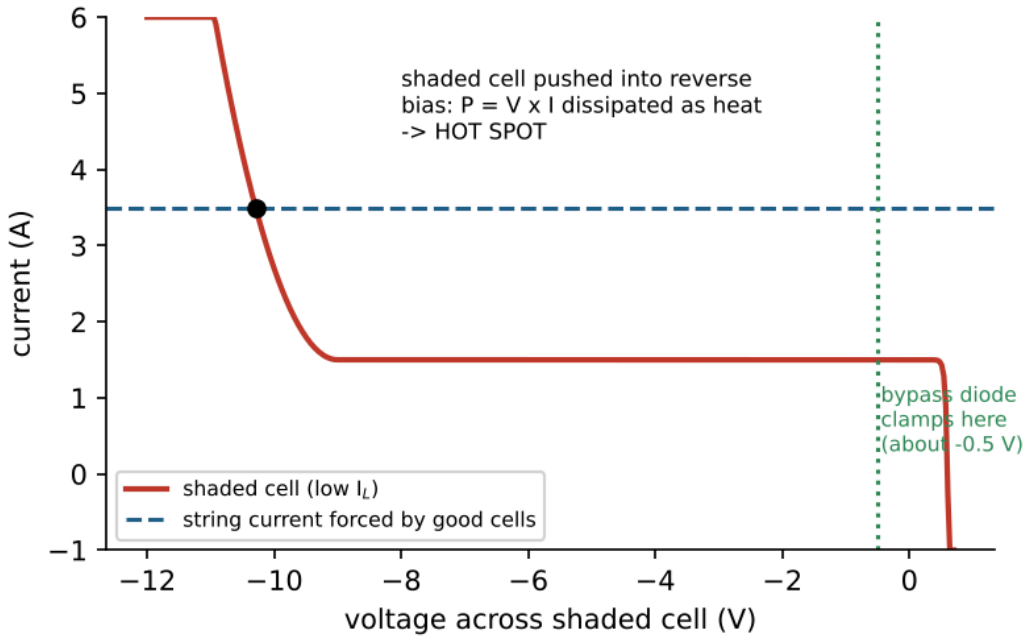
In a series string every cell must carry the same current. If one cell gets less light (shade, dirt, crack) or is simply weaker, it limits the current of all the others, and it can even be driven into reverse bias and heat up. Source: Muhit sir L4 (shading effect, bypass and blocking diodes); Messenger sec. 3.7; Smets ch. 19.

Shading of a PV system and its performance during shading (2021 5b, 2020 5b)



One shaded cell in a series string: the string current falls to the shaded cell's current

- **Shading** is the condition in which incident solar radiation on a cell, module or array is partly or completely blocked, reducing the photo-generated current. **Static** shading: fixed objects (buildings, trees, poles, other rows). **Dynamic** shading: moving or changing conditions (clouds, birds, dust, snow, leaves).
- **Performance (justification):** cells are in series, so the current must be the same in each. If one cell of a 36-cell module is 50% shaded, it can only make half the photocurrent, and the whole string current falls to about that level: **shading 1 of 36 cells (3% of area) can cut module power by 50% or more**, i.e. the loss is far out of proportion to the shaded area.
- If the other cells push more current than the shaded cell can produce, the shaded cell is driven into **reverse bias**: the voltage of the good cells appears across it and it **dissipates** the power of the unshaded cells ($P = V_{reverse} \times I$) as heat. This makes a **hot spot**, which can reach 100-150 °C, burn the back sheet, crack glass or break down the cell (avalanche).
- The I-V curve of a shaded string with bypass diodes shows **steps** and the P-V curve has **several local maxima**, so a simple MPPT may lock on the wrong peak.
- **Remedies:** bypass diodes (every 18-24 cells), blocking diodes/string fuses, avoid shade in layout (row spacing, site survey), MPPT with global scan, module-level power electronics (micro-inverters, DC optimisers), half-cut cells and split junction boxes, regular cleaning.



Hot-spot formation and protection by a bypass diode

Why bypass and blocking diodes are used (2023 5c)

- **Bypass diode:** connected **in parallel** with a cell or group of cells (typically 3 diodes per 60-cell module) with **opposite polarity**. Normally (no shade) it is reverse biased and carries no current. When a cell in its group is shaded, the reverse voltage across that group forward-biases the diode, which then carries the string current **around** the shaded cells. This limits the reverse voltage across the bypassed group to one diode drop (about 0.6-0.7 V), **prevents hot spots and damage**, and lets the rest of the string keep producing power (only the bypassed group is lost).
- **Blocking diode:** connected **in series** with a module or string. (i) In a battery system at night the PV modules are idle and the battery would discharge back through them; the blocking diode prevents this **reverse current from the battery**. (ii) In parallel strings it stops a healthy string pushing current back into a shaded or faulty string. Cost: a forward drop of ~0.5-0.7 V (Schottky diodes ~0.4 V); modern charge controllers often do this job electronically.

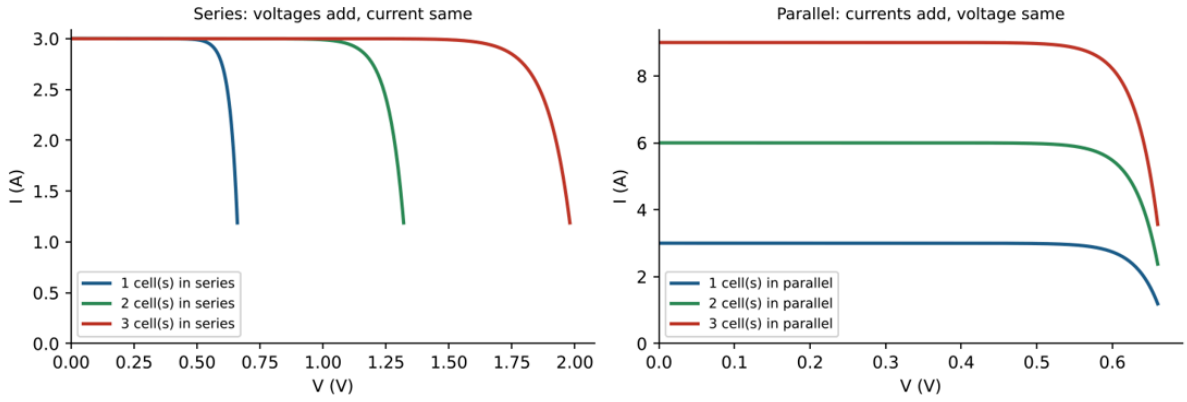
> **Memory trick: By**pass = **by** the side (parallel), saves the module from shade. **Block**ing = in the line (series), blocks the battery at night.

Types of mismatch in solar power systems (2023 5d)

Mismatch occurs when interconnected cells or modules do not have identical properties or do not experience identical conditions, so the combination gives less than the sum of its parts.

Type	Description and effect
By cause: manufacturing (intrinsic)	tolerance in I_{sc}, V_{oc}, power (±3%), different resistances; modules of different makes or ages
By cause: operating (extrinsic)	partial shading, soiling (dust, bird droppings), non-uniform temperature, different orientation/tilt within a string, uneven degradation, cracks, PID (potential-induced degradation), broken bypass diode
Current mismatch in series	the most serious: the string current is limited by the lowest-current cell; the excess current of good cells is lost and the weak cell may be reverse biased (hot spot)
Voltage mismatch in parallel	strings at a common voltage; the lower-V_{oc} string drags the voltage down and a higher string may push current into it; loss is usually smaller
Series-parallel (array) mismatch	combination of both; also caused by different string lengths or cable drops
Module-inverter/MPPT mismatch	string voltage outside the inverter MPPT window, or several strings with different MPPs on one tracker

Effect of current mismatch in series modules and mitigation (2022 6a)



Series string: the I-V curves add along the voltage axis at a common current

In a series string the common current cannot exceed what the weakest module can carry at its operating voltage. The string operates at a current set by the poorest module, so all other modules work below their MPP (maximum power point) and the **whole string power falls** to roughly N × V × I_{weak}. If the forced current exceeds the weak module's I_{sc}, it goes into reverse bias and dissipates power (hot spots, accelerated degradation). Also the P-V curve gets several peaks and the inverter may track a local peak.

Strategies and technologies: (1) **bypass diodes** in every module (cut loss to the affected sub-string); (2) **sorting and binning** of cells and modules by current at the factory and in the field ("current matching"), same make and model in a string; (3) **layout**: avoid shading, same orientation

and tilt per string, short strings, landscape vs portrait choice; (4) **multi-MPPT inverters** and **string inverters** (one string per tracker); (5) **module-level power electronics**: DC power optimisers (each module at its own MPP, DC-DC converter) and micro-inverters; (6) **global MPPT algorithms** that scan the whole P-V curve; (7) half-cut cells, multi-busbar and split junction boxes (two half-modules in parallel); (8) monitoring and cleaning, replacing faulty modules; (9) blocking diodes or string fuses for parallel strings.

Remember: Series string: current = weakest cell. 3% shading → up to 50% loss; shaded cell reverse biased → hot spot. Bypass diode in parallel (anti-parallel), blocking diode in series. Mismatch: manufacturing vs operating; current in series, voltage in parallel. Fix: bypass diodes, binning, multi-MPPT, optimisers, micro-inverters.

B6 Designing and sizing a PV system (solar home system numericals)

Pass set | Very likely · 100% | Medium

PAST-PAPER QUESTIONS

- **2023 Q6(c)** [20] SHS (solar home system) load profile: (i) laptop 2 × 20 W, 3 h/day; (ii) lamp 4 × 30 W, 8 h/day; (iii) colour TV 2 × 50 W, 6 h/day; (iv) 1 × 800 W, 2 h/day (geyser). Inverter pf unity, DOD (depth of discharge) 60%, autonomy 4 days, battery charging efficiency 75%, peak sun hours 5, battery 12 V. Design a PV module. Also calculate (i) battery capacity, (ii) charge controller, (iii) DC to AC inverter capacity, (iv) switch capacity, (v) wire size.
- **2022 Q6(d)** [6] Design a solar PV system for a house with 3 fans of 70 W each running 4 h/day, 3 tube lights of 35 W each running 8 h/day, and a refrigerator of 250 W running 12 h/day (compressor runs 6 h on and 6 h off). Battery autonomy 2 days, DOD 50%, battery charging efficiency 80%, peak sun hours 6, battery voltage 12 V.
- **2021 Q7(c)** [13] How to design and size solar photovoltaic systems, and installation of a solar PV system.
- **2020 Q6(a)** [12] Write down the design guidelines of all components of a 3 kW off-grid PV system.
- **2019 Q6(d)** [15] Same load profile and data as 2023 6(c).

UNDERSTAND **Read this to understand; you do not need to write it**

Size an off-grid system in one chain: **Load → PV → Battery → Controller → Inverter**. Energy per day E = Σ(power × count × hours). PV peak watts = E/(efficiency × peak sun hours). Battery Ah = E × autonomy days/(V × DOD). Controller current = 1.25 × array current. Inverter = 1.25 × total running load.

CALCULATOR **Solar home system sizing (2022 6(d), 2023 6(c))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Daily energy: E = Σ P × n × h = **70** × **3** × **4** + **35** × **3** × **8** + **250** × **1** × **6** + **0** × **0** × **0** = **3180** Wh/day
2. Total running load: **70** × **3** + **35** × **3** + **250** × **1** + **0** × **0** = **565** W
3. PV size: P_{PV} = E/(η × PSH) = **3180** / (**0.8** × **6**) = **662.5** Wp
4. Battery capacity: C = E × days/(V × DOD) = **3180** × **2** / (**12** × **0.5**) = **1060** Ah
5. Charge controller: 1.25 × P_{PV}/V = 1.25 × **662.5** / **12** = **69.0** A
6. Inverter: 1.25 × load = 1.25 × **565** = **706** W

Round each answer up to a standard size (100 Wp modules, 200 Ah batteries, 80 A controller, 1 kVA inverter). The fridge running 12 h at 50% duty = 6 h.

Before you start: sizing step by step

Sizing means choosing how many watts of modules, how many ampere-hours of battery, and what rating of controller, inverter, switches and cables are needed so the load is always met. The order is always: load → PV array → battery → controller → inverter → switches and cables. Source: standard SHS method (Muhit sir's course, Messenger ch. 4). Notation: E = daily energy (Wh/day), PSH (peak sun hours) = peak sun hours (h/day of 1000 W/m^2), DOD = depth of discharge.

How to design and size a solar PV system and install it (2021 7c); guidelines for a 3 kW off-grid system (2020 6a)

1. **Load assessment:** list every appliance with power (W), number and hours of use; daily energy E = Σ P × n × h (Wh/day); also the total connected (peak) load and surge loads; add inverter losses for AC loads.
2. **Site and solar resource:** peak sun hours of the worst month (Bangladesh ~4-5 h), tilt ~latitude, shading.
3. **PV array size:** P_{PV} (Wp) = E/(PSH × η_{system}), where the system efficiency includes battery charging efficiency, controller, wiring, temperature and dust losses (a factor of 1.3 is often used: E × 1.3/PSH). Number of modules = P_{PV}/module rating, arranged as N_s (system voltage/module voltage) × N_p.
4. **Battery bank:** Ah = E × days of autonomy/($V_{battery}$ × DOD × $\eta_{battery}$); choose system voltage (12 V up to ~1 kW, 24 V up to ~3 kW, 48 V above), number in series = $V_{system}/V_{battery}$, in parallel = Ah total/Ah per battery. Use deep-cycle batteries.
5. **Charge controller:** current rating = 1.25 × array short-circuit current (N_p × I_{sc}); voltage = system voltage; maximum PV voltage > array V_{oc} at low temperature; MPPT (maximum power point tracking (tracker)) preferred for bigger systems.
6. **Inverter:** continuous rating ≥ 1.25-1.3 × total simultaneous AC load, surge rating for motors (3-5 ×), input voltage = battery bank voltage, pure sine wave, η > 90%.
7. **Switches, fuses and breakers:** 1.25 × maximum current of each circuit.
8. **Cable size:** A = 2ρLI/ΔV (two conductors) for a voltage drop ΔV ≤ 2-3% (ρ_{Cu} = 0.0175 Ωmm^2/m); choose the next standard size and check the current rating.
9. **Installation:** mount array facing south at the tilt; place battery in a ventilated area close to controller and inverter; wire battery → controller → PV → inverter → loads; earthing and lightning protection; test V_{oc}, I_{sc}, polarity, set points and loads; commission and train users.

Guidelines for a 3 kW off-grid system (2020 6a): 3 kW is the AC load (peak). Use a **48 V** (or 24 V) DC bus to keep currents moderate (3000/48 = 62.5 A). Inverter: pure sine, ~3.75-4 kVA continuous (1.25 ×), 48 V input, surge capacity for motor starting, η ≥ 90%, with low-voltage cut-off. Array: from the daily energy, e.g. 3 kW × 5 h = 15 kWh/day → 15000 × 1.3/4.5 ≈ 4.3 kWp (e.g. 11 × 400 Wp, strings matched to MPPT input window). Battery: 15 kWh × 2 days/(48 × 0.5 × 0.85) ≈ 1470 Ah at 48 V (24 × 2 V cells, or Li-ion with 80% DOD ≈ 780 Ah). Charge controller: MPPT, 48 V, current 1.25 × array I_{sc} (~1.25 × 4300/48 ≈ 110 A → 120 A or two 60 A units). DC cables for ≤ 2% drop, battery fuses and DC breakers, AC breakers, SPDs, earthing, ventilated battery room, monitoring.

Numerical: SHS design (2023 6c, 2019 6d)

Given: laptop 2 × 20 W × 3 h; lamp 4 × 30 W × 8 h; colour TV 2 × 50 W × 6 h; geyser 1 × 800 W × 2 h (the paper says "gizer"); pf = 1, DOD 60%, autonomy 4 days, charging efficiency 75%, PSH 5 h,

battery 12 V.

Load	Power (W)	Hours	Energy (Wh/day)
Laptop 2 × 20	40	3	120
Lamp 4 × 30	120	8	960
TV 2 × 50	100	6	600
Geyser 1 × 800	800	2	1600
Total	**1060**		**3280**

Step (write this)	Why / what it means
PV energy needed = E/n_{charge} = 3280/0.75 = 4373 Wh/day	the battery charging efficiency means more energy must come from the array than the load uses
PV size = 4373/PSH = 4373/5 = 874.7 Wp → **9 modules of 100 Wp = 900 Wp** (all in parallel for a 12 V system, V_m ≈ 17-18 V each)	each peak sun hour gives 1 Wh per Wp
Battery capacity = E × autonomy/(V × DOD) = 3280 × 4/(12 × 0.6) = **1822 Ah** → 10 batteries of 200 Ah, 12 V in parallel (2000 Ah)	4 days without sun, only 60% of the capacity may be used. If the 75% efficiency is applied to the battery as well: 1822/0.75 = 2430 Ah (13 × 200 Ah)
Charge controller: array current = 900/12 = 75 A; rating = 1.25 × 75 = 93.75 A → **100 A, 12 V** (or two 50 A)	25% margin for irradiance above 1000 W/m^2 and I_{sc} > I_m
Inverter = 1.25 × connected load = 1.25 × 1060 = 1325 W → **1500 VA, 12 V DC/230 V AC** pure sine (pf = 1, so VA = W)	all loads could run together; margin for surges
Switch: DC current battery → inverter = 1060/12 = 88.3 A; 1.25 × 88.3 = 110 A → **125 A DC switch/breaker** (AC side: 1060/230 = 4.6 A → 6-10 A switch)	switches are rated for 1.25 × the maximum current
Wire size, A = 2ρLI/ΔV with ρ = 0.0175 Ωmm^2/m, ΔV = 3% × 12 = 0.36 V: battery-inverter (L = 2 m, I = 88.3 A): A = 2 × 0.0175 × 2 × 88.3/0.36 = 17.2 mm^2 → **25 mm^2**	factor 2: current goes out and back; at 12 V even a short cable needs a large cross-section
PV-controller (L = 5 m, I = 75 A): A = 36.5 mm^2 → **50 mm^2**; controller-battery (L = 2 m, I = 75 A): A = 14.6 mm^2 → **16 mm^2**	use the next standard size above

Side note: The paper gives no cable lengths, so 2 m and 5 m are assumed and stated; with your own lengths only L changes. Note how a 12 V system with an 800 W geyser needs huge currents; in practice the geyser would make you choose 24 or 48 V, which divides the currents and cable areas by 2 or 4.

Numerical: house system design (2022 6d)

Load	Power (W)	Hours	Energy (Wh/day)
Fan 3 × 70	210	4	840
Tube light 3 × 35	105	8	840
Refrigerator 250 W, 12 h with 50% duty (6 h running)	250	6	1500
Total	**565**		**3180**

Step (write this)	Why / what it means
PV size = $E/(n_{charge} \times PSH) = 3180/(0.8 \times 6) = 662.5$ Wp → **700 Wp** (7 × 100 Wp in parallel at 12 V)	fridge compressor cycles 6 h on, 6 h off, so it runs 6 h per day
Battery = $E \times autonomy/(V \times DOD) = 3180 \times 2/(12 \times 0.5) =$ **1060 Ah** → 6 × 200 Ah, 12 V in parallel (1200 Ah)	with 80% charging efficiency also applied: 1325 Ah
Charge controller = $1.25 \times 700/12 = 1.25 \times 58.3 = 72.9$ A → **80 A, 12 V**	
Inverter = $1.25 \times 565 = 706$ W → **800 VA-1 kVA** pure sine wave (choose 1 kVA for the fridge's starting surge); DC current 565/12 = 47 A	

Memory trick: Sizing chain **L-P-B-C-I-S-W**: Load, PV ($E/\eta PSH$), Battery ($E \cdot days/V \cdot DOD$), Controller ($1.25\ I_{array}$), Inverter (1.25 load), Switch (1.25 I), Wire ($2\rho LI/\Delta V$).

Remember: 2023/2019 SHS: E = 3280 Wh, 875 Wp → 900 Wp; 1822 Ah; 100 A controller; 1500 VA inverter; 125 A switch; 25/50/16 mm^2 cables. 2022: E = 3180 Wh, 700 Wp, 1060 Ah, 80 A, 1 kVA.

B3 Battery, charge controller (set points, PWM, shunt and series regulators), MPPT (P&O, incremental conductance), DC-DC converters Pass set Very likely · 87% Medium

PAST-PAPER QUESTIONS

- **2023 Q6(a)** [7] Why is a charge controller needed in a solar PV system?
- **2023 Q6(b)** [8] Illustrate the key set points of a charge controller and explain their behaviour using a labelled diagram.
- **2023 Q8(b)** [10] What is the function of an MPPT (maximum power point tracking (tracker)) device and how does it ensure maximum power transfer from a PV module?
- **2023 Q8(d)** [10] What are the basic parameters of a DC to DC converter? Write the basic working principles of DC-DC conversion of a solar PV system.
- **2022 Q6(b)** [12] What role does a charge controller play in a solar system? How does the incremental conductance algorithm function in MPPT charge controllers and what advantages does it offer over simpler algorithms like perturb and observe?
- **2022 Q8(a)** [10] Same question as 2023 8(d).
- **2021 Q6(b)** [12] What is considered the most important feature in a charge controller? How do solar MPPT charge controllers work? Explain in detail.
- **2021 Q8(c)** [8] Same question as 2023 8(d).
- **2019 Q8(b)** [10] Why is a charge controller important in a solar PV system? Write the operation principle of the shunt regulator and series regulator.
- **2019 Q8(c)** [5] What are the basic parameters of a charge controller?
- **2019 Q5(c)** [10] Explain the basic working principle of a PV solar battery charging technique using the PWM (pulse width modulation) method.
- **2019 Q6(a)** [8] Write short notes on (i) specific gravity, (ii) autonomy days, (iii) luminous efficiency, and (iv) total harmonic distortion (THD).

★ REMEMBER IT

- Charge controller protects the battery: stops **overcharge** (regulation voltage set point, then float) and **over-discharge** (low-voltage disconnect, then reconnect when voltage recovers). Set points: VR (voltage regulation (set point)), ARV (array reconnect), LVD (low-voltage disconnect), LVR.
- Shunt regulator: diverts (shorts) the array current through a switch when the battery is full; simple, but heats. Series regulator: opens a switch in series with the array. **PWM**: switches rapidly so the average current tapers as the battery fills.
- **MPPT**: a DC-DC converter that keeps the array at its maximum power point. **P&O (perturb and observe)** (perturb and observe): nudge V, if P rises keep going, else reverse. **Incremental conductance**: at the MPP (maximum power point) $dP/dV = 0$, so $dI/dV = -I/V$; compare and move.
- DC-DC: buck (steps down), boost (steps up), buck-boost. Parameters: input and output voltage range, current rating, efficiency, switching frequency, ripple.

Before you start: keeping the battery alive and the panel at its best point

A battery stores the daytime energy for the night, but lead-acid batteries are ruined by overcharging (gassing, water loss, heat) and by deep discharge (sulfation). The **charge controller** sits between the PV array, the battery and the DC load and keeps the battery between safe voltage limits. An **MPPT** controller also keeps the array at its maximum power point. Source: Muhit sir L5 (battery), L6-7 (charge controller, MPPT); Messenger ch. 3.

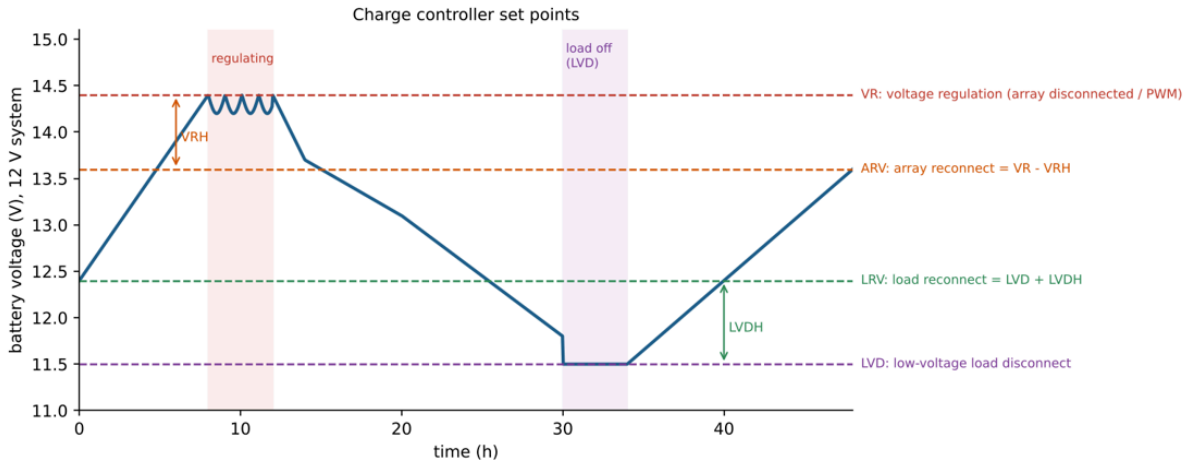
Why a charge controller is needed; its role and functions (2023 6a, 2019 8b, 2022 6b)

- **Prevents overcharging:** when the battery is full the PV current is cut or reduced; overcharging causes gassing, electrolyte loss, grid corrosion and overheating.
- **Prevents over-discharge (low-voltage disconnect):** the load is disconnected when the battery voltage falls to the LVD level; deep discharge sulfates the plates and shortens life.
- **Blocks reverse current** from the battery to the array at night.
- **Load control:** switches loads on/off (e.g. dusk-to-dawn street lights), overload and short-circuit protection.
- Also: multi-stage charging (bulk, absorption, float, equalisation), temperature compensation of set points, MPPT, display of state of charge.

A good controller extends battery life (the battery is the most expensive replacement item); a poor one causes battery failure and shuts the whole off-grid system down.

Basic parameters (features) of a charge controller (2019 8c): nominal system voltage (12/24/48 V); nominal PV array (input) current, sized 1.25 × array I_{sc}; nominal load current; maximum PV open-circuit voltage; charge regulator set points (VR, VRH (voltage regulation hysteresis), LVD, LVDH (low-voltage disconnect hysteresis)); regulation type (PWM/MPPT, shunt/series); temperature compensation; self-consumption and efficiency; protections. **Most important feature (2021 6b):** the ability to regulate charging correctly at the right set points (temperature compensated) so the battery is never overcharged or deeply discharged, and, for MPPT controllers, the tracking efficiency.

Key set points of a charge controller (2023 6b)

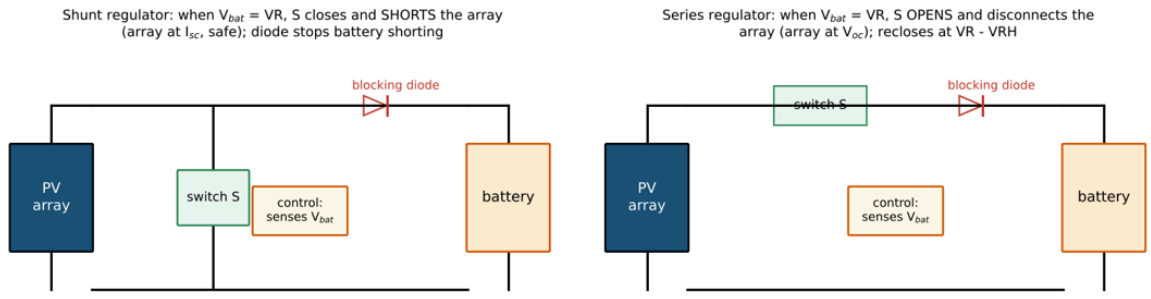


Charge controller set points on a battery voltage-time plot (12 V lead-acid)

Set point	Meaning	Typical (12 V lead-acid)
Voltage regulation VR (high-voltage disconnect, HVD (high-voltage disconnect))	maximum battery voltage; at VR the controller stops or reduces array current to prevent overcharge	14.1-14.6 V
Voltage regulation hysteresis VRH	the voltage difference between VR and the array reconnect voltage (ARV); after disconnection, charging resumes only when voltage drops to VR − VRH. A small VRH keeps the battery fuller but causes rapid cycling	ARV (array reconnect voltage) ≈ 13.0-13.5 V (VRH 1-1.5 V)
Low-voltage disconnect LVD	minimum battery voltage; at LVD the load is disconnected to stop deep discharge (it sets the allowable DOD (depth of discharge))	11.4-11.7 V
LVD hysteresis LVDH	the difference between LVD and the load reconnect voltage (LRV): the load is reconnected only after the battery has been recharged to LVD + LVDH, to avoid oscillation	LRV (load reconnect voltage) ≈ 12.5-12.8 V

Behaviour: during the day the voltage rises while charging; at VR the array is disconnected, the voltage relaxes, and at ARV it is connected again (the saw-tooth); at night the load discharges the battery; at LVD the load is cut, and only after charging back to LRV is it reconnected. Set points are temperature-compensated (about −5 mV/°C per cell, i.e. −30 mV/°C for 12 V).

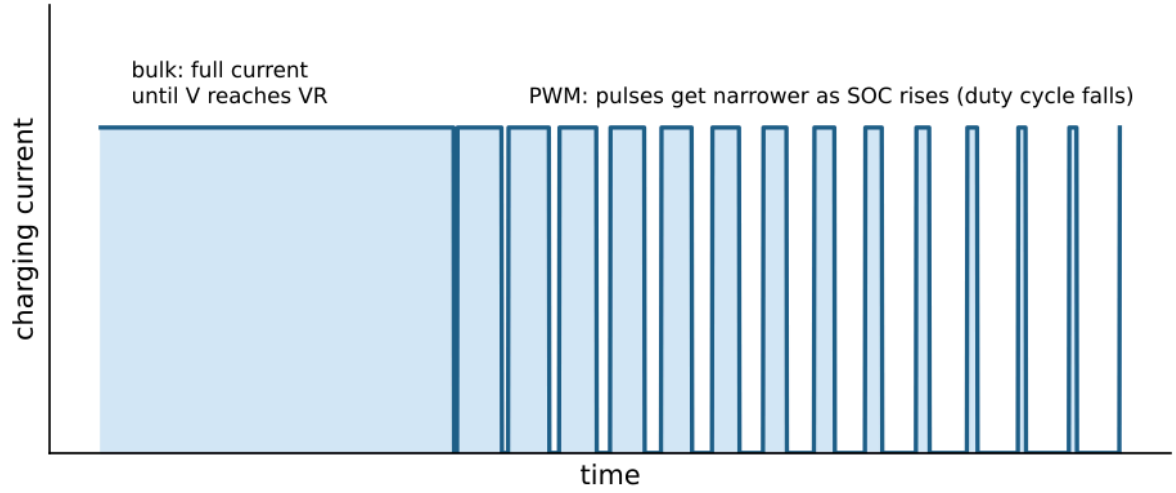
Shunt and series regulators (2019 8b)



Shunt and series charge regulators

- **Shunt regulator:** a switch (transistor/MOSFET) is connected **across (in parallel with) the PV array**, with a blocking diode in series between array and battery. When the battery reaches VR, the switch closes and **short-circuits the array**; the current flows through the switch instead of the battery (a short-circuited PV module is not harmed because its current is limited to I_{sc}). The blocking diode prevents the battery from being shorted. When the voltage falls to ARV the switch opens. Simple, low loss in the charging path; but the switch must carry full I_{sc} and dissipates heat; used for small systems (< 20 A).
- **Series regulator:** the switch is **in series** between the array and the battery. At VR the switch **opens** (on/off type) or its resistance is varied (linear/constant-voltage type, PWM), interrupting or limiting the charging current; the array is then at open circuit. No large heat at full charge, can also act as the blocking device (no diode needed), suitable for larger currents; but the series switch has a voltage drop and loss while charging.

PWM battery charging (2019 5c)

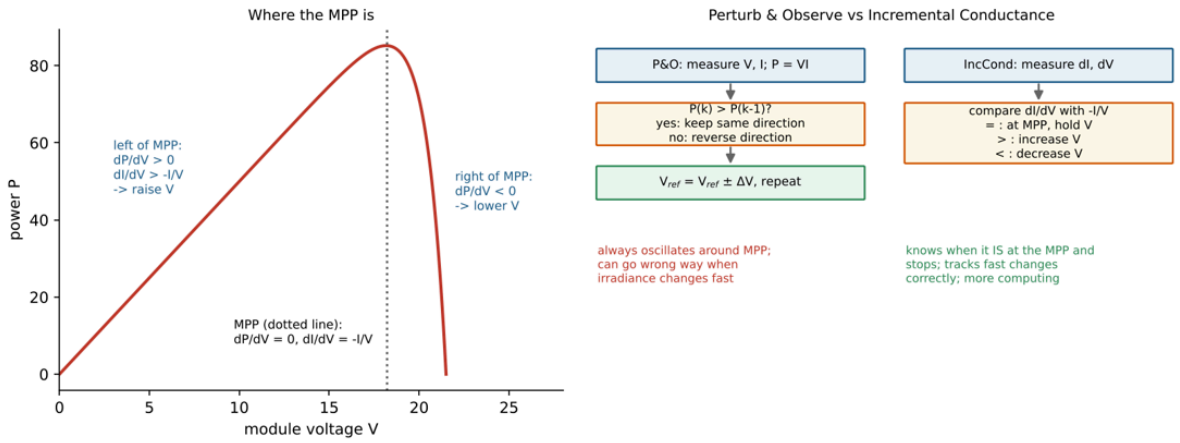


PWM charging: the duty cycle shrinks as the battery approaches full charge

1. **Bulk stage:** when the state of charge is low, the controller switch is fully ON and the full array current flows into the battery; the battery voltage rises.
2. **Absorption (PWM) stage:** when the voltage reaches the regulation set point (VR/HVD), the controller does not allow continuous charging any more; it rapidly switches the charging current ON and OFF (hundreds of Hz to kHz). The **duty cycle** $D = t_{on}/(t_{on} + t_{off})$ is adjusted so the battery voltage stays at the set point: as the SOC (state of charge) rises, the pulses become **narrower** and the gaps longer, so the average current $I_{avg} = D \times I_{array}$ tapers off.
3. **Float stage:** when the battery is full, short pulses keep it at the float voltage (13.5-13.8 V), compensating self-discharge.

Benefits: battery is charged fully without gassing, less sulfation, temperature compensation possible, simple and cheap. Limitation: the array is tied to the battery voltage (e.g. module V_m 17 V working at 13 V), so 20-30% of the available power is wasted compared with MPPT.

MPPT: function, how it works, P&O and incremental conductance (2023 8b, 2021 6b, 2022 6b)



Maximum power point tracking: P&O moves the operating point toward dP/dV = 0

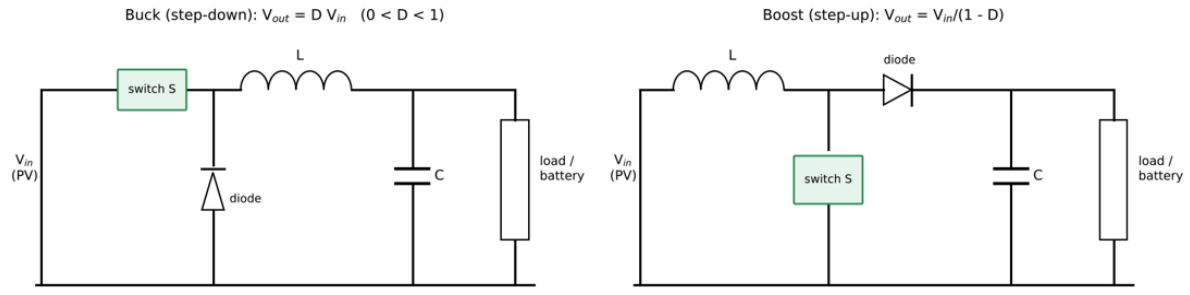
- **Function:** the power from a module depends on the load: the operating point is where the load line crosses the I-V curve. Only one point, the MPP, gives maximum power, and it moves with irradiance and temperature during the day. A **maximum power point tracker** is a DC-DC

converter plus a control algorithm that continuously adjusts the operating voltage so that the array always works at its MPP, whatever the load or battery voltage.

- **How it ensures maximum power transfer:** by **impedance matching**. Maximum power transfer occurs when the load resistance seen by the array equals $R_{mp} = V_m/I_m$. The DC-DC converter transforms the actual load resistance: for a buck converter $R_{in} = R_{load}/D^2$; changing the duty cycle D changes the resistance the array sees until $R_{in} = R_{mp}$. Power is conserved (minus ~2-5% loss), so the converter takes the array at e.g. 17.5 V × 5.7 A = 100 W and gives the battery 13 V × 7.4 A: the battery current is higher than the array current. Gain over PWM: 10-30%, most in cold weather and at low battery voltage.
- **Indirect methods:** fixed (constant) voltage method (operating voltage set seasonally) and fractional open-circuit voltage ($V_m \approx 0.76$-$0.8\ V_{oc}$, measured periodically): simple but not exact.
- **Perturb and observe (P&O, hill climbing):** the controller perturbs (changes) the operating voltage by a small step ΔV and measures the new power. If the power increased, it keeps moving in the same direction; if the power decreased, it reverses direction. The operating point thus climbs to the MPP and then oscillates around it. Simple and cheap, but it oscillates around the MPP (losing a little power) and can go the wrong way when irradiance changes quickly.
- **Incremental conductance (IncCond):** uses the fact that at the MPP dP/dV = 0. Since P = VI, dP/dV = I + V dI/dV, so: **dI/dV = −I/V at the MPP**; dI/dV > −I/V → left of MPP, increase V; dI/dV < −I/V → right of MPP, decrease V. The controller compares the incremental conductance ΔI/ΔV with the instantaneous conductance −I/V and stops perturbing when they are equal.
- **Advantages of IncCond (incremental conductance) over P&O:** it can tell when it has reached the MPP and stops (no steady-state oscillation); it tracks rapidly changing irradiance correctly because it compares conductances at the same instant rather than power before and after; higher tracking efficiency (~98-99%); disadvantage: more computation and sensitive to sensor noise.

> **Memory trick:** P&O: “step, look, keep or turn”. IncCond: “at the top, dI/dV = −I/V”.

DC-DC converters: parameters and working principle (2023 8d, 2022 8a, 2021 8c)



Buck and boost DC-DC converters

Basic parameters: input voltage range ($V_{in,min}$-$V_{in,max}$, must cover the array V_{oc} at low temperature); output voltage and its regulation; rated output current and power; conversion efficiency $\eta = P_{out}/P_{in}$ (90-98%); duty cycle D range; switching frequency (20-200 kHz); voltage ripple and current ripple; conversion ratio V_{out}/V_{in}; transient response; protection (over-voltage, over-current, reverse polarity, temperature).

Working principle: a DC-DC converter (chopper) changes one DC voltage into another by switching a semiconductor (MOSFET/IGBT) on and off at high frequency and smoothing the pulses

with an **inductor** and **capacitor**, which store and release energy; a diode (or second switch) gives the inductor current a path when the switch is off. The ratio is set by the duty cycle D:

- **Buck (step-down):** switch ON: input supplies the inductor and load; OFF: inductor current freewheels through the diode. $\mathbf{V_{out} = D\,V_{in}}$. Used in MPPT chargers to charge a 12 V battery from a 17-36 V array.
- **Boost (step-up):** switch ON: inductor is charged from the input (diode blocks); OFF: inductor voltage adds to input and charges the output. $\mathbf{V_{out} = V_{in}/(1 - D)}$. Used to raise array voltage to the inverter DC link (300-400 V).
- **Buck-boost:** $V_{out} = -D\,V_{in}/(1 - D)$: step up or down.
- In a PV system the converter's duty cycle is controlled by the MPPT algorithm so that the input (array) side stays at the MPP while the output matches the battery or inverter; in power optimisers it is used at module level.

Short notes: specific gravity, autonomy days, luminous efficiency, THD (2019 6a)

- **Specific gravity:** the ratio of the density of the battery electrolyte (H_2SO_4 solution) to that of water. In a lead-acid cell it indicates the state of charge: about **1.265-1.28 fully charged** and about 1.10-1.12 discharged; measured with a hydrometer (temperature-corrected).
- **Autonomy days:** the number of consecutive days the battery can supply the full load without any charging from the PV array (cloudy days); typically 2-5 days. Battery capacity ∝ daily load × days of autonomy/(V × DOD).
- **Luminous efficiency (efficacy):** the light output of a lamp per watt of electric input, in lumens per watt (lm/W): incandescent ~15, CFL ~60, LED 100-150 lm/W. High-efficacy loads (LED) reduce the size of the PV system.
- **Total harmonic distortion:** a measure of how much a waveform deviates from a pure sine wave: THD (total harmonic distortion) = $\sqrt{(V_2^2 + V_3^2 + \ldots)}/V_1 \times 100\%$ (see B4).

Battery: teacher's problems (L5)

Step (write this)	Why / what it means
`100 Ah discharged in 20 h: I = 100/20 =` **`5 A`**`; in 5 h: I = 100/5 =` **`20 A`**	C/20 and C/5 rates; in practice the usable capacity falls at higher rates
`C-rating for 1, 2, 5, 10, 100 h: C/1 = 100 A, C/2 = 50 A, C/5 = 20 A, C/10 = 10 A, C/100 = 1 A`	C-rate = capacity/time
`Ah efficiency = Ah out/Ah in = (10 × 10)/(12 × 10) =` **`83.3%`**	charge 12 A × 10 h, discharge 10 A × 10 h
`Wh efficiency = (10 × 10 × 12)/(12 × 10 × 14.5) = 1200/1740 =` **`69%`**	also includes the lower discharge voltage

Remember: Controller: stop overcharge, stop deep discharge, block reverse current, control load. Set points VR 14.4 V, ARV (VR − VRH) 13.2 V, LVD 11.5 V, LRV (LVD + LVDH) 12.6 V. Shunt shorts the array; series opens the line. PWM: duty cycle shrinks. MPPT = impedance matching: P&O vs IncCond ($dI/dV = -I/V$). Buck $V_o = DV_i$; boost $V_o = V_i/(1 - D)$.

B4 Inverters: types by waveform and by mode, THD, converter efficiency and inverter selection numericals

Pass set · Possible · 47% · Easy

PAST-PAPER QUESTIONS

- **2022 Q6(c)** [7] A power converter has a specified input power of 300 W and a manufacturer-specified efficiency of 88%. How much is the power loss within the converter? What is the output power?
- **2021 Q6(c)** [8] A house has 3 × 60 W lights, 2 × 90 W fans, 1 × 60 W radio, AC loads rated 230 V. All loads are powered simultaneously. A 12 V battery is available. Choose the appropriate inverter, assuming inverter efficiency 90%.
- **2021 Q8(b)** [9] What is total harmonic distortion in solar inverters? How can total harmonic distortion in inverters be reduced? Discuss the different types of solar inverters based on output waveform.

★ REMEMBER IT

- Inverter waveforms **S-M-P**: Square (cheap, high THD (total harmonic distortion)), Modified sine (middle), Pure sine (clean, needed for motors, fridges, grid). THD = (harmonic content)/(fundamental); grid limit < 5%; reduce with SPWM (sinusoidal pulse width modulation), LC filters, multilevel inverters.

CALCULATOR **Converter loss and inverter selection (2022 6(c), 2021 6(c))**

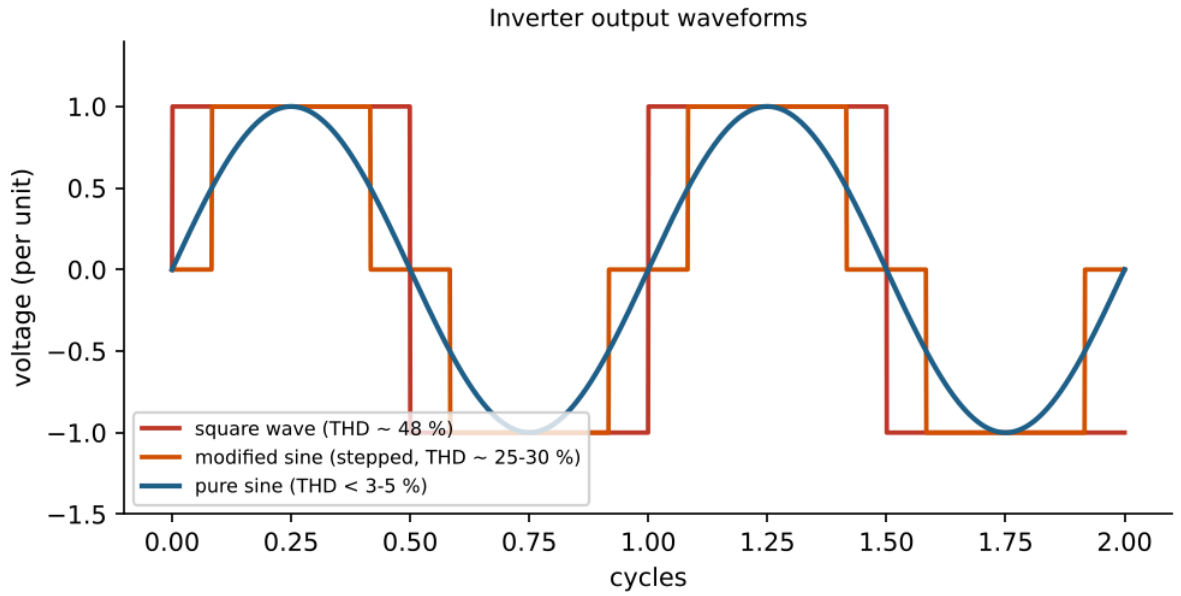
Change any number; every step is recalculated and shown with the numbers put in.

1. Converter output: **300** × **88** / 100 = **264.0** W
2. Converter loss: loss = $P_{in}(1 - \eta)$ = **300** - **264** = **36.0** W
3. Inverter input power: P_{in} = load/η = **420** / (**90** / 100) = **466.7** W
4. Battery current: **466.7** / **12** = **38.9** A
5. Inverter rating (25% margin): 1.25 × **420** = **525** W
6. AC current: **420** / **230** = **1.83** A

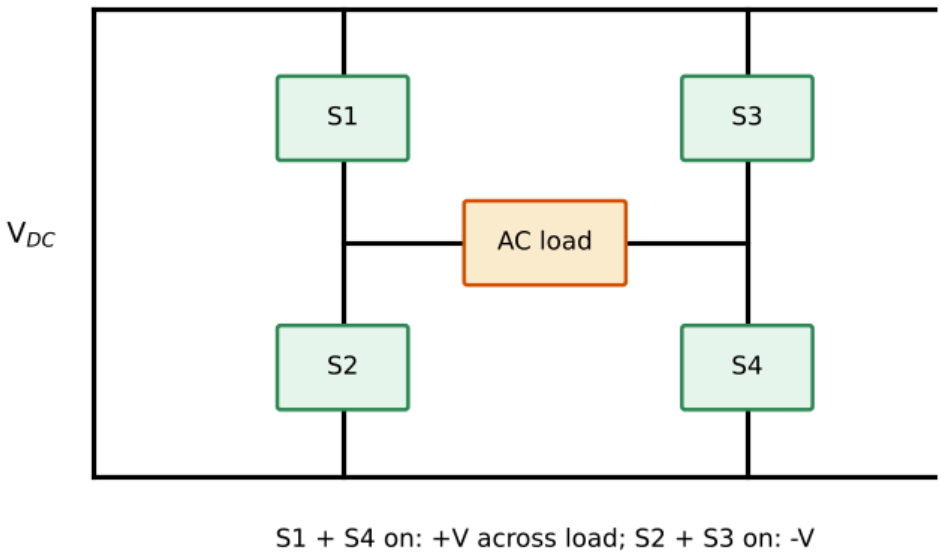
Before you start: DC in, AC out

PV modules and batteries produce DC, but most appliances and the grid use 230 V, 50 Hz AC. An **inverter** converts DC to AC, and also regulates voltage and frequency. Source: Muhit sir L8 (inverters, THD, converter problem); Messenger ch. 6.

Inverter types by output waveform, THD and how to reduce it (2021 8b)



Square, modified sine and pure sine inverter outputs



H-bridge inverter: switch pairs S1-S4 and S2-S3 alternate the polarity on the load

Type	Waveform and working	THD	Use
Square wave	H-bridge: S1-S4 on for half a cycle (+V), S2-S3 on for the next half (−V); output jumps between +V and −V	very high (~40-48%)	cheapest; only resistive loads (heaters, incandescent lamps); motors and electronics overheat or hum
Modified (quasi) sine wave	stepped wave: +V, 0, −V, 0 (the zero step is added by a short dead time); approximates the sine RMS	~20-25%	moderate cost; most simple loads, but some electronics, motors and chargers run hot or noisy
Pure sine wave	high-frequency PWM (sinusoidal pulse width modulation) of the bridge, then an LC filter smooths it to a near-perfect sine	< 3-5%	all loads, required for grid-tie; more expensive

Total harmonic distortion (THD) is the measure of the unwanted harmonic frequencies (multiples of 50 Hz) in a signal that make it deviate from an ideal sine: **THD = $\sqrt{(V_2^2 + V_3^2 + V_4^2 + \ldots)}/V_1$**, the ratio of the RMS of all harmonics to the fundamental. High THD heats motors and transformers, causes noise, malfunction of electronics, low power factor, and is not allowed on the grid (IEEE 519/1547 limit 5% current THD).

How to reduce THD: use pure sine wave topology with sinusoidal PWM (pulse width modulation) at high switching frequency (harmonics pushed to high frequency, easy to filter); output **LC/LCL filters**; multilevel inverters (more voltage steps); selective harmonic elimination by choosing switching angles; closed-loop current control; good dead-time compensation; correct sizing (no overloading); active power filters.

By operating mode: standalone (off-grid) inverters power loads from a battery independent of the grid; **grid-tied** inverters synchronise frequency and phase with the grid and feed PV power into it (MPPT, anti-islanding); **bimodal/hybrid** inverters work both ways: feed the grid and also supply loads from batteries when the grid fails. By structure: central, string, multi-string and micro-inverters.

Numerical: converter loss (2022 6c; L8)

Step (write this)	Why / what it means
$P_{out} = nP_{in} = 0.88 \times 300 =$ **264 W**	efficiency is output over input
Loss = $P_{in} - P_{out} = 300 - 264 =$ **36 W**	dissipated as heat in switches, inductors and transformer

Numerical: choosing an inverter (2021 6c)

Step (write this)	Why / what it means
AC load = 3 × 60 + 2 × 90 + 1 × 60 = **420 W**	all loads run at the same time, so add them
Inverter input power = 420/0.90 = **466.7 W**	the inverter loses 10%
DC current from battery = 466.7/12 = **38.9 A**	battery cables and fuse must carry this
Inverter rating with 25% safety margin = 1.25 × 420 = 525 W → choose a standard **600 VA (≥ 500 W), 12 V DC input, 230 V 50 Hz pure sine wave** inverter	margin for fan starting surge and future load; pure sine because fans (motors) and radio need low THD
AC output current = 420/230 = 1.83 A	unity power factor assumed

Memory trick: Inverter waveform = **S-M-P**: Square (cheap, dirty), Modified (middle), Pure (clean, grid). THD = harmonics ÷ fundamental.

Remember: Inverter rating = 1.25 × simultaneous load, input = load/η, DC current = input/$V_{battery}$. 300 W at 88% → 264 W out, 36 W loss. THD < 5% for grid; reduce by SPWM + LC filter + multilevel.